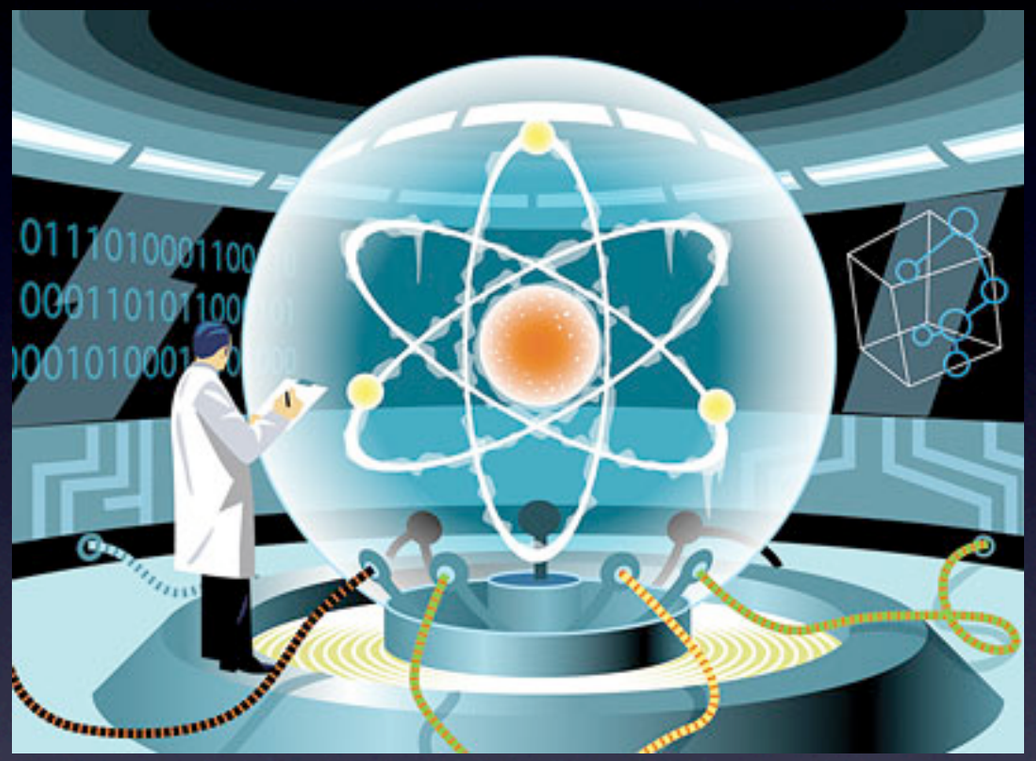
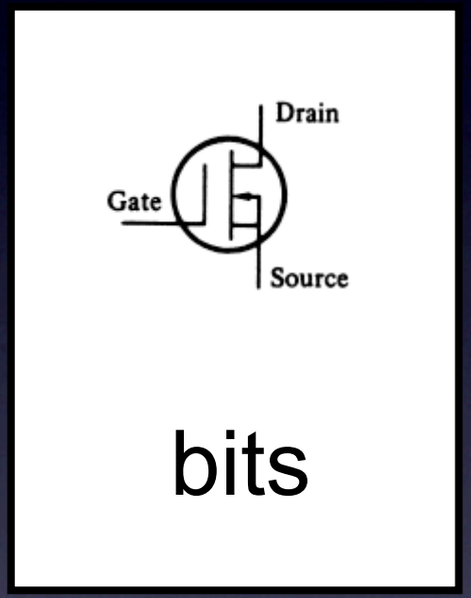
Quantum Computing



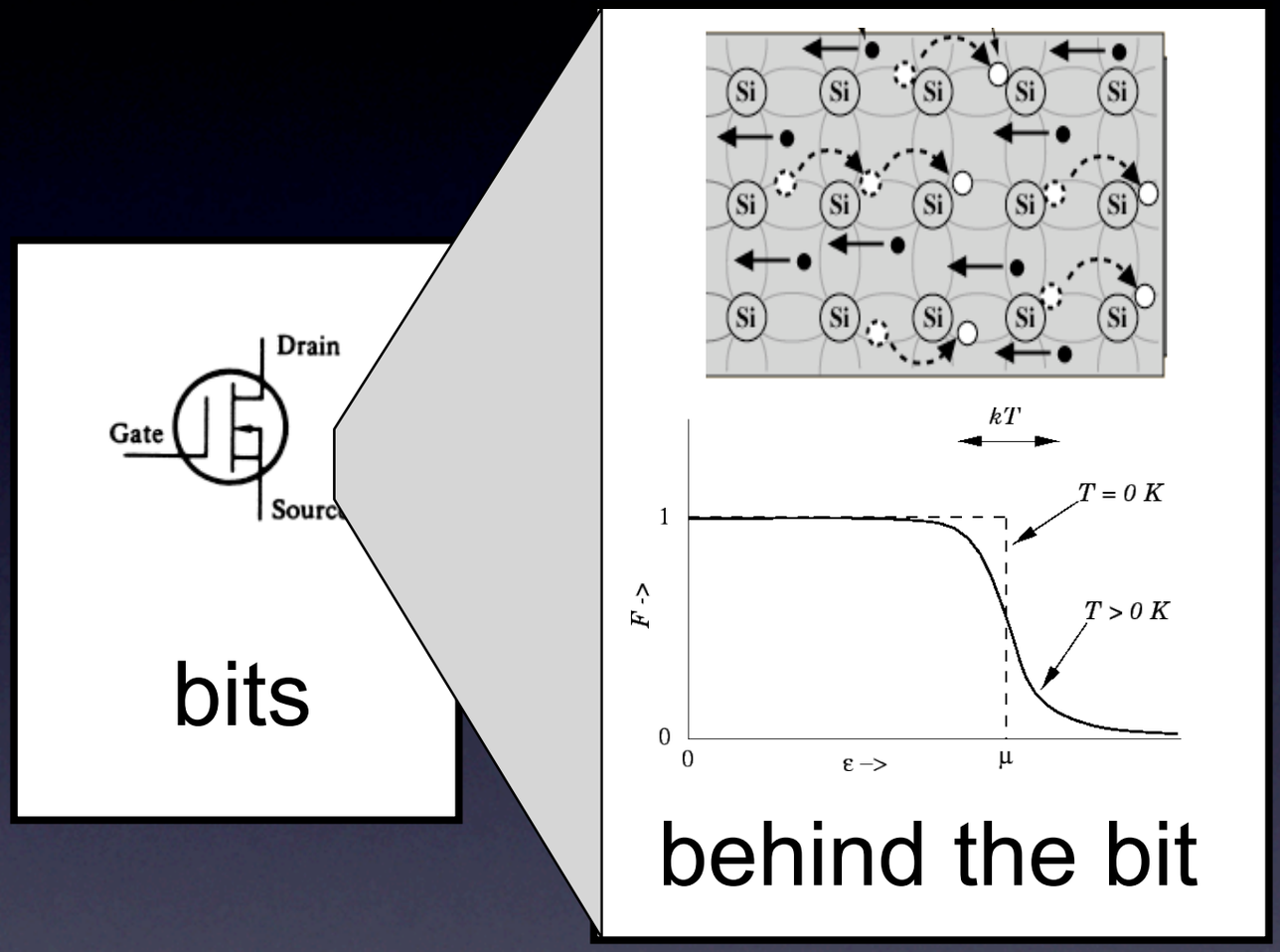
Dave Bacon
University of Washington

A Mystery



bits

A Mystery



How is $\int \rightarrow \{0, 1\}$ possible?

Category Errors

Analog / Digital



vs



0.9982131123....

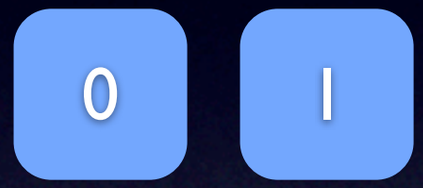
0

Category Errors

Analog / Digital



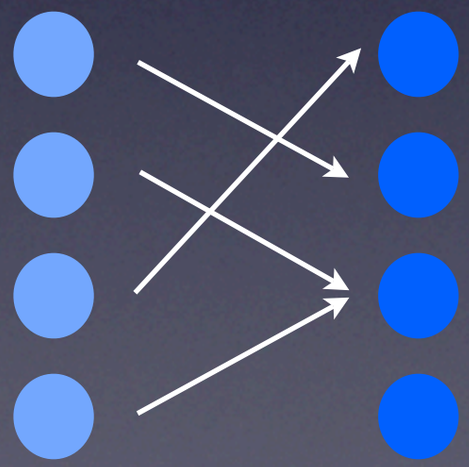
vs



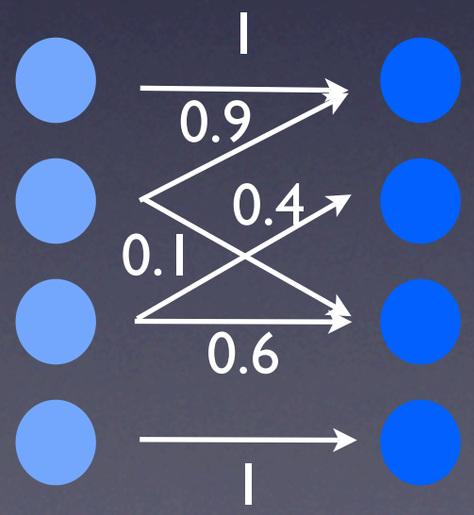
0.998213123....

0

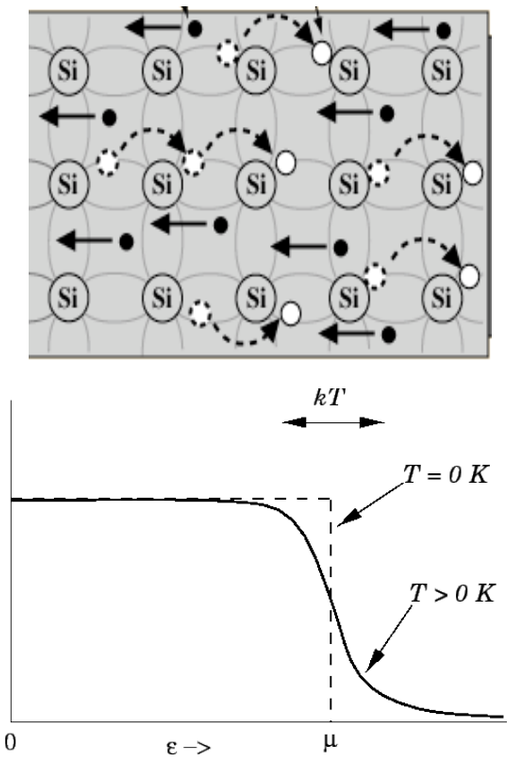
Probabilistic / Deterministic



vs



Analog Probabilistic
Digital Deterministic



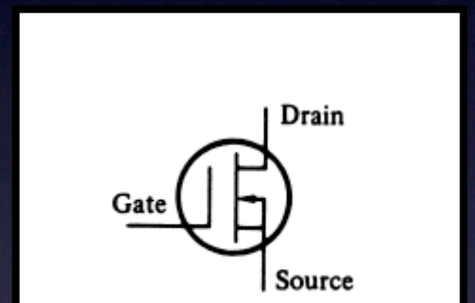
behind the bit

Analog
Voltages

Probabilistic
Electrons

Digital
Signal

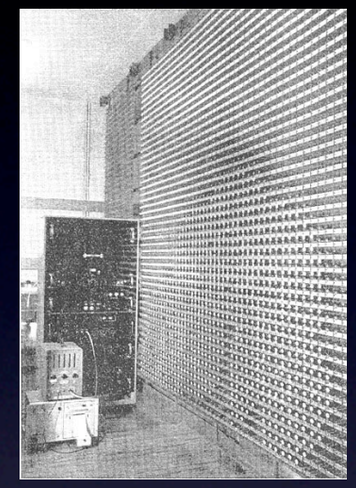
Deterministic
Gates



bits

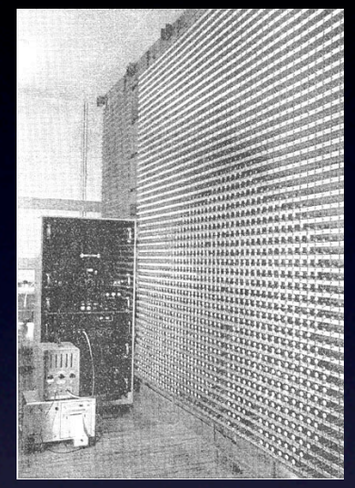
Alternative Models

- Analog Computers
  - Infinite precision analog computers can efficiently solve NP-complete problem



Alternative Models

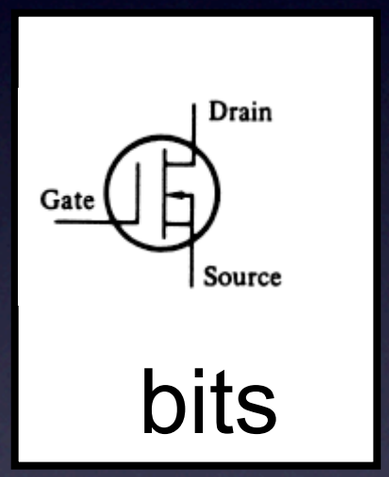
- Analog Computers
  - Infinite precision analog computers can efficiently solve NP-complete problem



- Probabilistic Computers
  - Does randomness help build more efficient computers?
  - Practice: of course!
  - Theory: open question

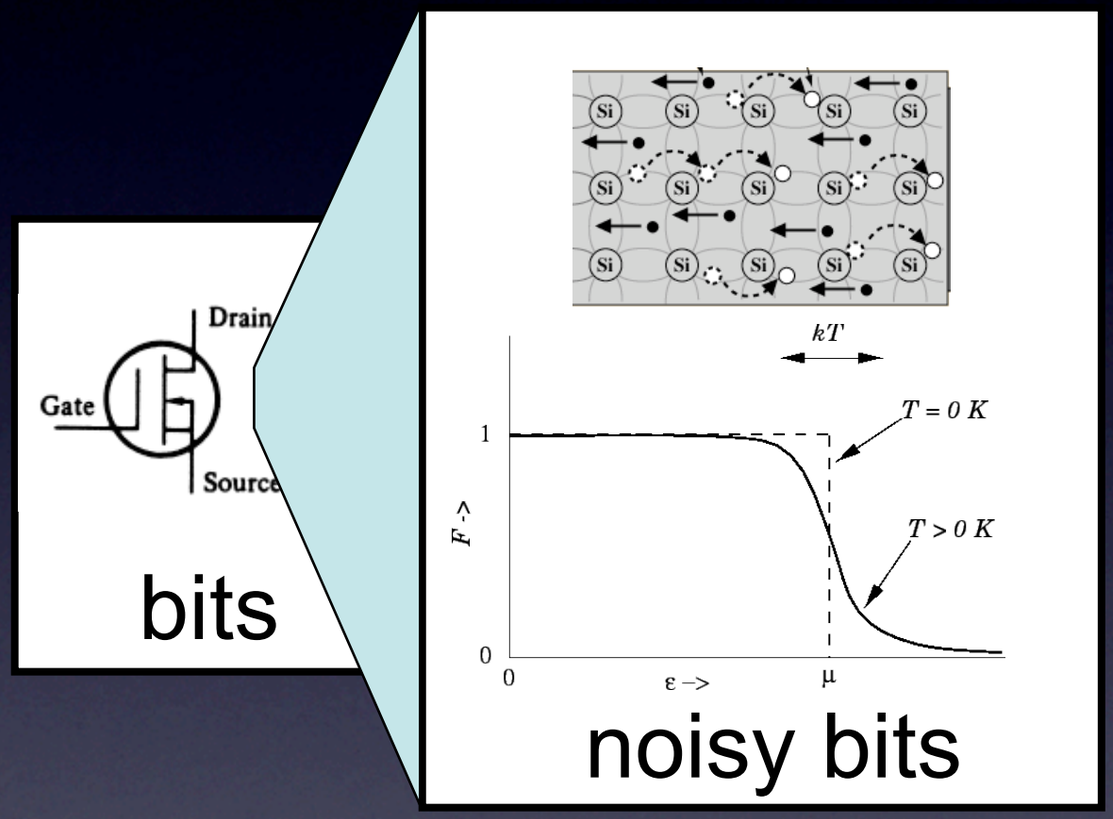


Deeper



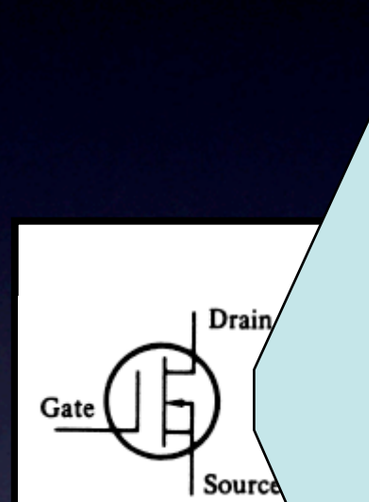
Deterministic / Probabilistic / Quantum

Deeper

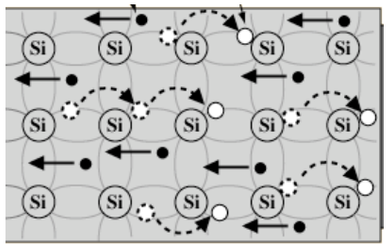


Deterministic / Probabilistic / Quantum

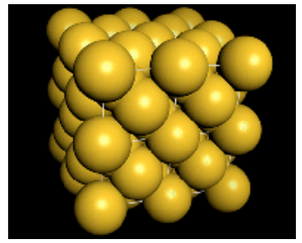
Deeper

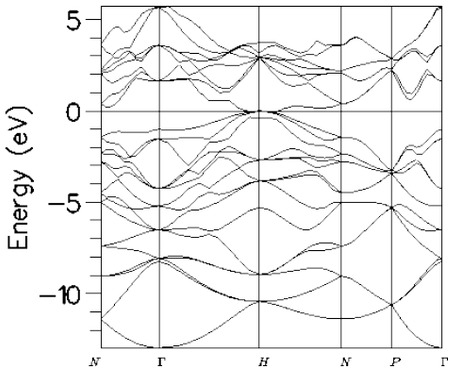


bits



noisy bits

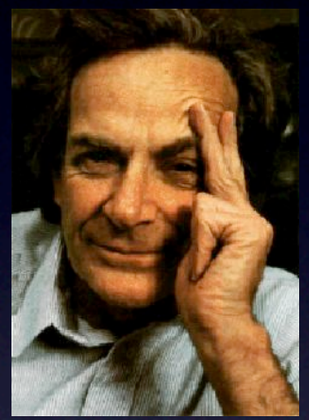




quantum bit

Deterministic / Probabilistic / Quantum

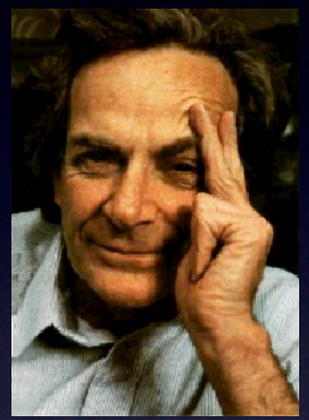
Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965

Quantum Propaganda



I think I can safely say that nobody understands quantum mechanics.

Richard Feynman
Nobel Prize 1965



Anyone who is not shocked by quantum theory has not understood it.

Niels Bohr
Nobel Prize 1922

Hogwash



- They understand it
- They use it
- But do they believe it?

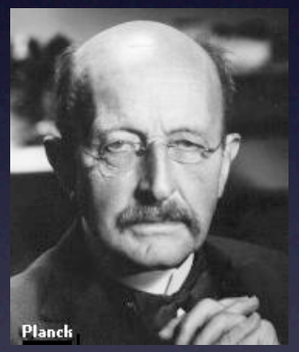
Quantum History



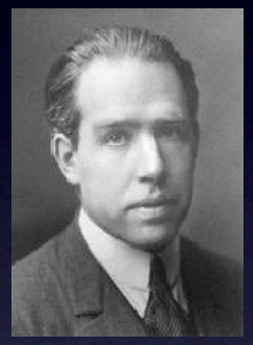
Quantum History

Inventing

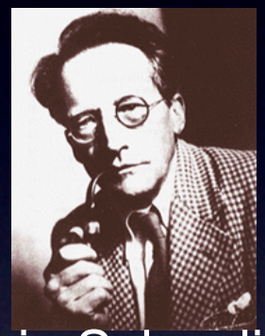
1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000



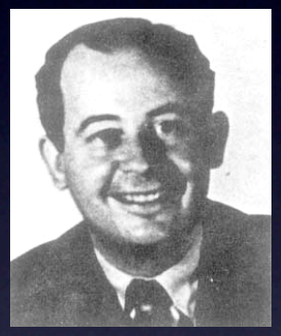
Max Planck



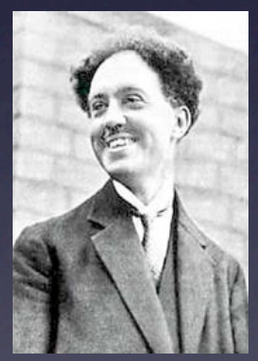
Niels Bohr



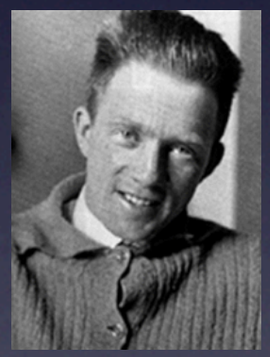
Erwin Schrodinger



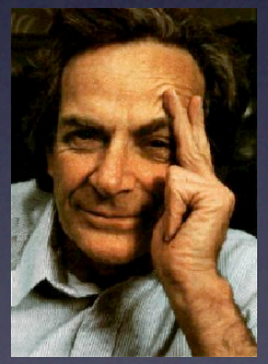
John Von Neumann



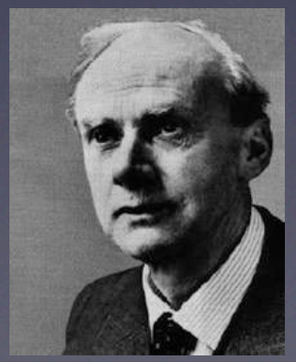
Louis de Broglie



Werner Heisenberg



Richard Feynman

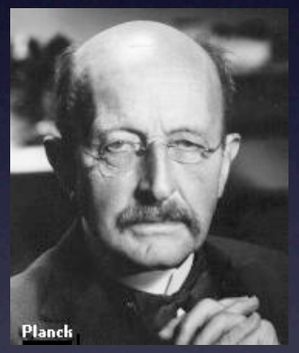


P.A.M. Dirac

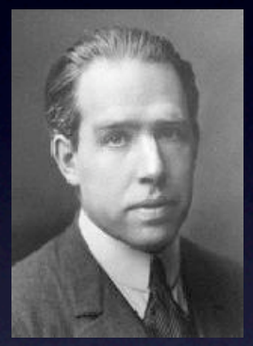
Quantum History

Inventing Worrying Accepting?

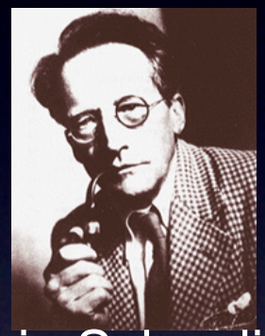
1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000



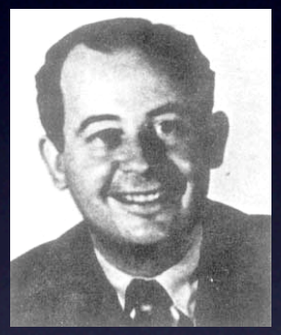
Max Planck



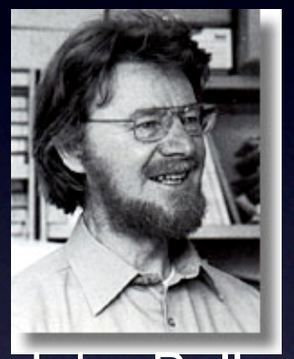
Niels Bohr



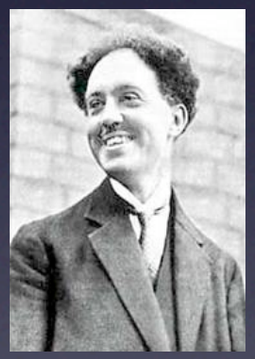
Erwin Schrodinger



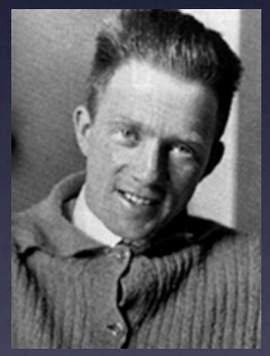
John Von Neumann



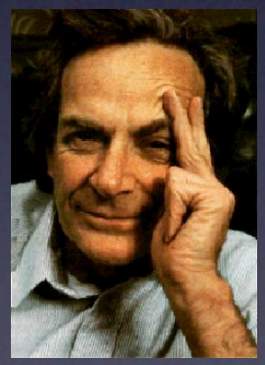
John Bell



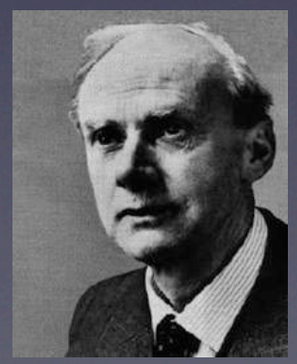
Louis de Broglie



Werner Heisenberg



Richard Feynman



P.A.M. Dirac

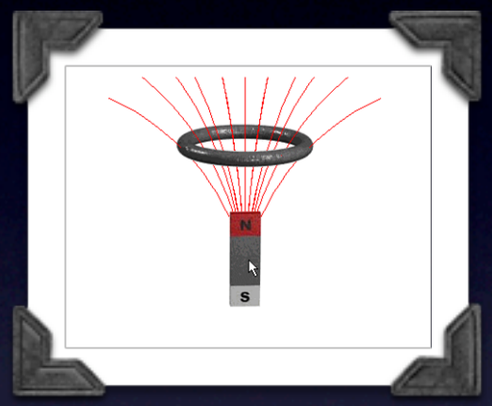
Worrying?



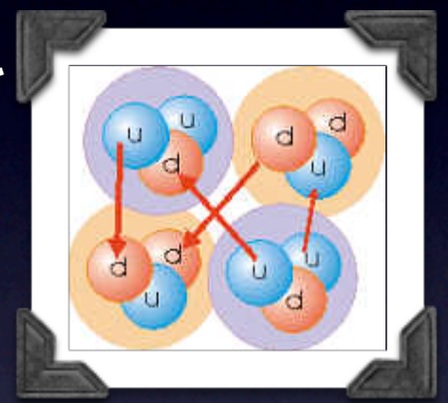
Quantum Theory

Worrying?

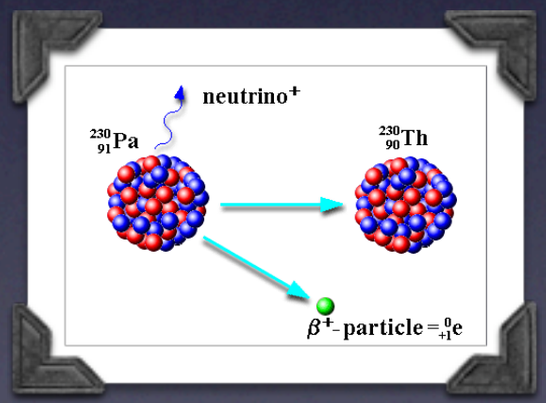
Electromagnetism



Strong force

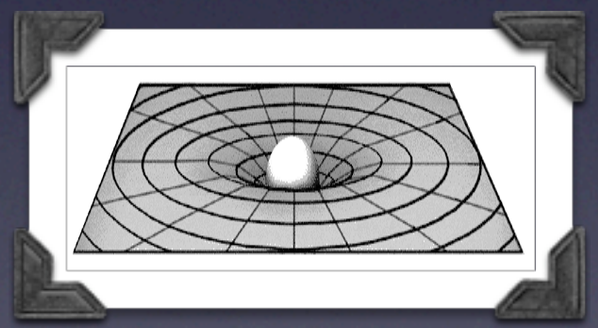


Weak force





Gravity ?!?!



Quantum Theory

Stop Worrying and Love Quantum Theory



Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

x

Quantum Computers

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199692321205734031879550656996221305168759307650257059

=

3980750864240649373971255005503864911990643623425267084063851895759463889572617685833172727214610743530253622307197304822463291469530209711645985217113052071125636359039752



Peter Shor, 1994:

- Quantum computers can efficiently factor
- No known classical efficient algorithm.
- Breaks RSA, Diffie-Hellman cryptosystems.

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:			
Observation:			
Evolution:			

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$ $p \in \mathbb{R}$	
Observation:			
Evolution:			

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:			
Evolution:			

Bits, Pbits, Qubits

	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	
		1 *1-p* percent	
Evolution:			

Bits, Pbits, Qubits

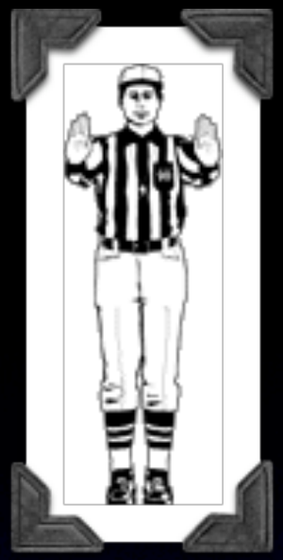
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	0 $\lvert\alpha\rvert^2$percent
		1 *1-p* percent	1 $\lvert\beta\rvert^2$percent
Evolution:			

Bits, Pbits, Qubits

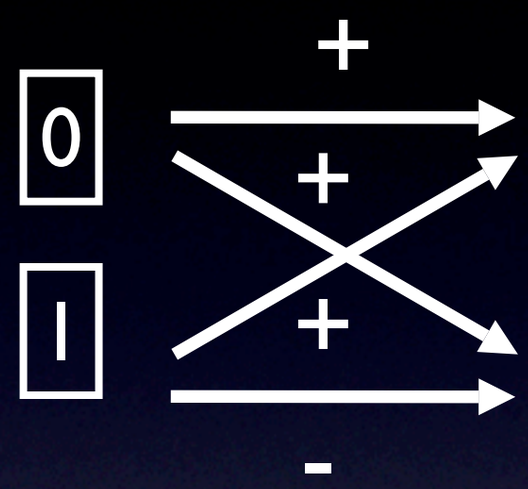
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	0 $\lvert\alpha\rvert^2$percent
		1 *1-p* percent	1 $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$	
	deterministic	stochastic	

Bits, Pbits, Qubits

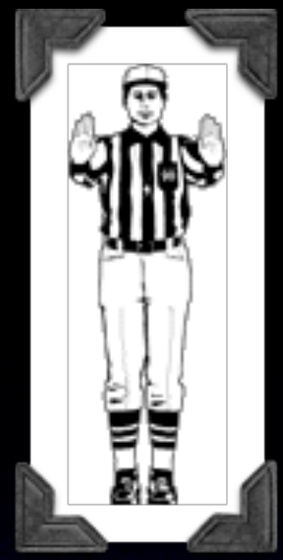
	bit	probabilistic bit	quantum bit
Configurations:	0 1	0 1	0 1
Description:	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} p \\ 1-p \end{bmatrix}$	$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
		$p \in \mathbb{R}$	$\alpha, \beta \in \mathbb{C}$
Observation:	0 certainty	0 *p* percent	0 $\lvert\alpha\rvert^2$percent
		1 *1-p* percent	1 $\lvert\beta\rvert^2$percent
Evolution:	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1-q & r \\ q & 1-r \end{bmatrix}$	$\begin{bmatrix} u & v \\ w & x \end{bmatrix}$
	deterministic	stochastic	unitary



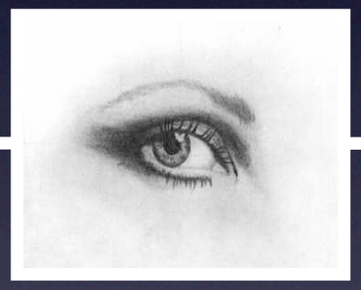
Interference



$$\begin{matrix} & 0 & 1 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{matrix}$$



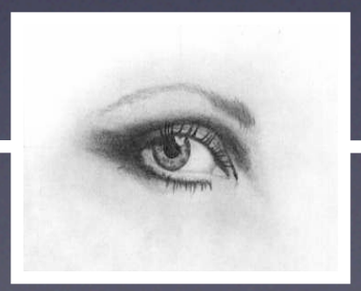
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$



0 50%

1 50%

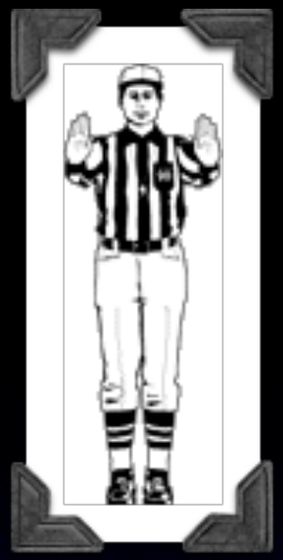
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$



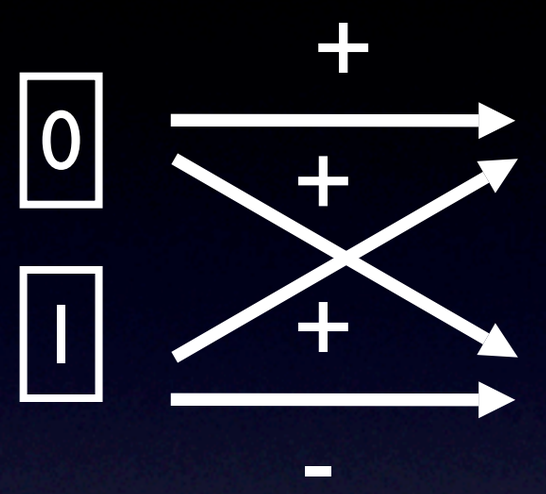
0 50%

1 50%

measure

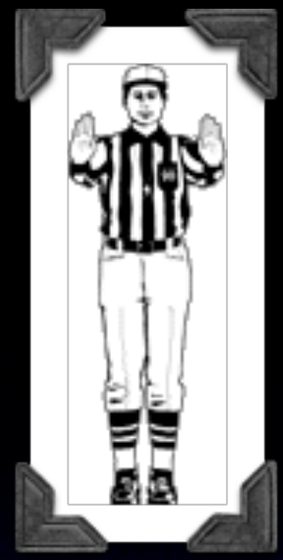


Interference

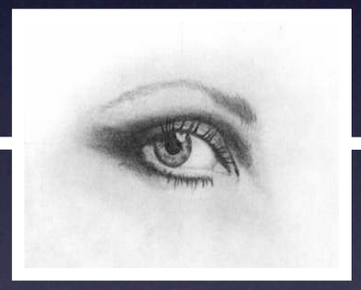


$$\begin{matrix} 0 & 1 \end{matrix}$$

$$\begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



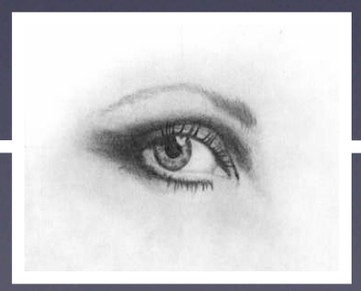
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



0 100%

1 0%

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

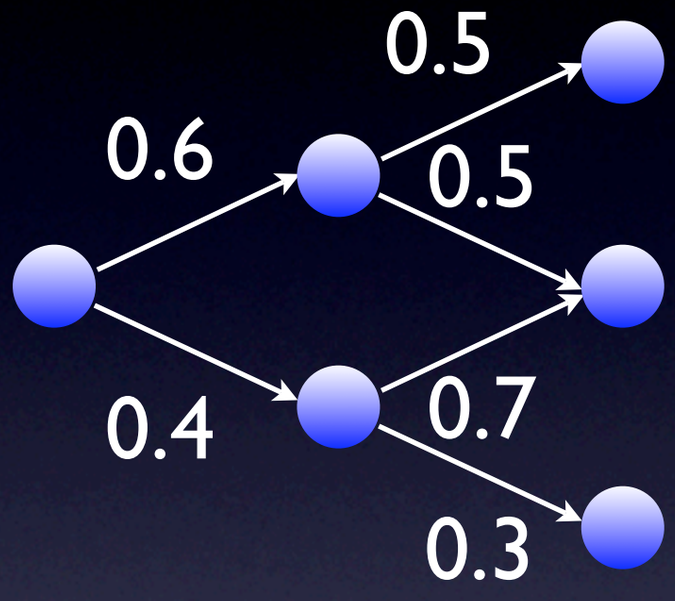


0 0%

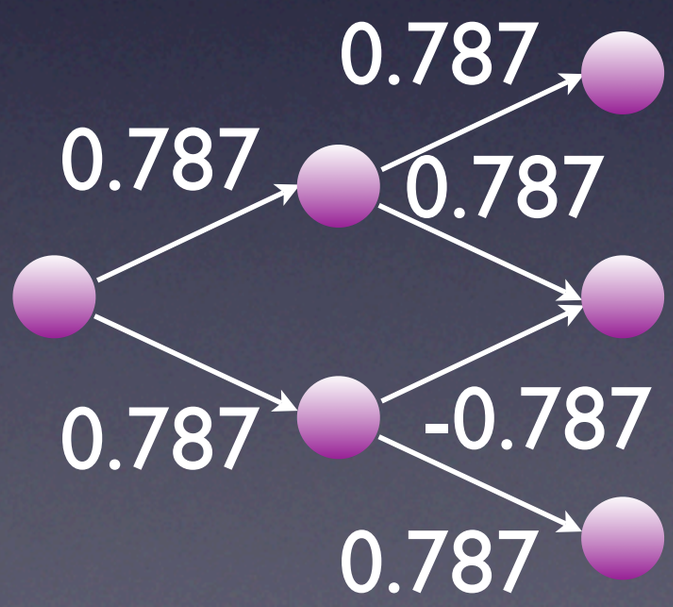
1 100%

evolve measure

Garden of Re-forking Paths



$(0.6)(0.5)+(0.4)(0.7)=0.58$



$|(0.787)(0.787)+(-0.787)\ (0.787)|^2=0$

Deutsch's Problem



David Deutsch

Dr. Falcon





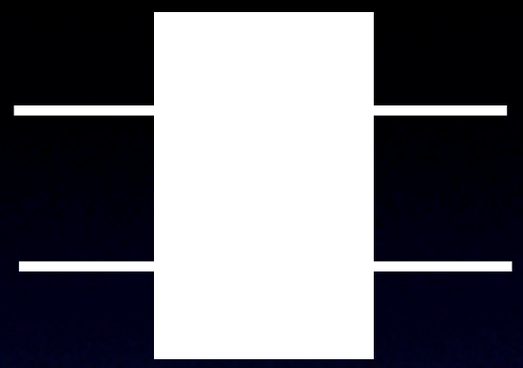
$xy \to uv$	$xy \to uv$	$xy \to uv$	$xy \to uv$
$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$

constant

balanced

Deutsch's Problem

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$



$00 \to 00$
$01 \to 01$
$10 \to 10$
$11 \to 11$

$00 \to 01$
$01 \to 00$
$10 \to 11$
$11 \to 10$

$00 \to 01$
$01 \to 00$
$10 \to 10$
$11 \to 11$

$00 \to 00$
$01 \to 01$
$10 \to 11$
$11 \to 10$

constant

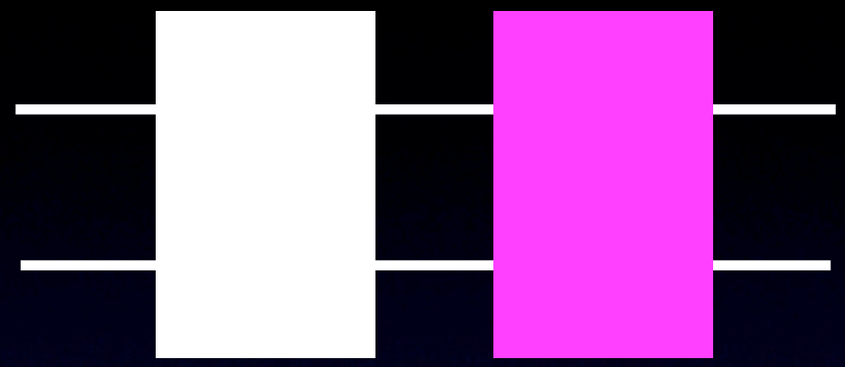
$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \qquad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

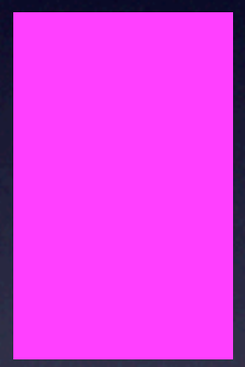
balanced

$$\begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \qquad \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Deutsch's Problem

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$





$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

constant

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

01

balanced

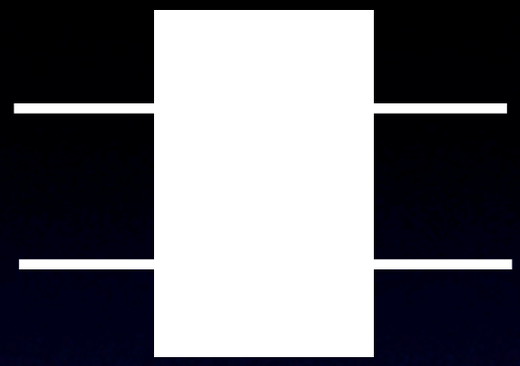
$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

11

Deutsch's Problem



1985



- Classically: 2 queries
- Quatumly: 1 query

Might the computational complexity depend on physics?

Beyond a Curiosity



Quantum computers offer superpolynomial query complexity speedups over classical computers [Bernstein, Vazirani 1993]

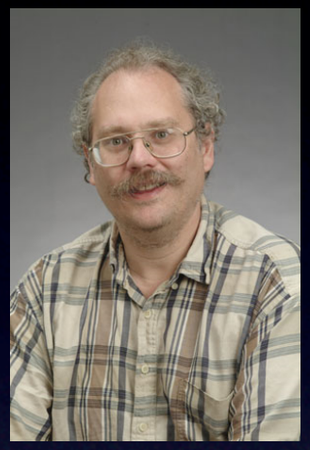
Umesh Vazirani
(Berkeley)

Quantum computers offer exponential query complexity speedups over classical computers [Simon 1994]



Dan Simon
(Microsoft)

Shor's Breakthrough



Peter Shor

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570599807508642406493739712550055038649119906436234252670840638518957594638895726176858331794727721461074353025362307197304822463291469530209711645985217113052071125636359039752747727214610743530253623071973048224632914695302097116459852171130520711256363590397527

=

Best classical algorithm	Shor's quantum algorithm
$O\left(\exp(n^{\frac{1}{3}})\right)$	$O\left(n^3\right)$

Hyperbole

Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world's most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security



- Simon Singh

Landscape

Quantum Entanglement

Quantum Error Correction

Experimental Quantum Computing

Quantum Information Theory

Quantum Computer Science

ion traps

quantum dots

superconducting circuits

linear optics

Quantum Cryptography

Quantum Algorithms

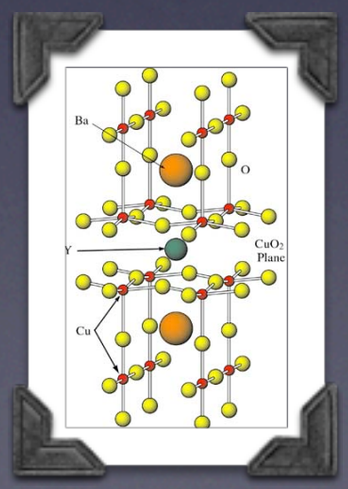
Quantum Algorithms

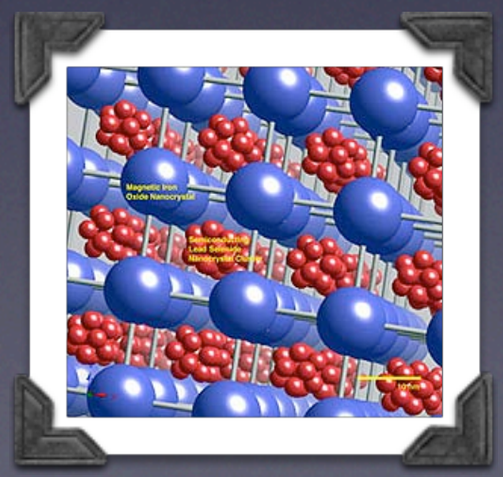
Exponential speedup

Shor's algorithm
- Breaks RSA, Diffie-Helman
- Only a few alternatives

which are note broken by quantum computers

Simulation of quantum physics





Polynomial speedup

Grover's algorithm
- quadratic speedup in needle-in-haystack search

NAND tree algorithm
- quadratic speedup in game algorithms

Amplitude amplifiction
- Ubiquitous quadratic speedups

Grover's Algorithm

x	f(x)
0	0
1	0
2	0
3	0
4	1
5	0
6	0
7	0

By querying f, find single marked item

Classical: O(N)

Quantum: $O(N^{1/2})$

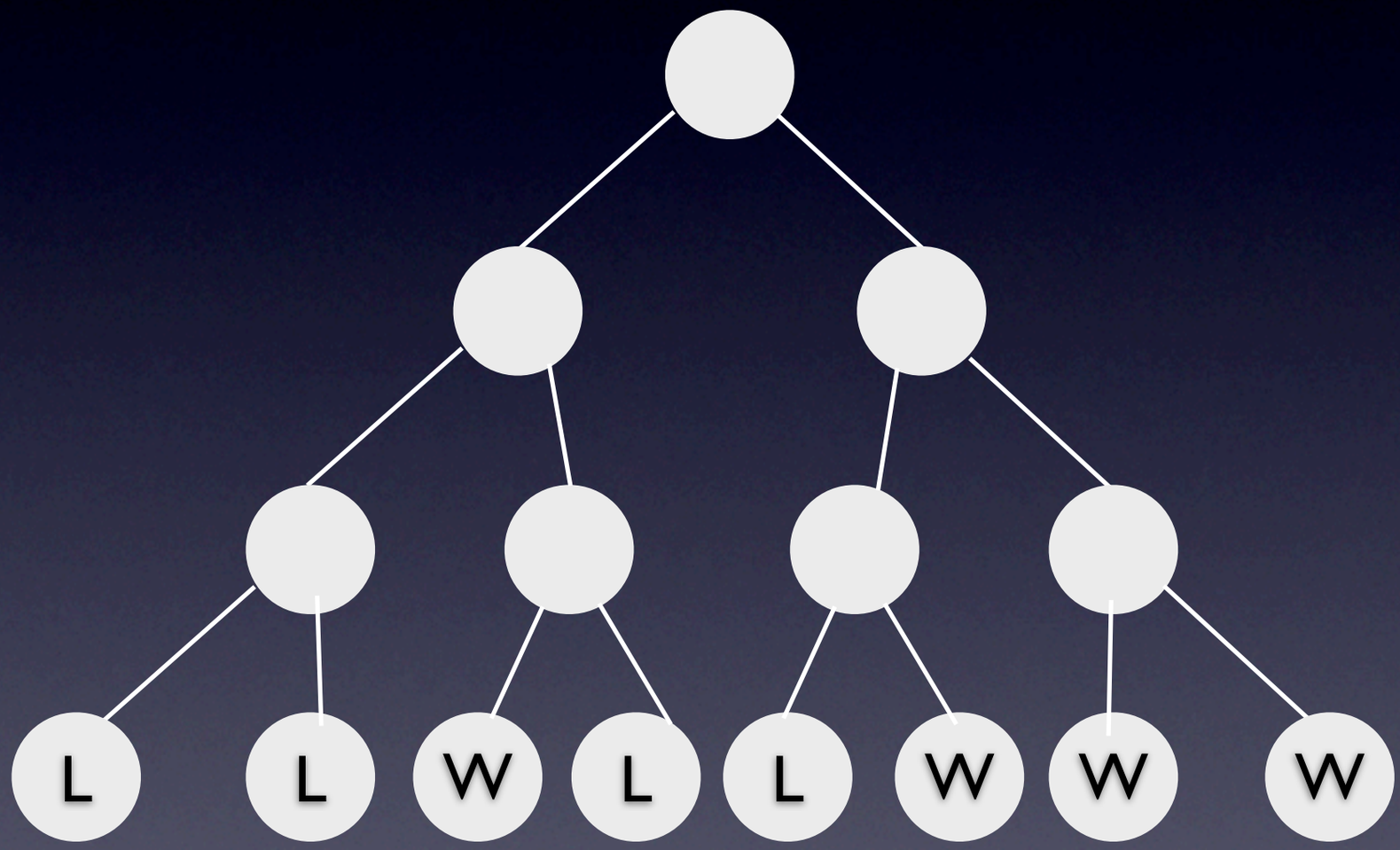
Sometimes erroneously called quantum database search

Games

player 1

player 2

player 1

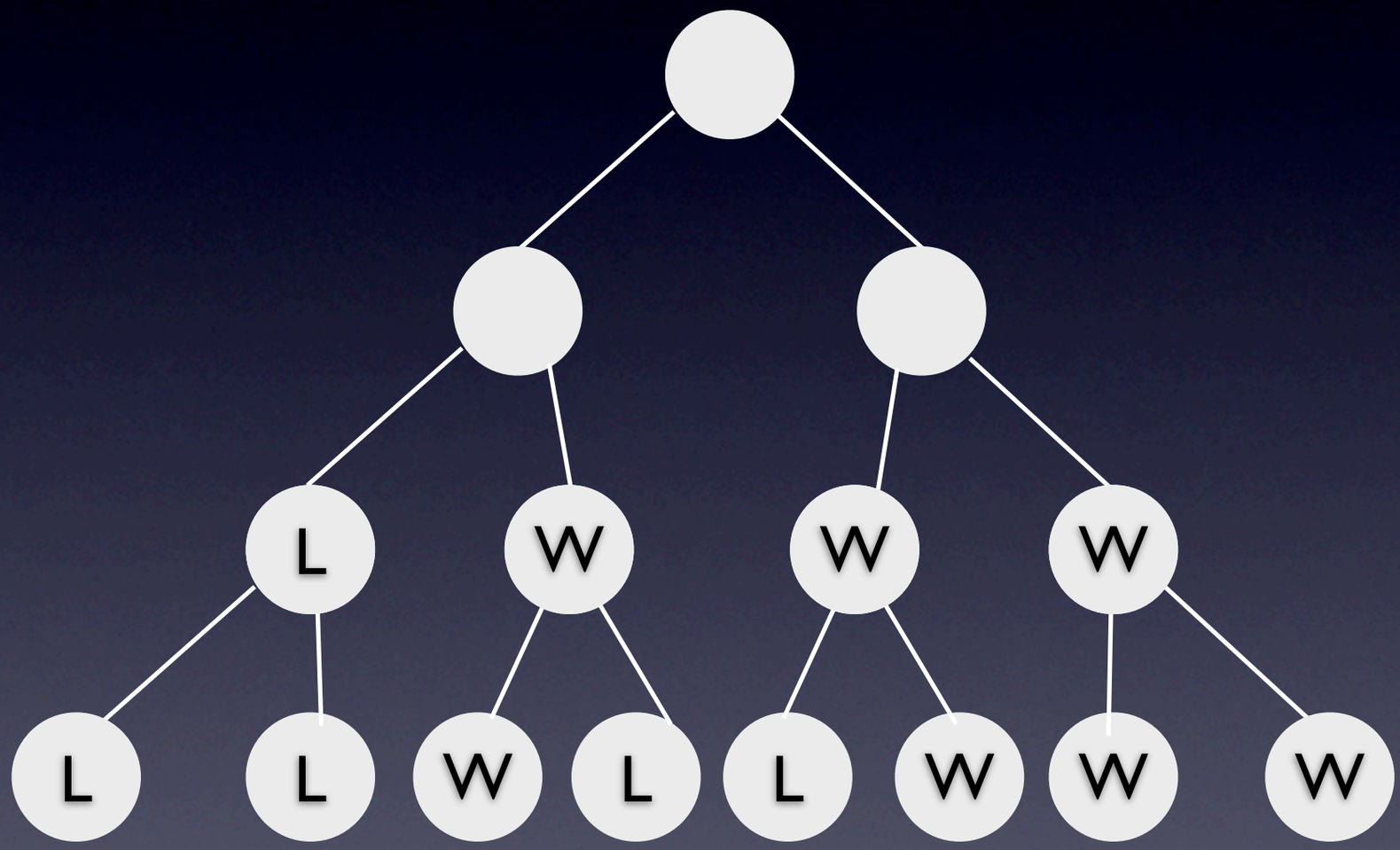


Games

player 1

player 2

player 1

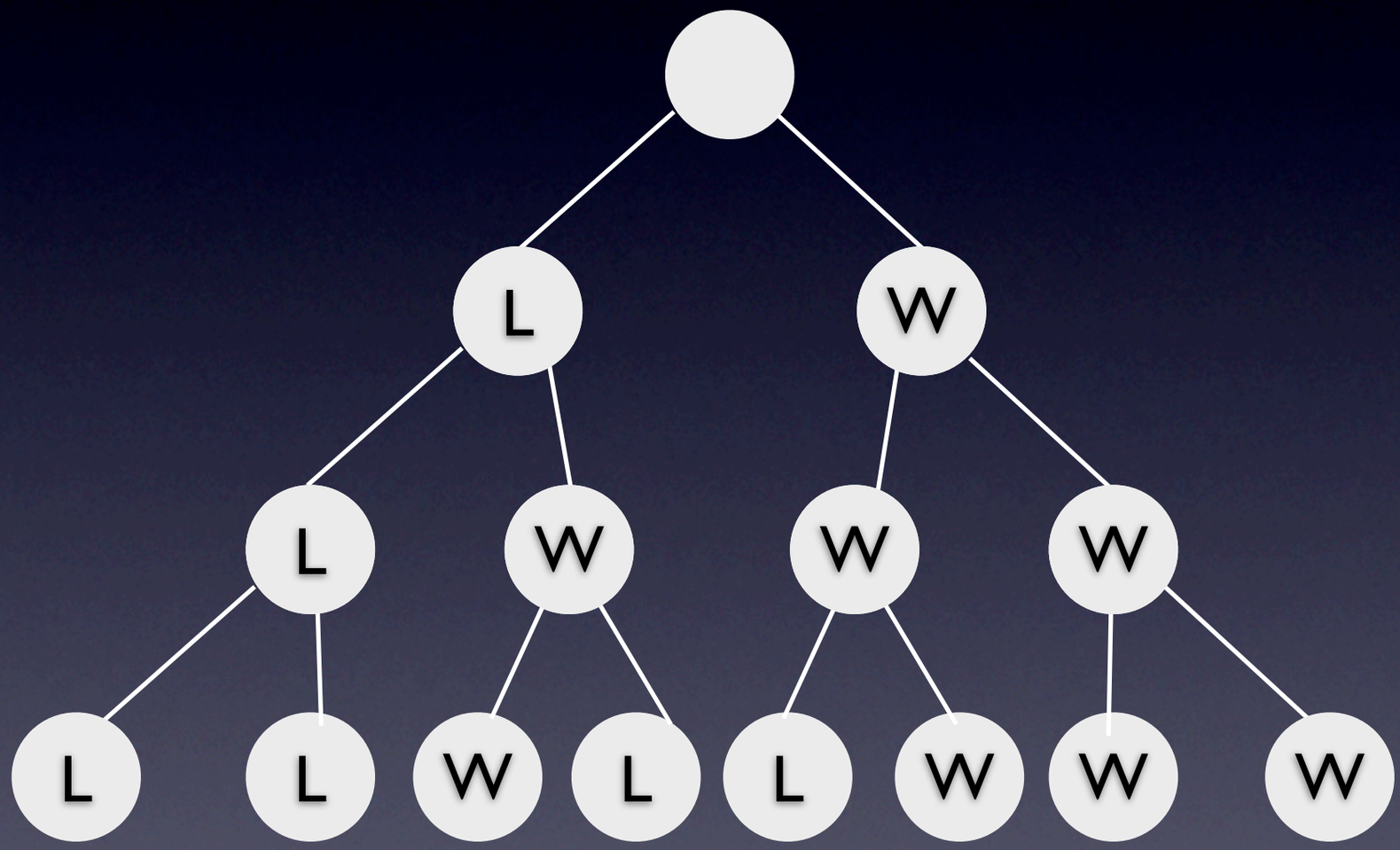


Games

player 1

player 2

player 1

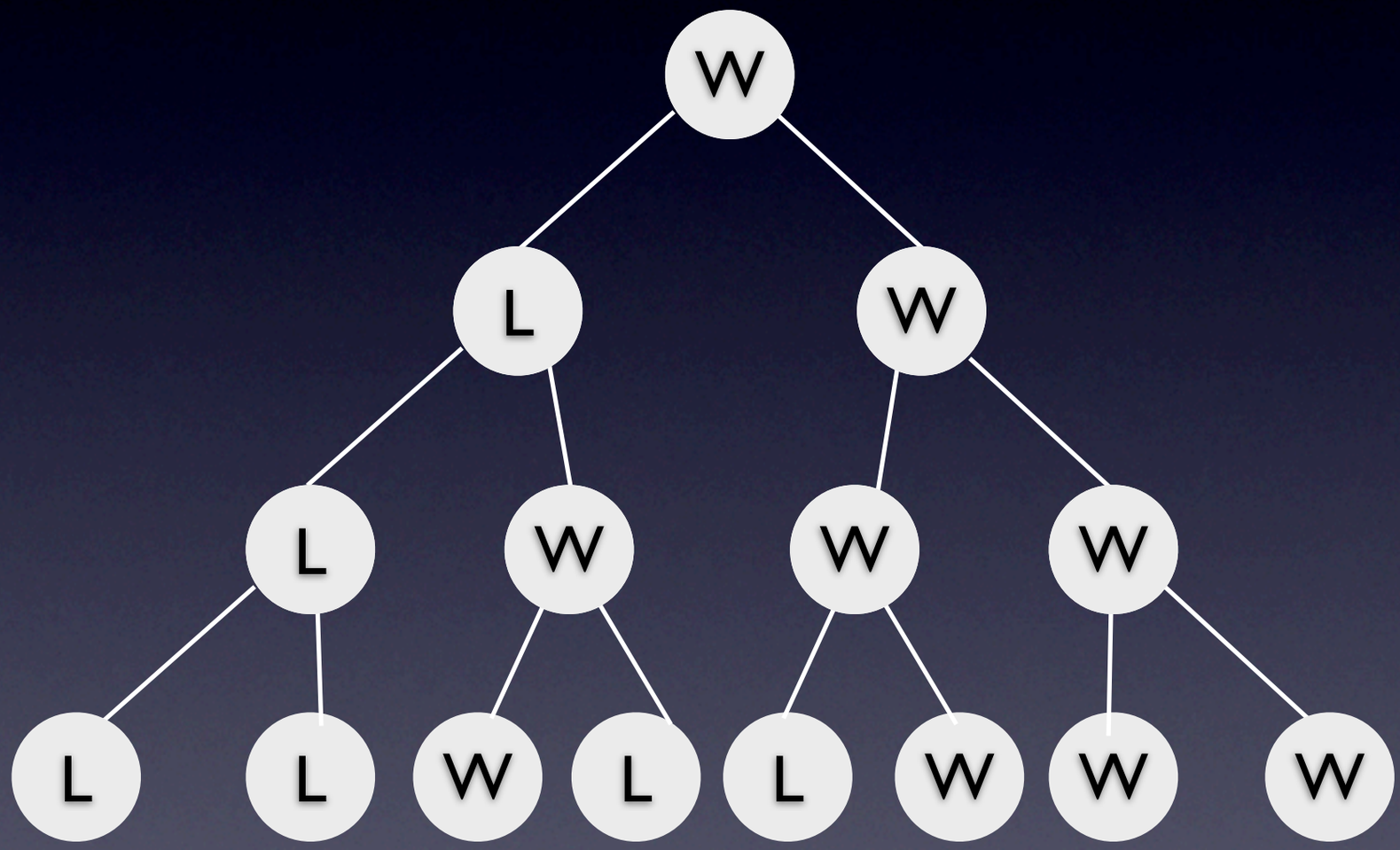


Games

player 1

player 2

player 1

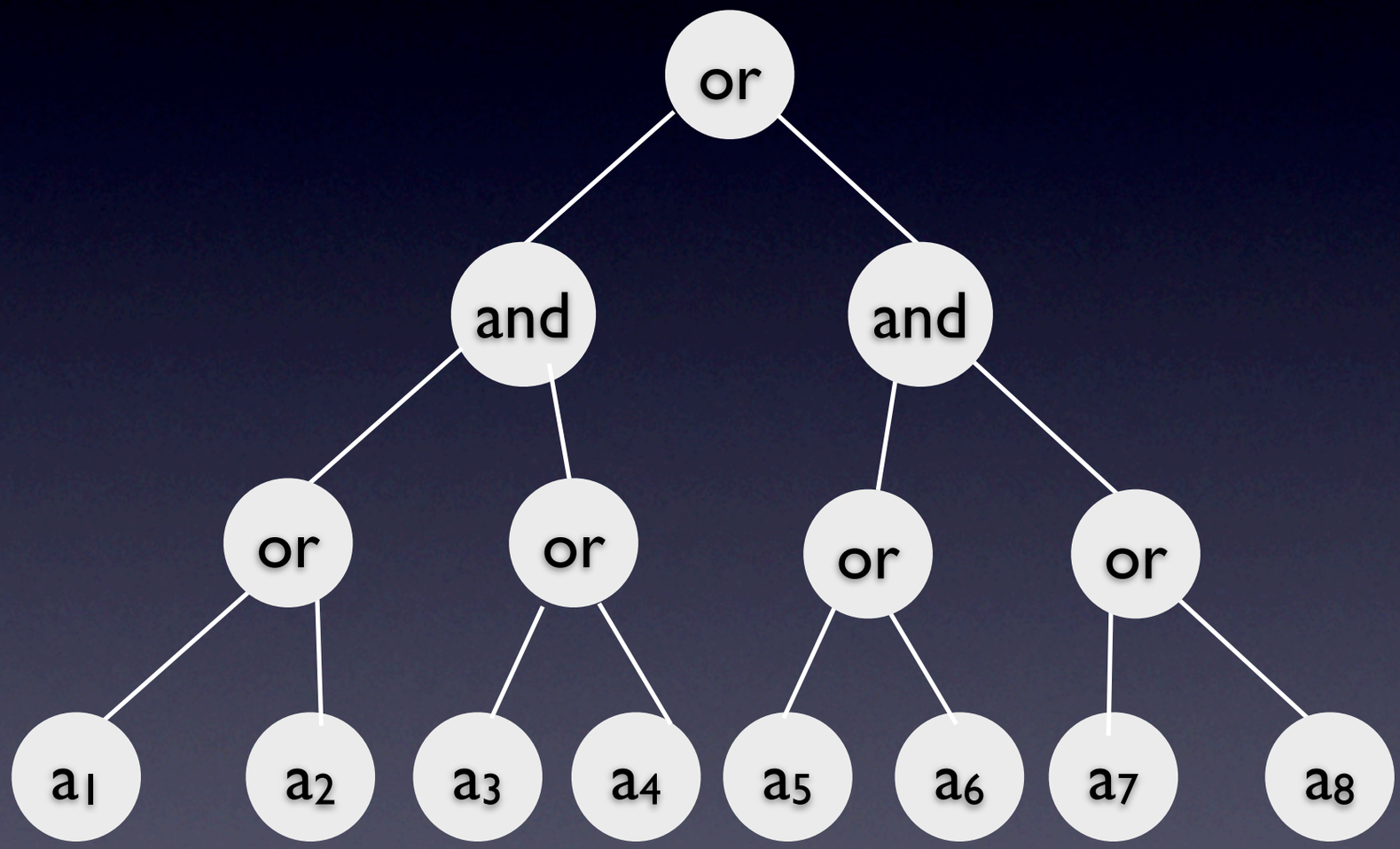


Games

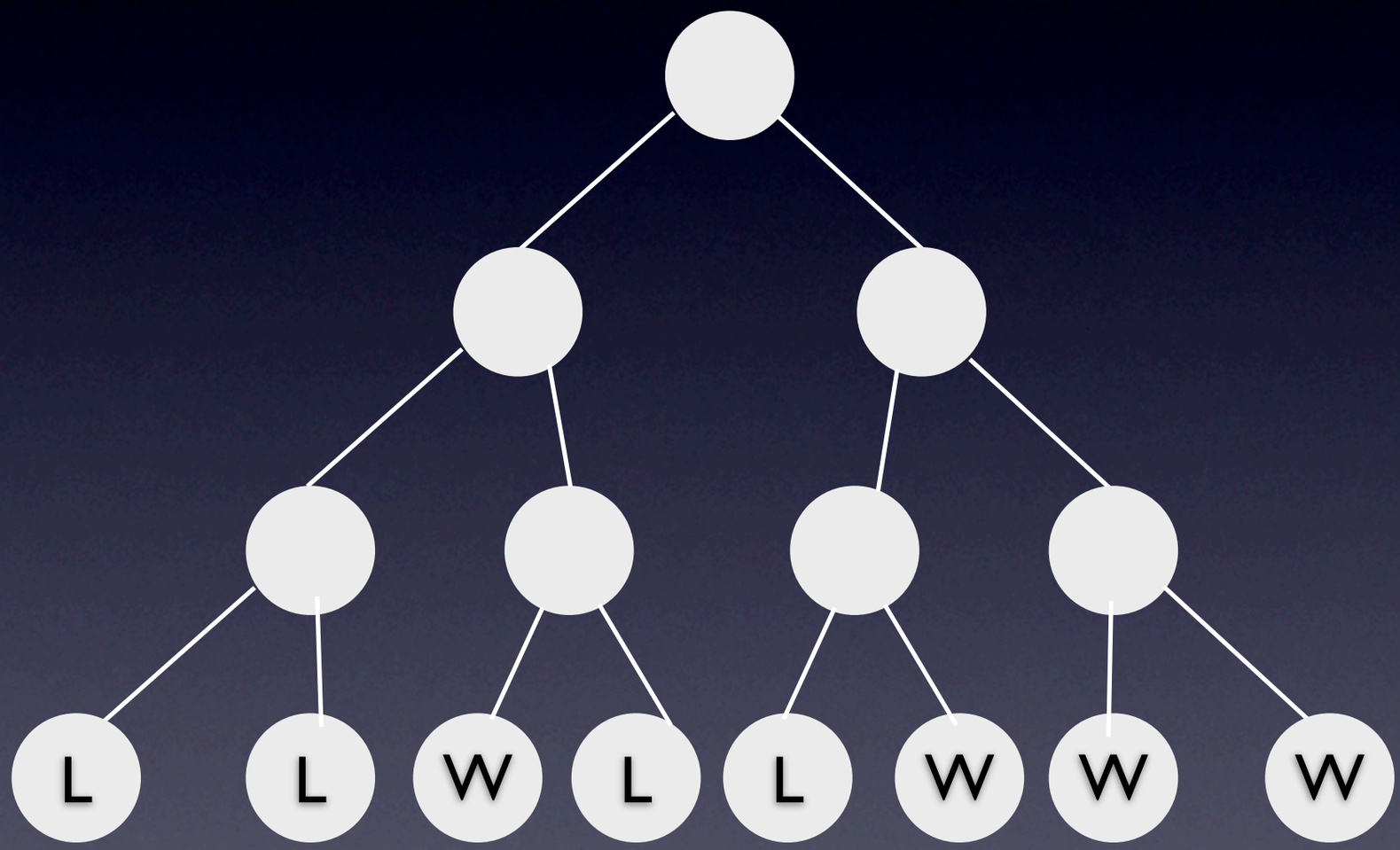
player 1

player 2

player 1

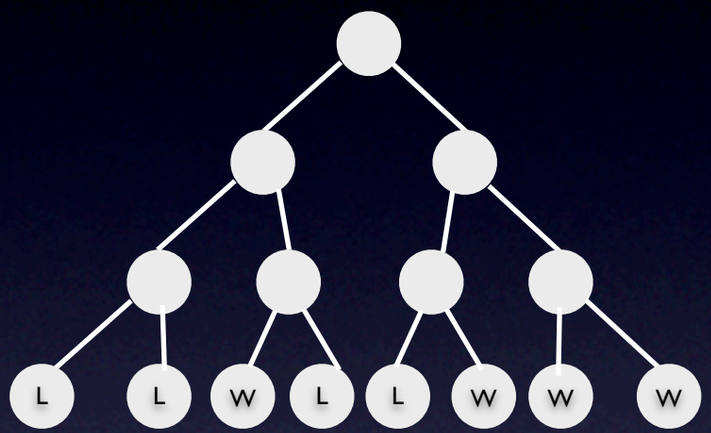


AND-OR Tree Evaluation



How many leaves do you need to query to compute value of root?

AND-OR Tree Evaluation



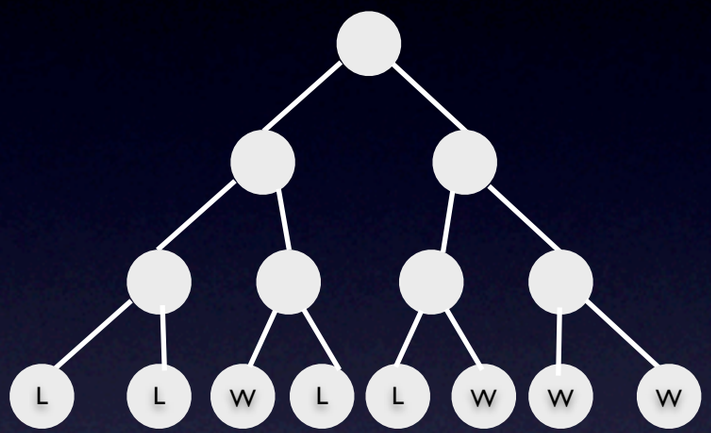
Classically (alpha-beta pruning)

$\Theta(N^\alpha)$

$\alpha = \log_2(1 + \sqrt{33}) - 2$

$\alpha \approx 0.75$

AND-OR Tree Evaluation



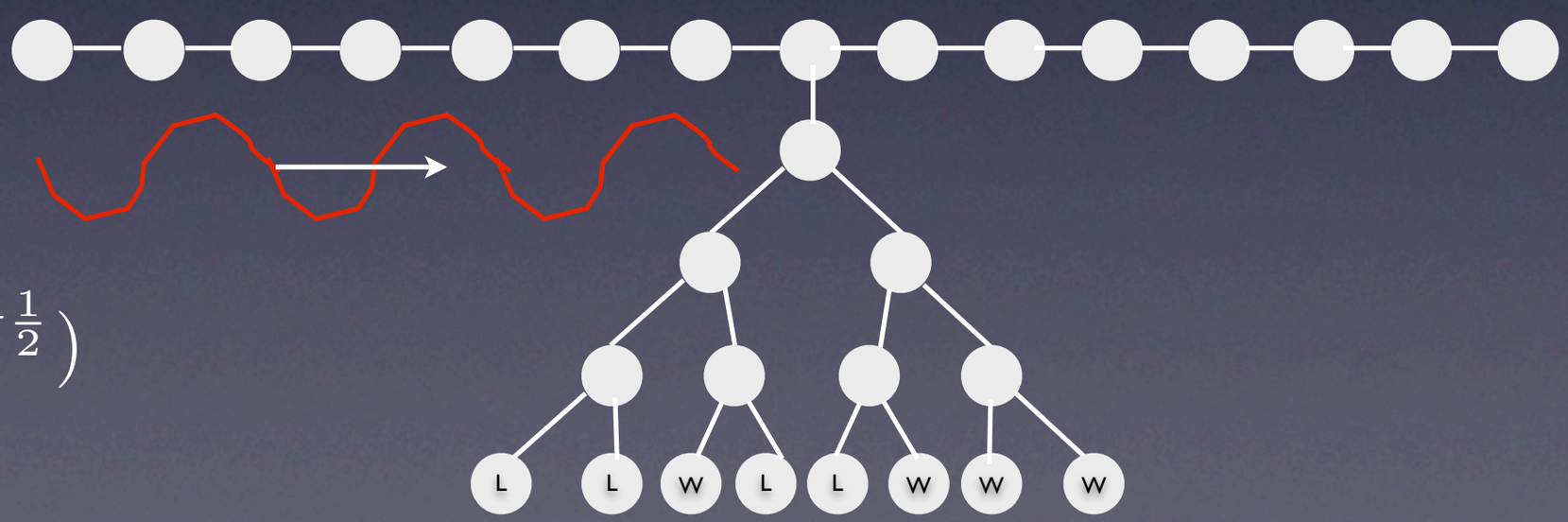
Classically (alpha-beta pruning)

$\Theta(N^{\alpha})$

$\alpha = \log_2(1+\sqrt{33}) - 2$

$\alpha \approx 0.75$

Quantum [Farhi, Goldstone, Gutman 2007]



$\Theta(N^{\frac{1}{2}})$

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ion traps

quantum dots

superconducting circuits

linear optics

Quantum Cryptography

Quantum Algorithms

Quantum Cryptography

What quantum computing taketh away it also giveth (sort of)

Key distribution system

- can detect if eavesdropped upon
- not secure to man-in-the-middle (i.e. you need some sort of authentication.)
- best thought of as key amplification
  - small initial key multiplied to nearly unlimited key

Quantum Cryptography



$$|+\rangle\ |1\rangle\ |1\rangle\ |0\rangle\ \cdots$$



Alice

Bob

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|+\rangle \quad |-\rangle \quad |1\rangle \quad |0\rangle$$

Quantum Cryptography





Alice

Bob

$$\left[\begin{array}{c}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{array}\right]\left[\begin{array}{c}\frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}}\end{array}\right]\left[\begin{array}{c}0\\ 1\end{array}\right]\left[\begin{array}{c}1\\ 0\end{array}\right]$$

$$|+\rangle \quad |-\rangle \quad |1\rangle \quad |0\rangle$$

$$\Sigma^x = \{|+\rangle, |-\rangle\}$$

$$\Sigma^z = \{|0\rangle, |1\rangle\}$$

Quantum Cryptography



1. Choses random state

$|+\rangle$



Alice

Bob

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$|+\rangle \quad |-\rangle \quad |1\rangle \quad |0\rangle$

$$\Sigma^x = \{|+\rangle, |-\rangle\}$$

$$\Sigma^z = \{|0\rangle, |1\rangle\}$$

Quantum Cryptography



1. Choses random state

$|+\rangle$



2. Randomly apply {I,H}

$|0\rangle$

Alice

Bob

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$|+\rangle \quad |-\rangle \quad |1\rangle \quad |0\rangle$

$$\Sigma^x = \{|+\rangle, |-\rangle\}$$

$$\Sigma^z = \{|0\rangle, |1\rangle\}$$

Quantum Cryptography



1. Choses random state

$|+\rangle$



2. Randomly apply {I,H}

$|0\rangle$

Alice

Bob

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$|+\rangle \quad |-\rangle \quad |1\rangle \quad |0\rangle$

$$\Sigma^x = \{|+\rangle, |-\rangle\}$$
$$\Sigma^z = \{|0\rangle, |1\rangle\}$$

If no H, and A's state from $\Sigma^z = \{|0\rangle, |1\rangle\}$ bits correlated

If H, and A's state from $\Sigma^x = \{|+\rangle, |-\rangle\}$ bits correlated

Quantum Cryptography



1. Choses random state

$|+\rangle$



2. Randomly apply {I,H}

$|0\rangle$

Alice

Bob

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$|+\rangle \quad |-\rangle \quad |1\rangle \quad |0\rangle$

$\Sigma^x = \{|+\rangle, |-\rangle\}$

$\Sigma^z = \{|0\rangle, |1\rangle\}$

If no H, and A's state from $\Sigma^z = \{|0\rangle, |1\rangle\}$ bits correlated

If H, and A's state from $\Sigma^x = \{|+\rangle, |-\rangle\}$ bits correlated

If A reveals Σ^x, Σ^z then random secret key

Quantum Cryptography



$|+\rangle\ |1\rangle\ |1\rangle\ |0\rangle \cdots$



Alice Bob

1. Alice sends states
2. Bob makes measurements
3. Alice reveals $\sum^{\alpha}$ for states
4. Alice reveals a few of her secret bits
5. Bob can detect if eavesdropping has occurred

Quantum Cryptography



$|+\rangle\ |1\rangle\ |1\rangle\ |0\rangle \cdots$



Alice

Bob



from the low, $100,000s

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linear optics

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Quantum Algorithms

A Game



Alice



Bob



Charlie

An Evil Game



Alice



Bob



Charlie

1. Each given slip **X** or **Y**

2. At midnight each party shouts "+1" or "-1"

Winning the Game







Alice	Bob	Charlie	Winning Product
X	X	X	+1
X	Y	Y	-1
Y	Y	X	-1
Y	X	Y	-1

A Game



Alice



Bob



Charlie

A_X B_X C_X

A_Y B_Y C_Y

All either +1 or -1

A Game







Alice	Bob	Charlie	Product
X	X	X	$A_X B_X C_X$
X	Y	Y	$A_X B_Y C_Y$
Y	Y	X	$A_Y B_Y C_X$
Y	X	Y	$A_Y B_X C_Y$

Winners?

Assuming winning strategy:

$A_X B_X C_X = +1$

$A_X B_Y C_Y = -1$

$A_Y B_Y C_X = -1$

$A_Y B_X C_Y = -1$

Multiply equations….contradiction!

LHS: $A_X^2 A_Y^2 B_x^2 B_Y^2 C_x^2 C_Y^2 = +1$

RHS: -1

Losers



Alice



Bob



Charlie

There is no strategy to win the game

In the Real World



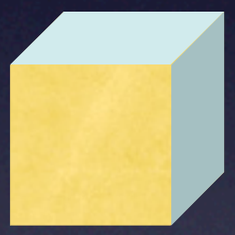
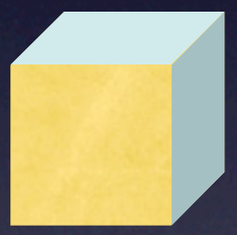
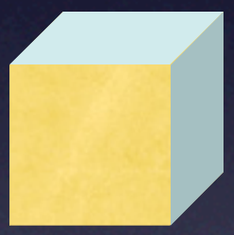
Alice



Bob



Charlie

This is wrong, there is a way to win

Quantum Theory

In the Real World



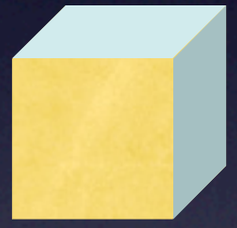
Alice



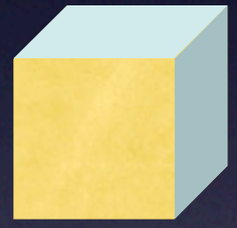
Bob



Charlie

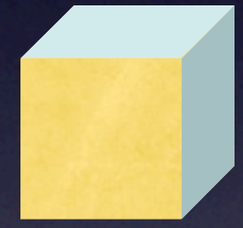


Share $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$



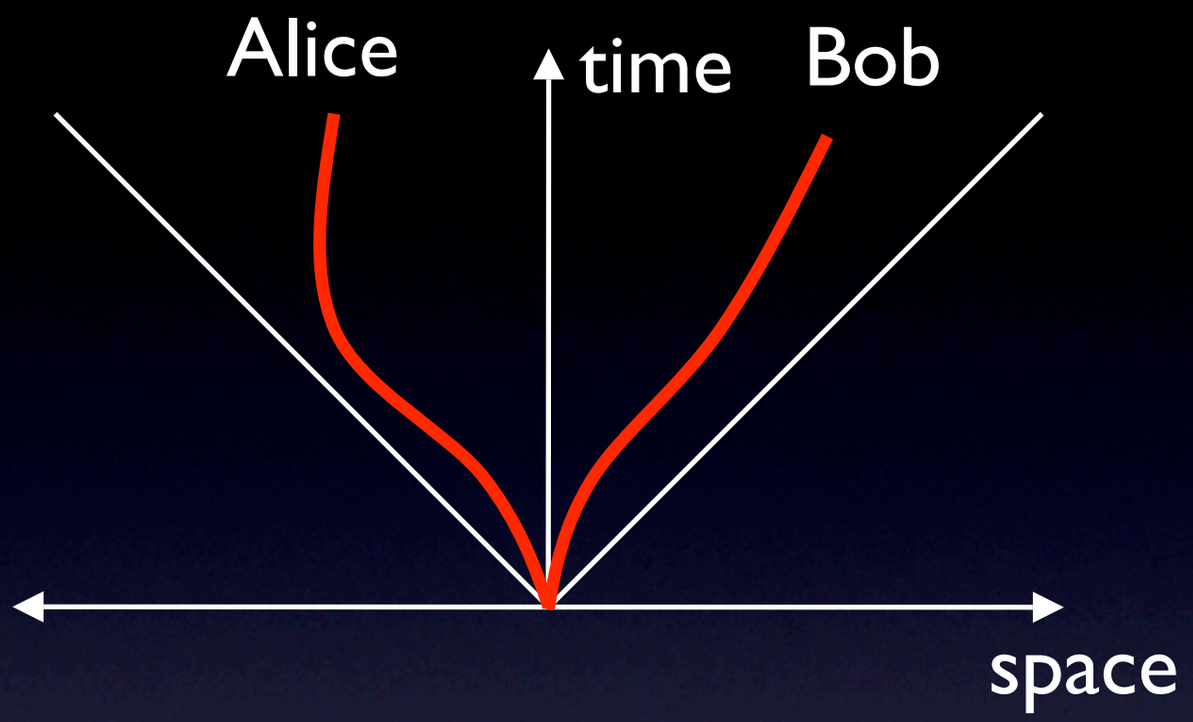
Apply $M_X = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

$M_Y = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$



Measure

Bell's Theorem





John Bell

Measurements on space-like separated entangled quantum systems cannot be explained by local classical communication.

Minimizing the Yakking



ALICE

a

$F_{eq}(a,b)=$

1 if a=b

0 if a≠b



BOB

b

0 1 0 1 … 0

n bits

0 1 1 1 … 0

n bits

How many bits do Alice and Bob need to exchange (in the worst case) to compute F(a,b)?

Minimizing the Yakking



ALICE

a

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1 if a=b

0 if a≠b



BOB

b

0 1 0 1 … 0

n bits

0 1 1 1 … 0

n bits

How many bits do Alice and Bob need to exchange (in the worst case) to compute F(a,b)?

With no error: n+1 bits

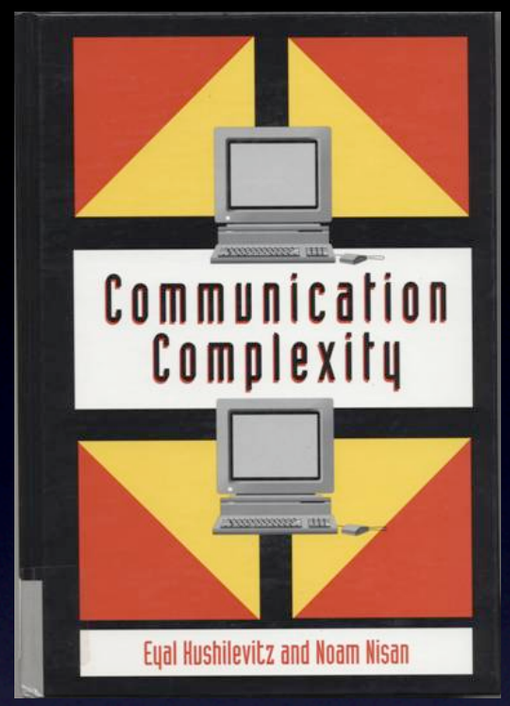
With one-sided error ε: O(log(n/ε)) bits

Communication Complexity



[Yao, STOC 79]

Classical communication costs of distributed classical computations



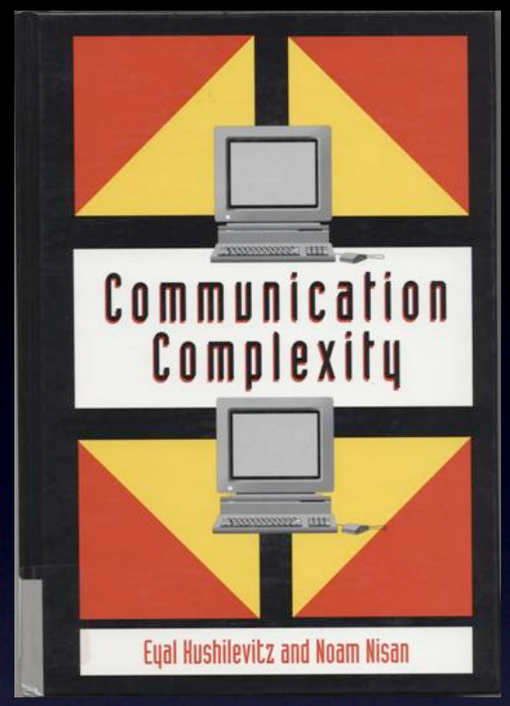
[Kushilevitz and Nisan 96]

Communication Complexity



[Yao, STOC 79]

Classical communication
costs of distributed
classical computations

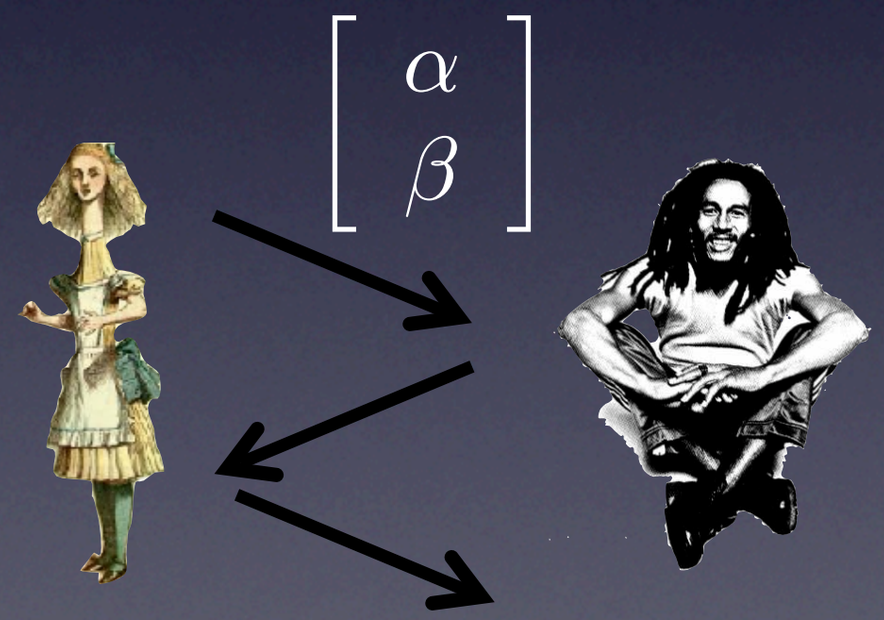


[Kushilevitz and Nisan 96]



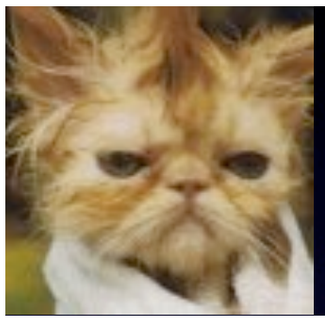
[Yao, FOCS 93]

Quantum communication
costs of distributed
classical computations



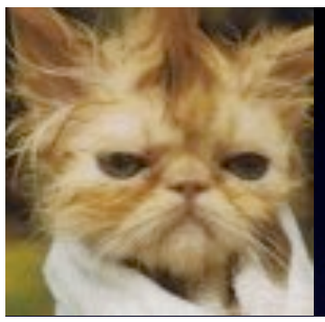
exchange qubits

Quantum Communication Complexity



Holevo's Theoerem: sending n classical bits over a quantum channel requires n qubits

Quantum Communication Complexity



Holevo's Theoerem: sending n classical bits over a quantum channel requires n qubits



[Raz, STOC 99]

There exist functions $F(a,b)$ which require $\Omega(m^{1/4}/\log(m))$ bits of communication to solve (with error) but can be solved using only $O(\log(m))$ qubits of communication.
(m is the length of a and b)



Exponential benefit of quantum over classical.

Landscape

Quantum Entanglement

Quantum Error Correction

Experimental Quantum Computing

Quantum Information Theory

Quantum Computer Science

ion traps

quantum dots

superconducting circuits

linear optics

Quantum Cryptography

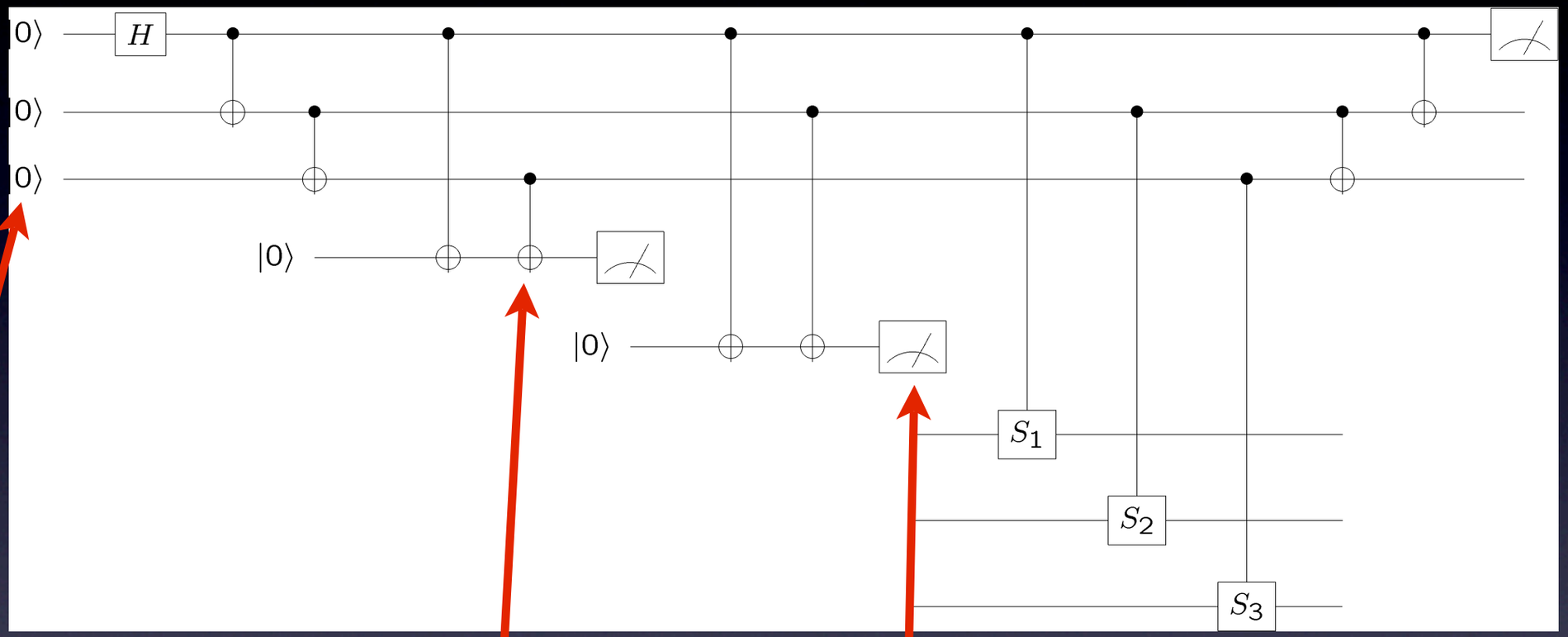
Quantum Algorithms

Towards a Real Quantum Computer

Experimental Quantum Computing

- ion traps
- quantum dots
- superconducting circuits
- linear optics

Quantum Circuit



Preparation

Gates

Measurement

Time

But What Will It Look Like?

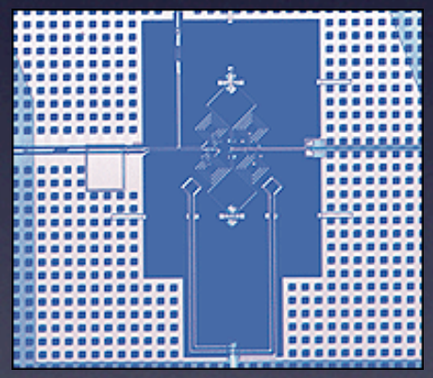
Solid State

superconducting circuits

electron spin in Phosphorus doped Silicon

quantum dots

defects in diamonds



Molecular

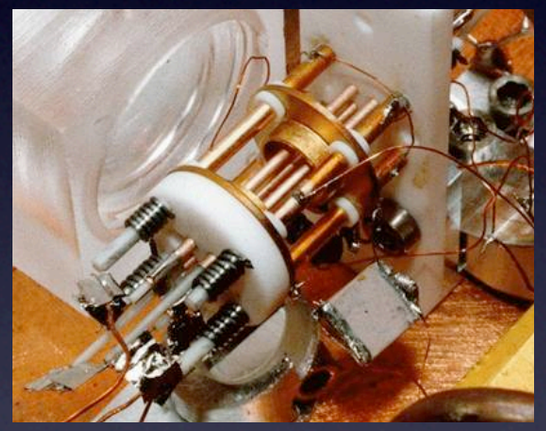
Liquid NMR (no longer?)

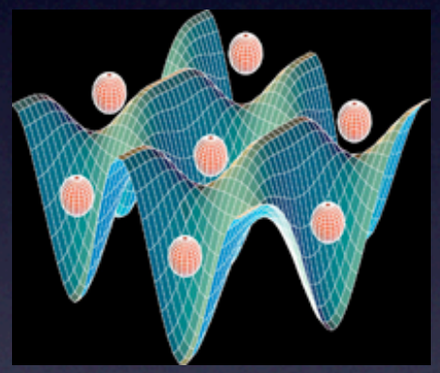
Atomic

cavity QED

neutral atoms in optical lattices

ion traps





Photon Based

linear optics + single photon devices

Pics: Martinis (UCSB), Wineland (NIST), Orlando (MIT)

DiVincenzo's Criteria



David DiVincenzo

1. Well defined qubits in a scalable architecture

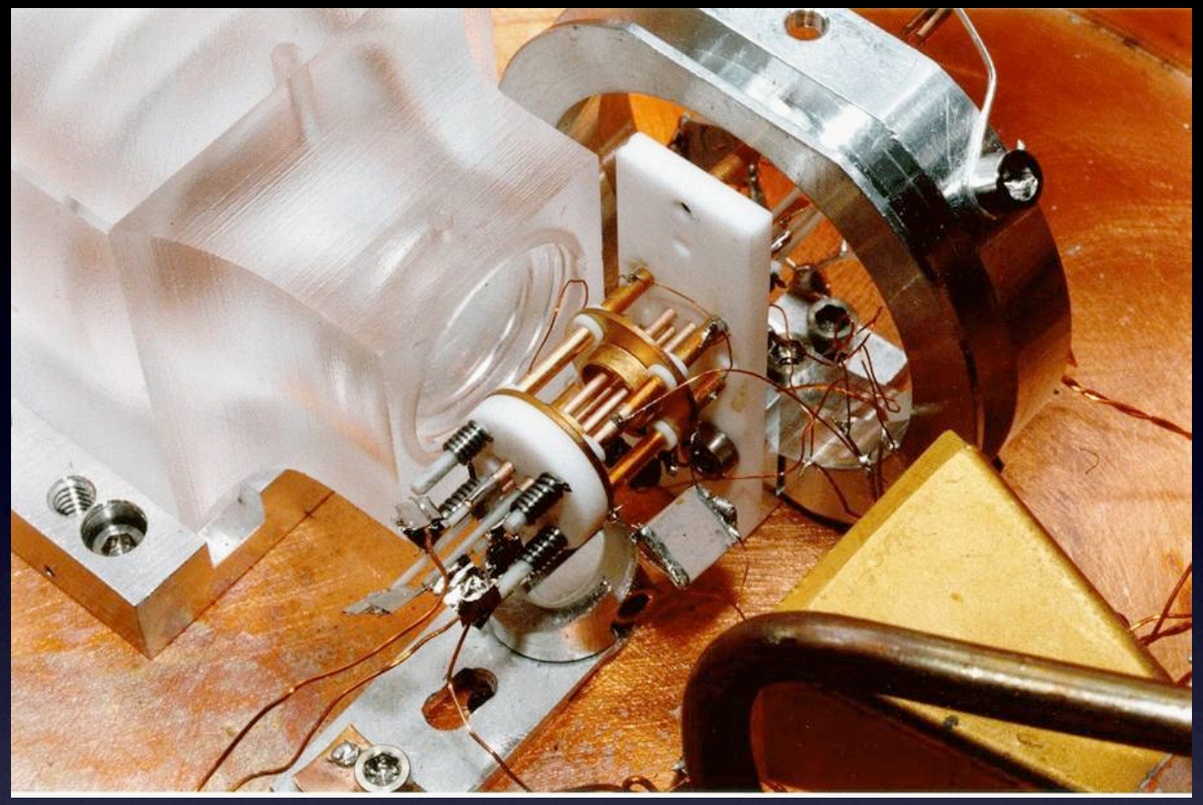
2. The ability to initialize the system to a fixed wave function.

3. Have faster control over the system than error processes in the system.

4. Have the ability to perform a universal set of quantum gates.

5. Have the ability to perform high quality measurements

Ion Trap

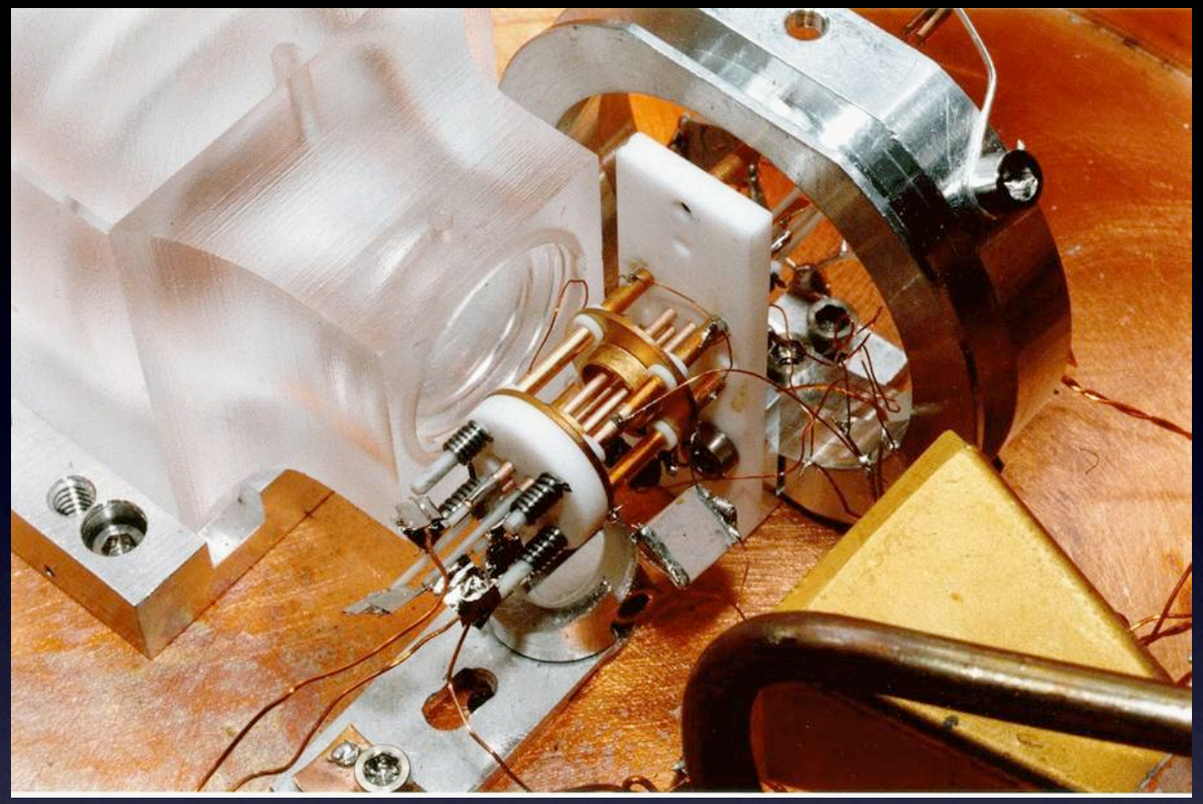


Oscillating electric fields trap ions

like charges repel



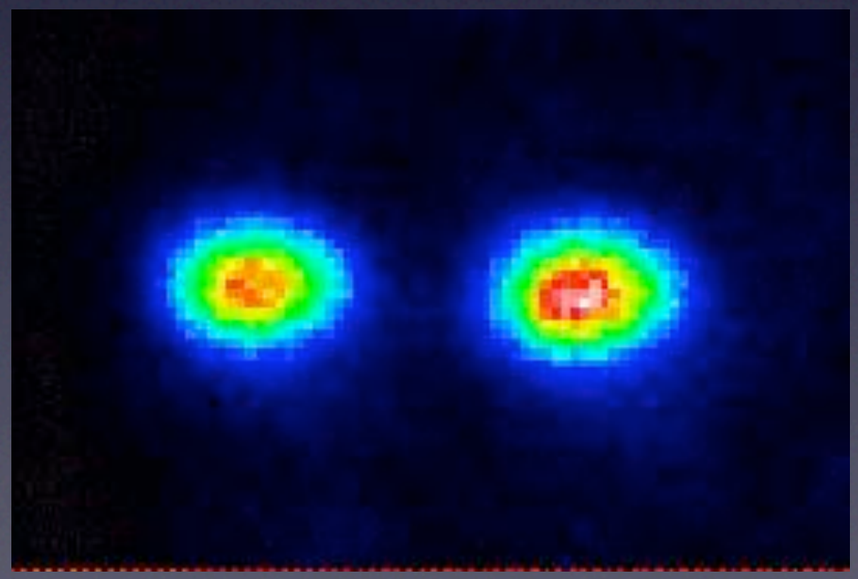
Ion Trap



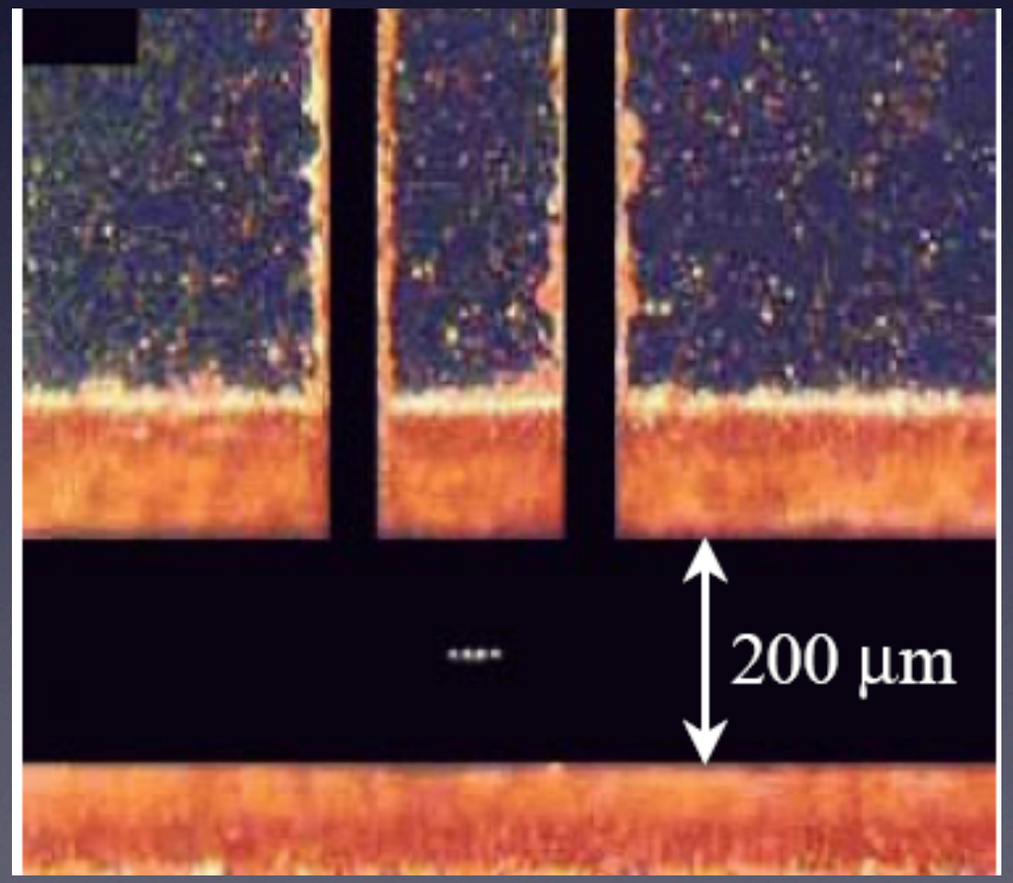
Oscillating electric fields trap ions

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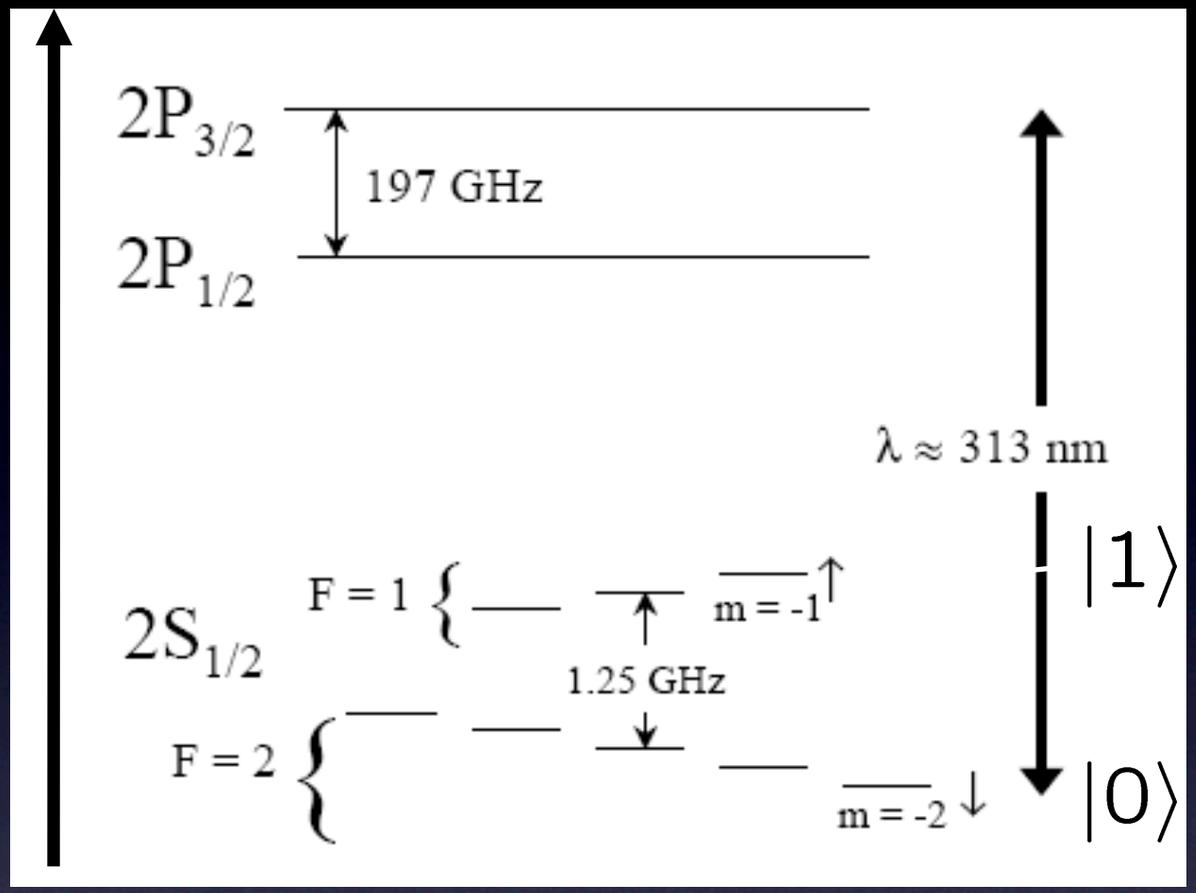


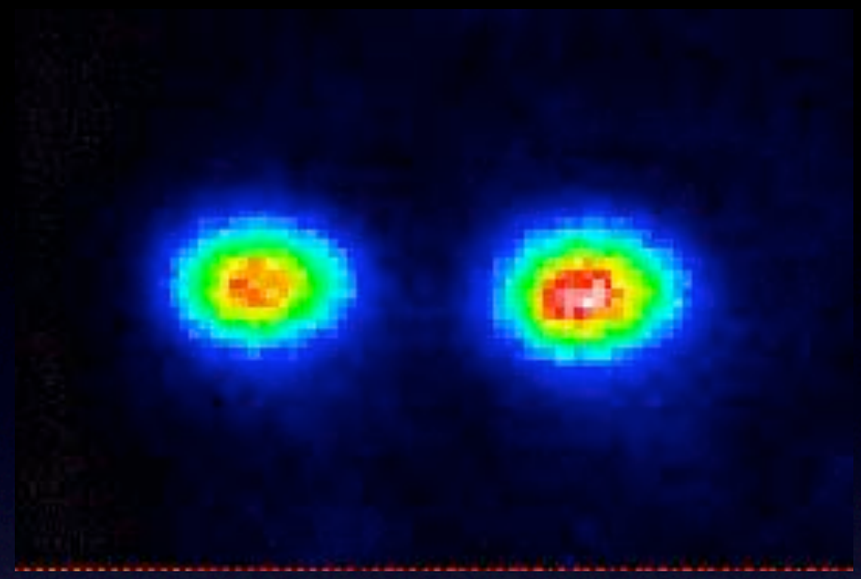


2 $^{9}Be^{+}$ Ions in an Ion Trap



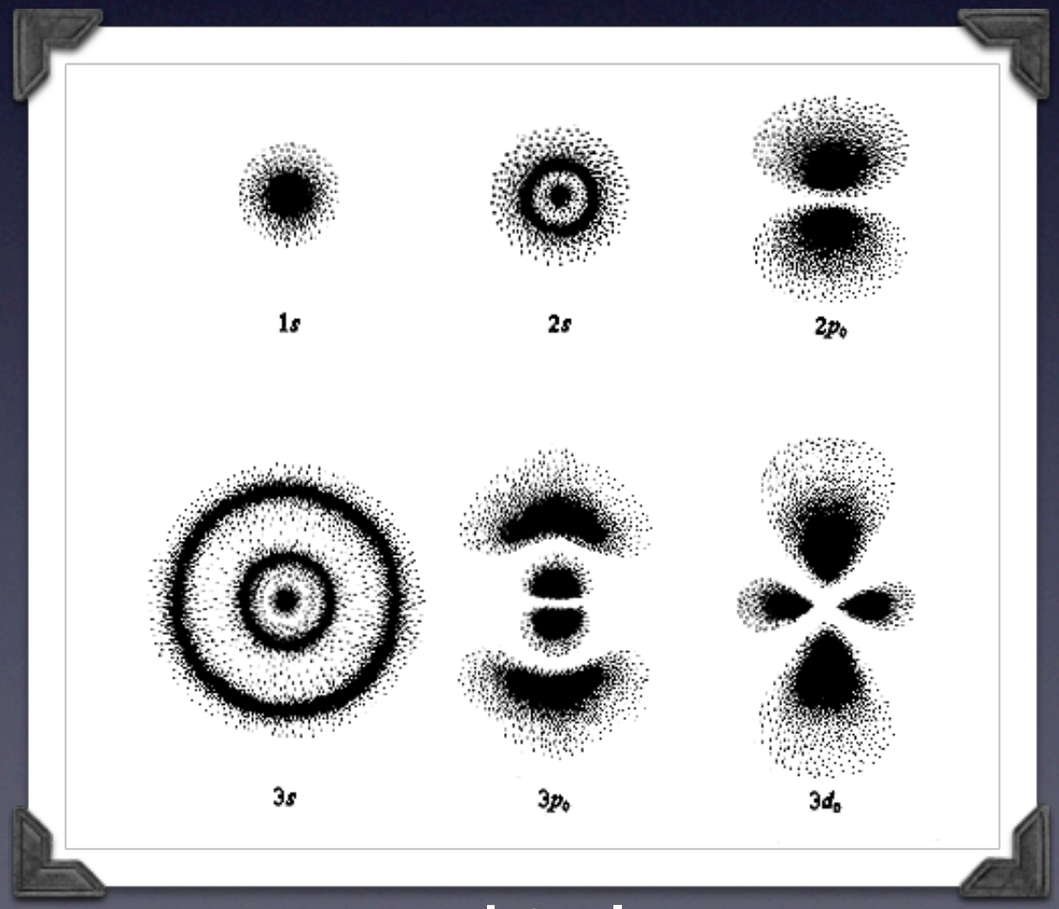
Where's the Qubit?





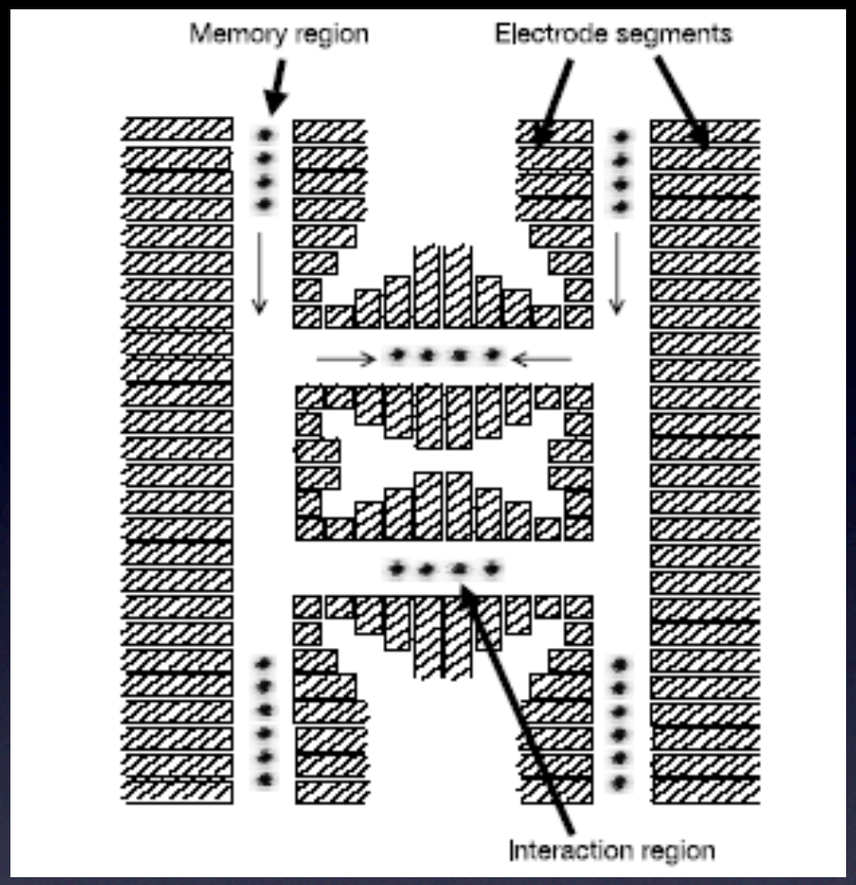
Each ion = 1 qubit

1. Well defined qubits



orbitals

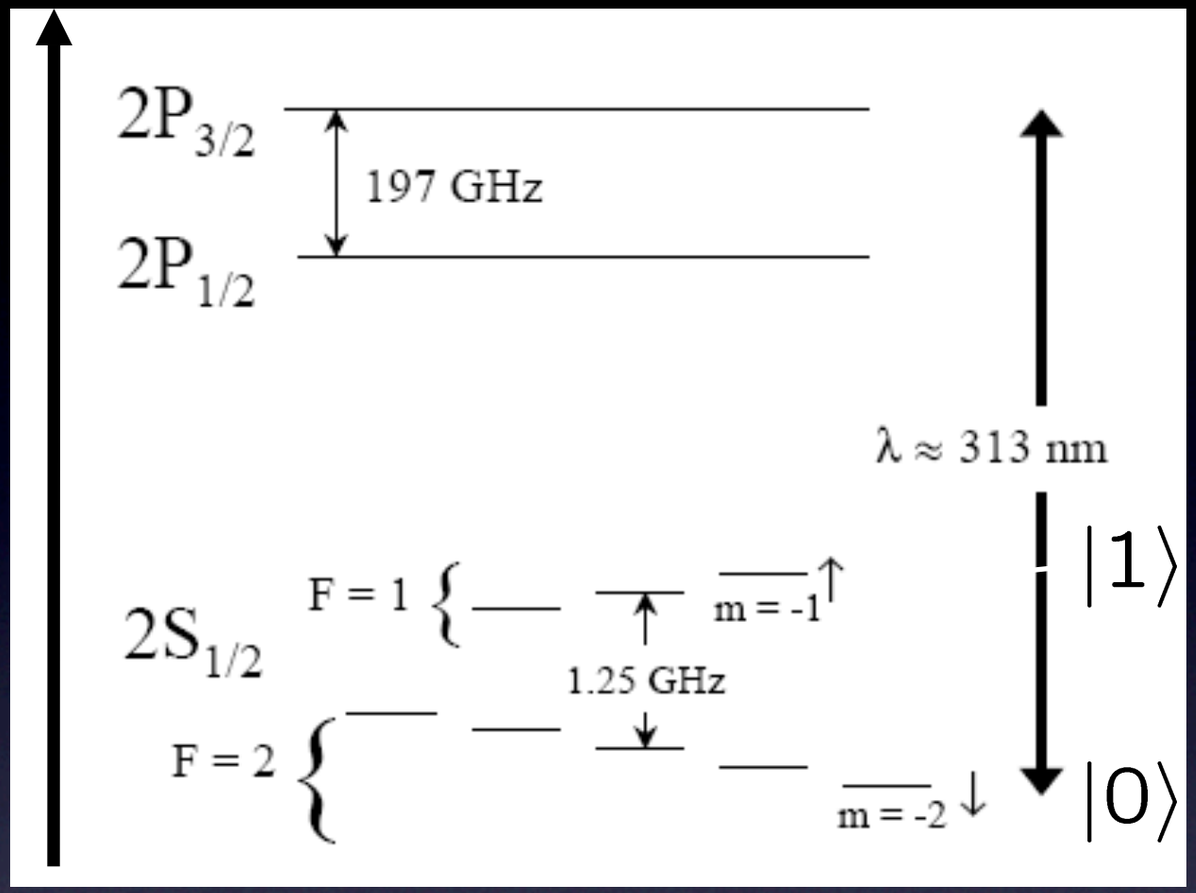
Scalable?

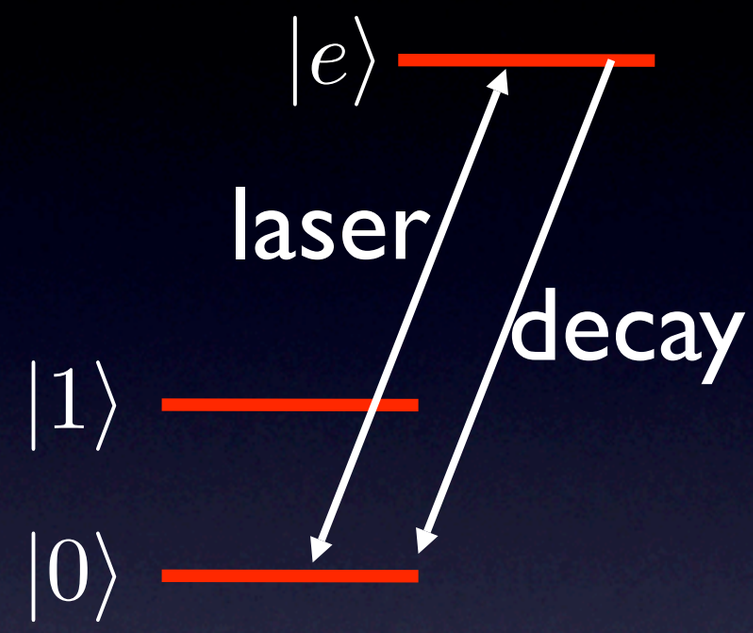


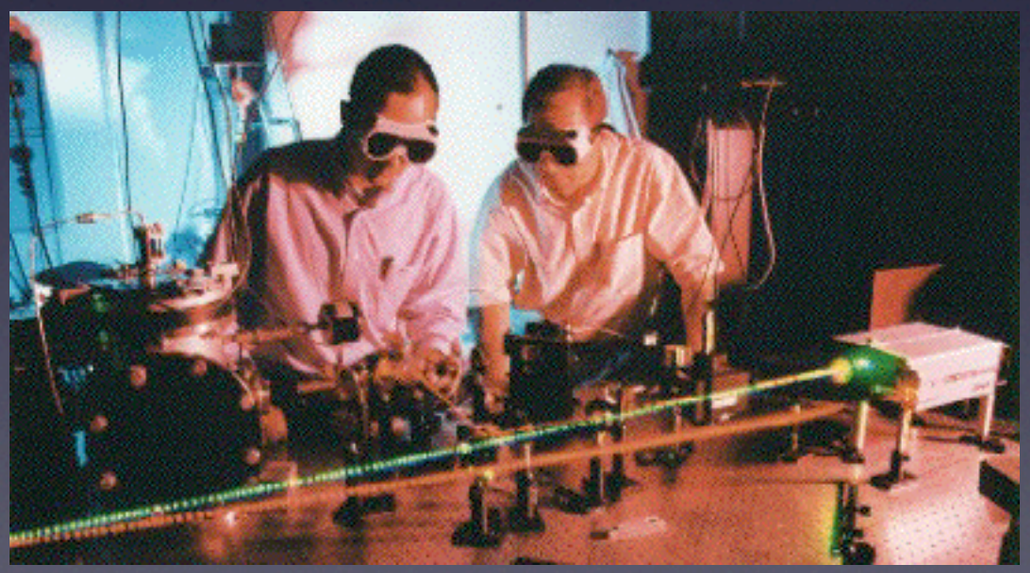
Well defined qubits in a scalable architecture?

Solid state qubits seem to have a huge advantage for scalability.

Measurement





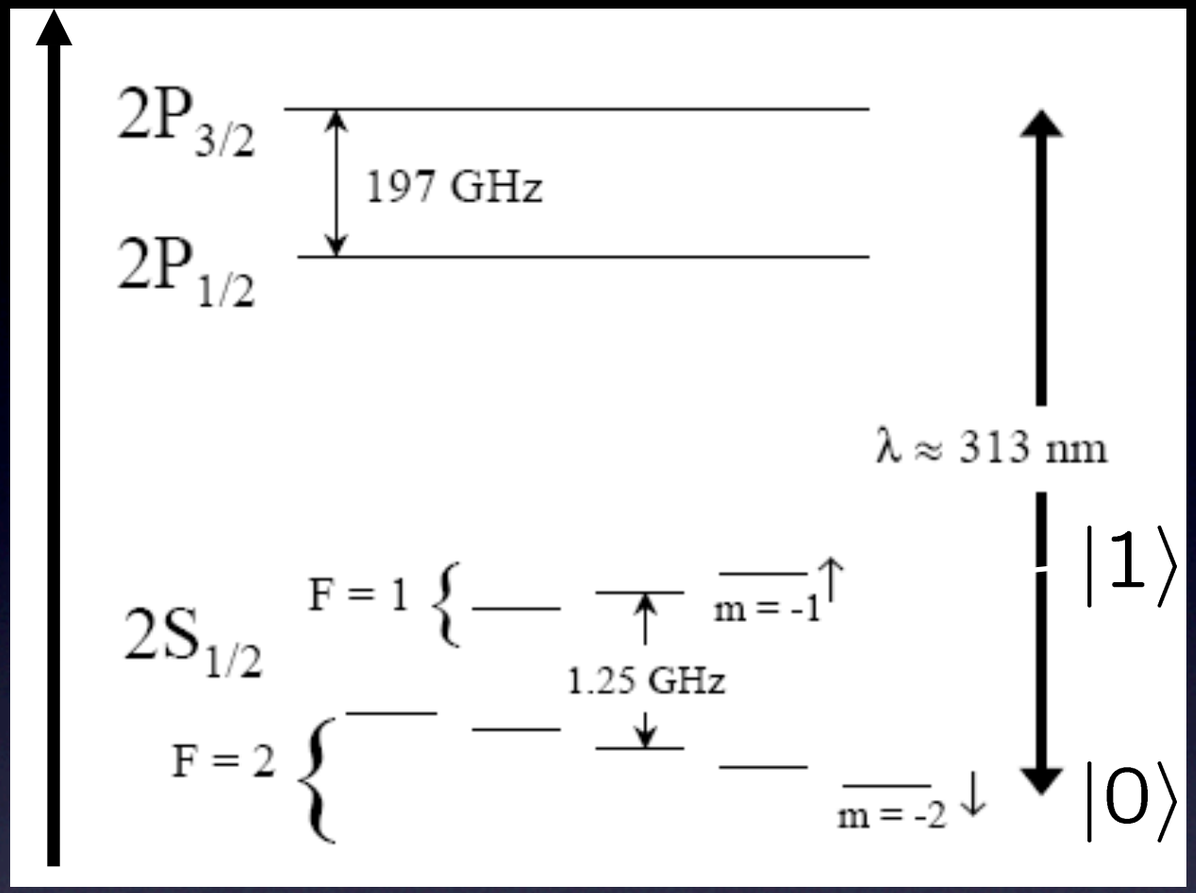


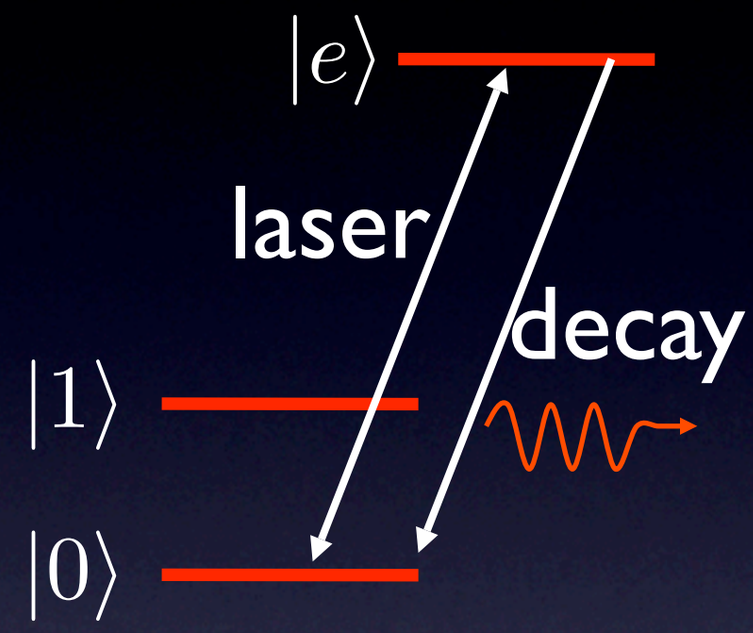
Detecting florescence implies in state 0

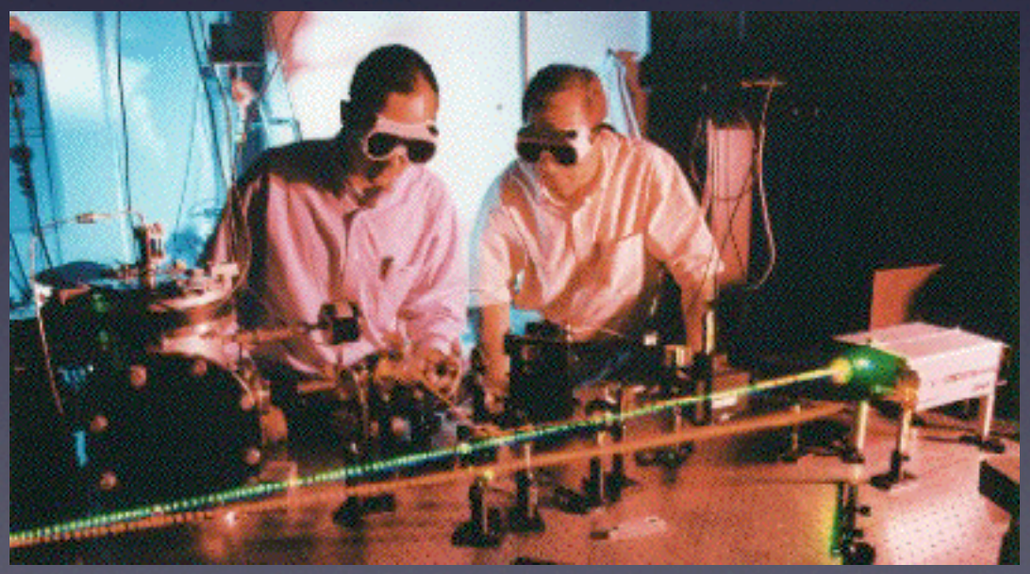
99.99% efficiency

5. Have the ability to perform high quality measurements

Measurement





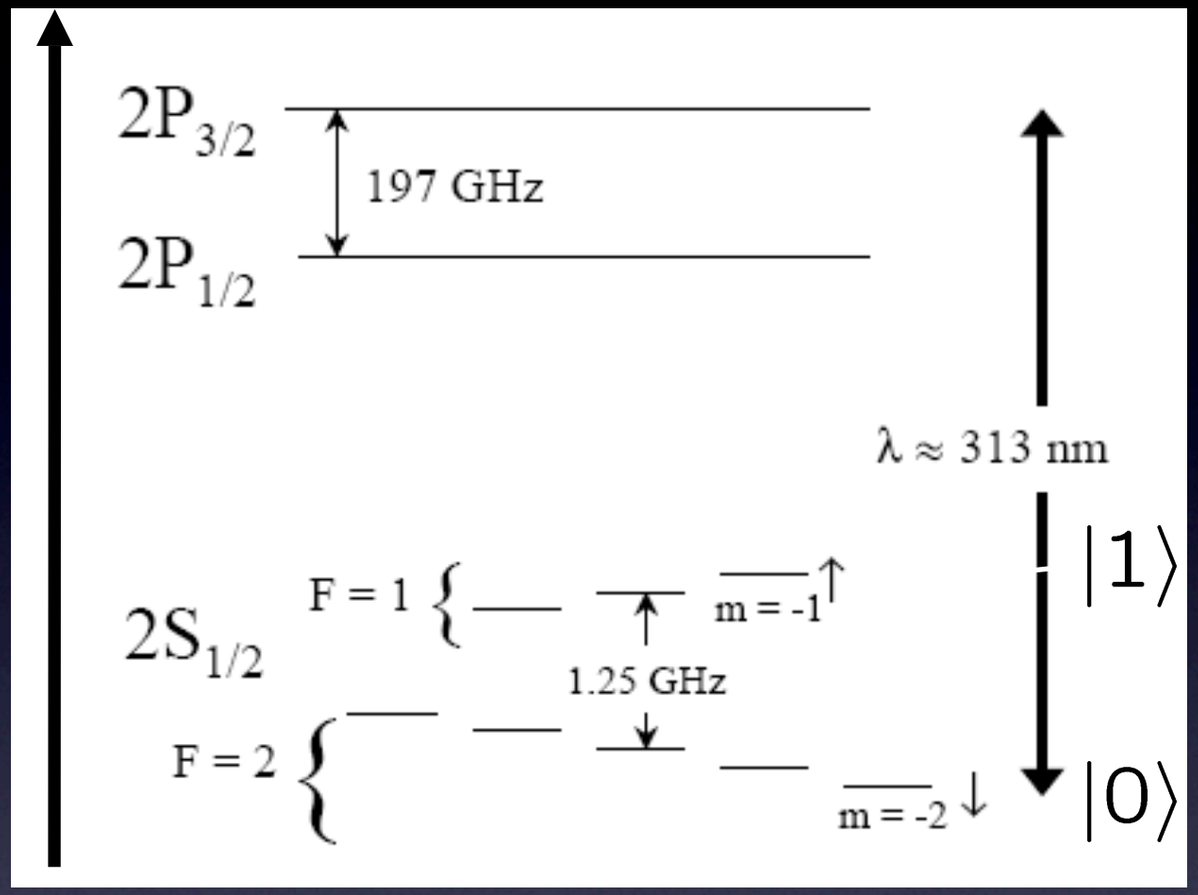


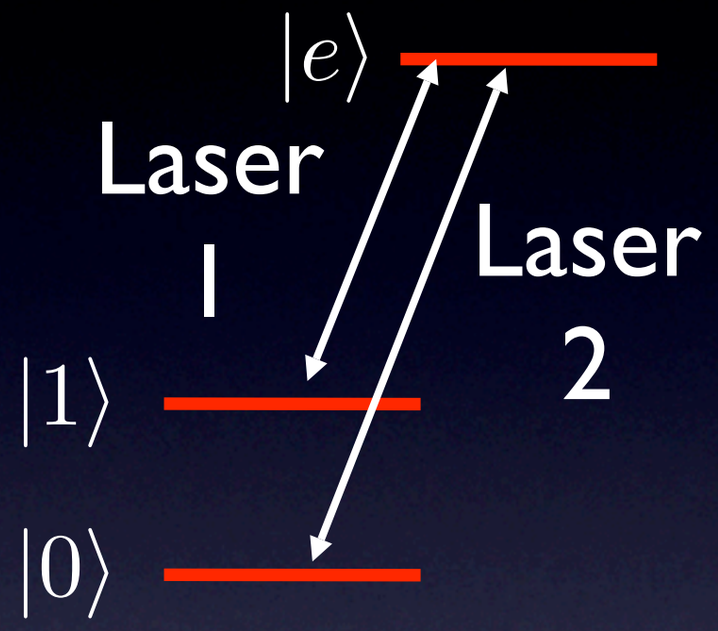
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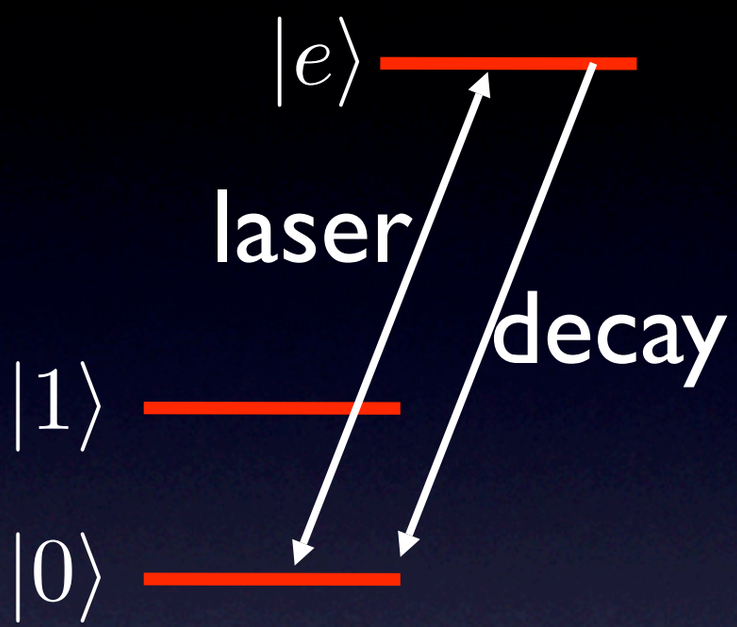
Single Qubit Operations

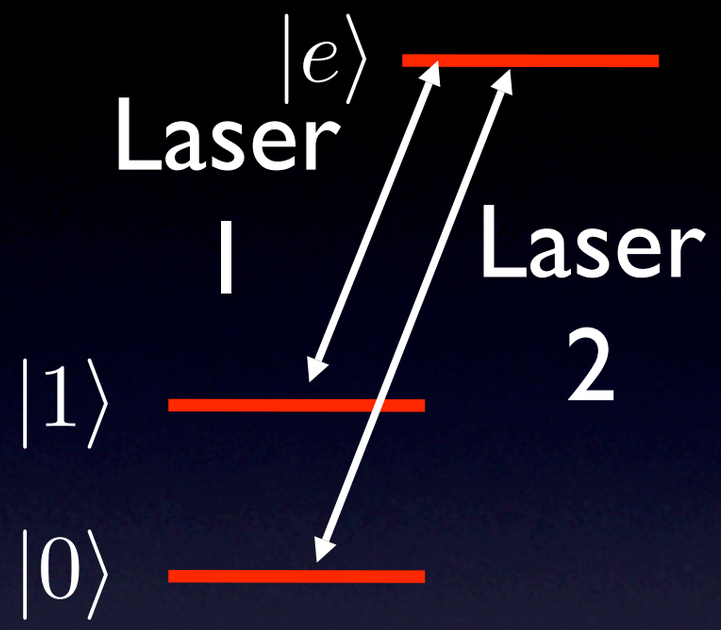




Allows any one qubit unitary operations $\begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix}$

Initialization



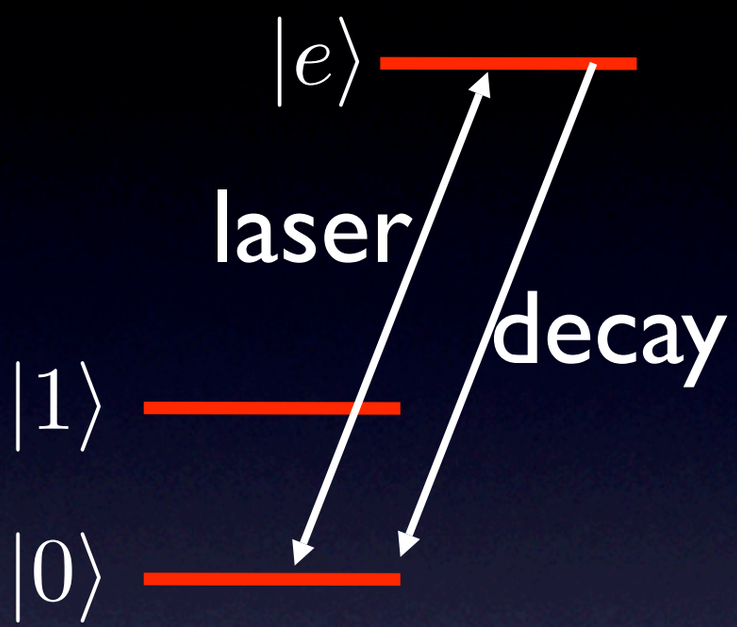


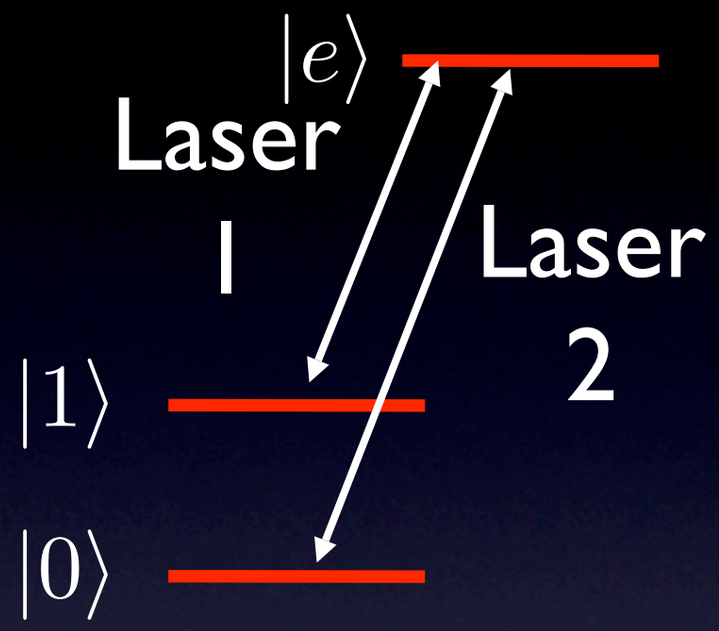
measure

If not in zero state, flip

2. The ability to initial the system to a deterministic state.

Initialization





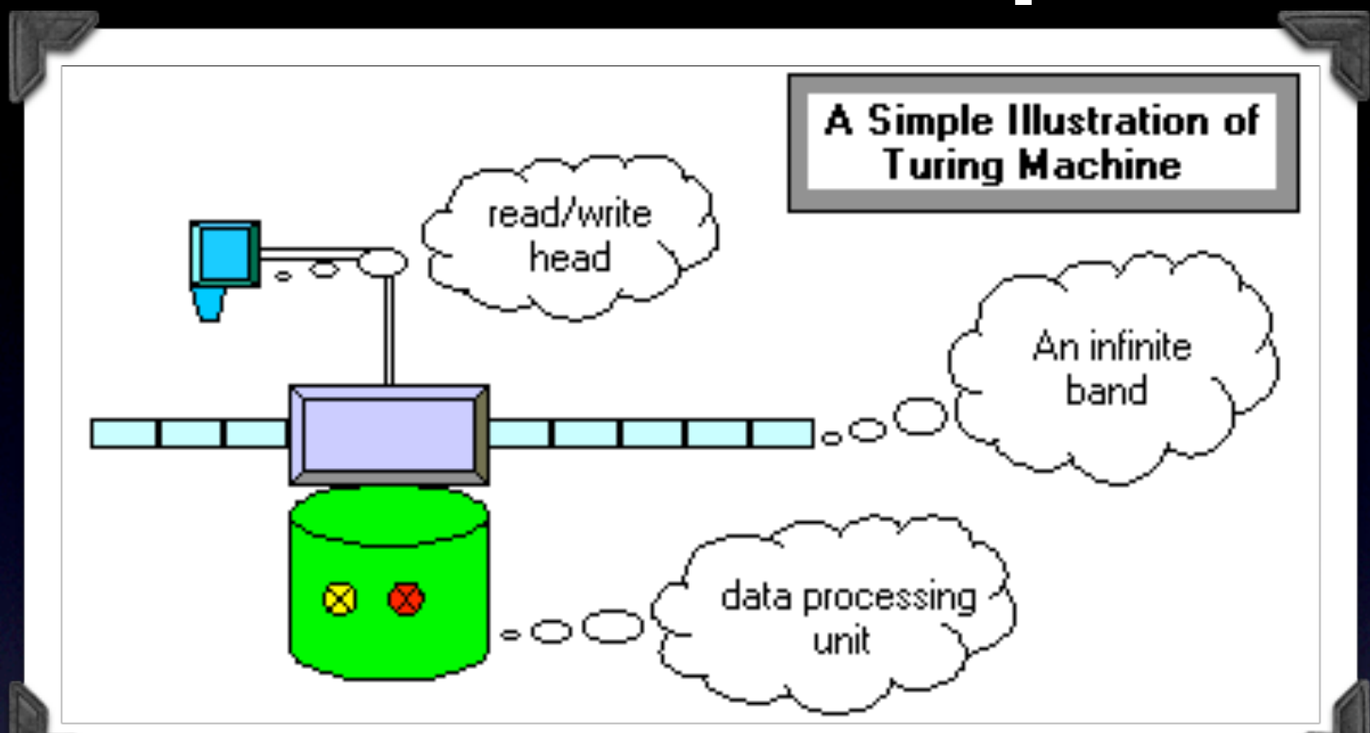
measure

If not in zero state, flip

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2. The ability to initial the system to a deterministic state.

Universal Computers



Turing machine reads state of tape at current position.

Based on this reading and state of machine, Turing machine writes new symbol at current position and possibly moves left or right, and updates state.

Certain Turing machines can perform certain tasks.

A Universal Turing Machine can act like any other possible Turing machine (i.e. it is programmable)

Universal Quantum Computer

Universal Quantum Computer

a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.

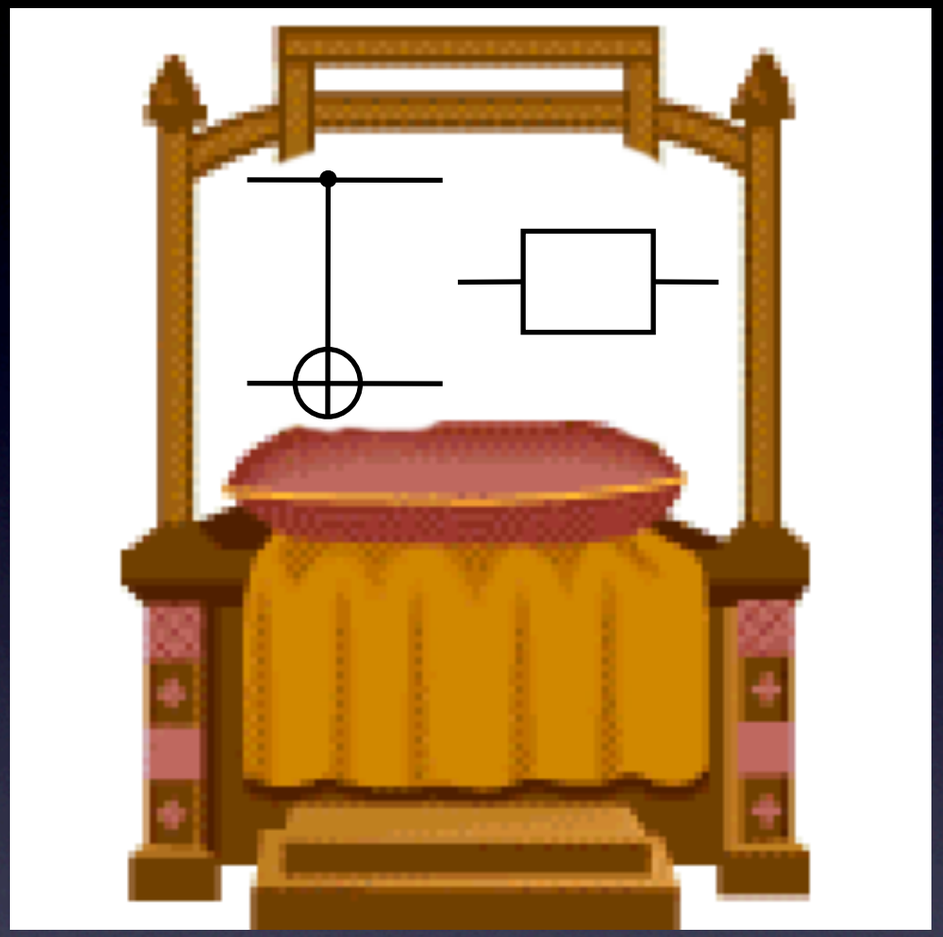
Universal Quantum Computer

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The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Coupling Two Qubits



stationary



sloshing mode

These modes can be used as a bus between the qubits.



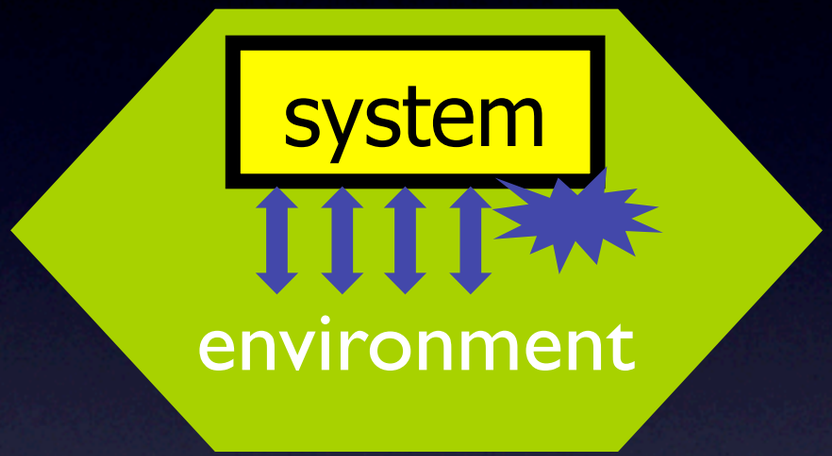
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

4. Have the ability to perform a universal set of quantum gates

What is the Problem?

3. Have faster control over the system than error processes in the system.

<u>Real quantum systems are open quantum systems!</u>



Quantum systems readily couple to an environment…

System decoheres: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ qubits → pbits **0**

$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}$ 50% **0** 50% **1**

Quantum **Classical**

The Decoherence Problem (1996)

More Obstructions

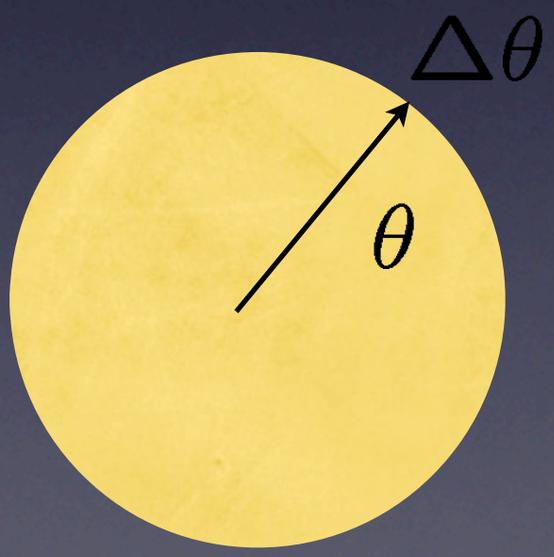
FAULTY CONTROL

System

Controller

Imprecise or faulty unitary gates



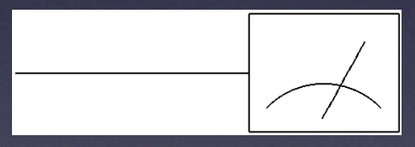


Imperfect state preparation



$$\rho \rightarrow (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$$

Imperfect measurements



$$Pr(0) = |\alpha|^2 - \epsilon$$

$$Pr(1) = |\beta|^2 + \epsilon$$

More Obstructions

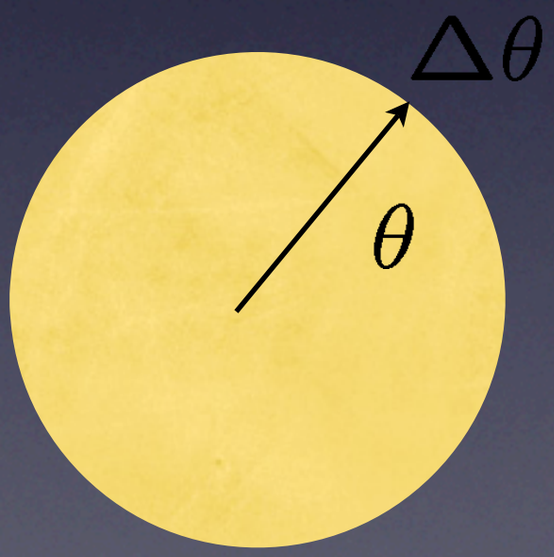
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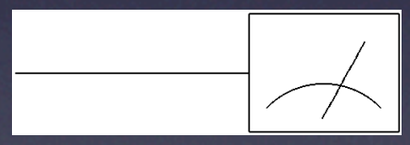


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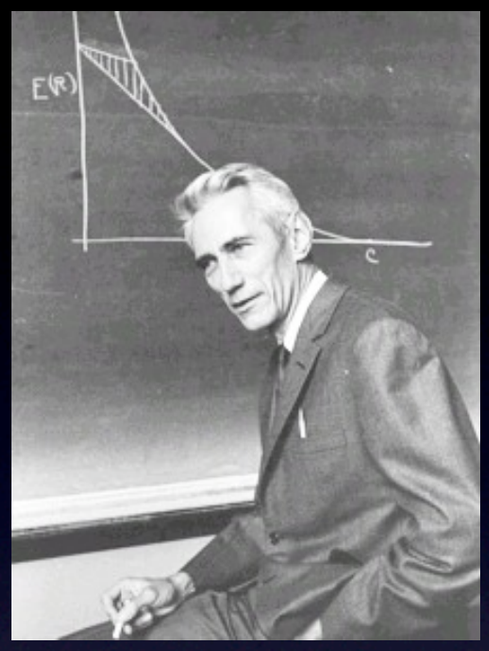


$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$Pr(0) = |\alpha|^2 - \epsilon$$

$$Pr(1) = |\beta|^2 + \epsilon$$

Noisy Pbits to Bits (Theory)



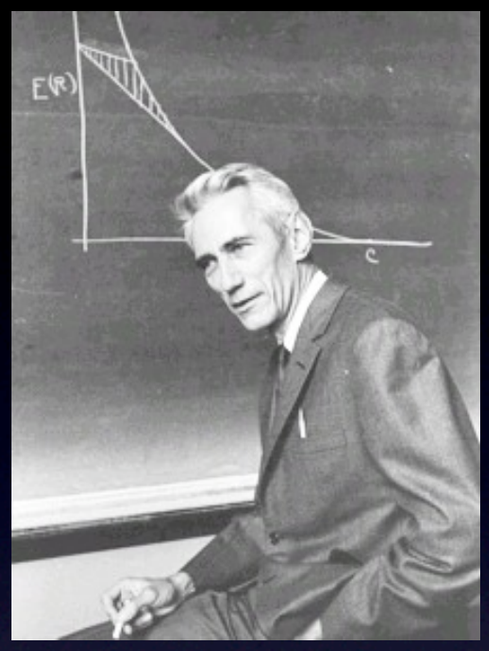
Claude Shannon

How to communicate in the presence of noise?

"A mathematical theory of communication" (1948)

Error correcting codes

Noisy Pbits to Bits (Theory)



Claude Shannon

How to communicate in the presence of noise?

"A mathematical theory of communication" (1948)

Error correcting codes



John von Neumann

How to compute when the subsystems of your computer can fail?

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)

Fault-tolerant architectures

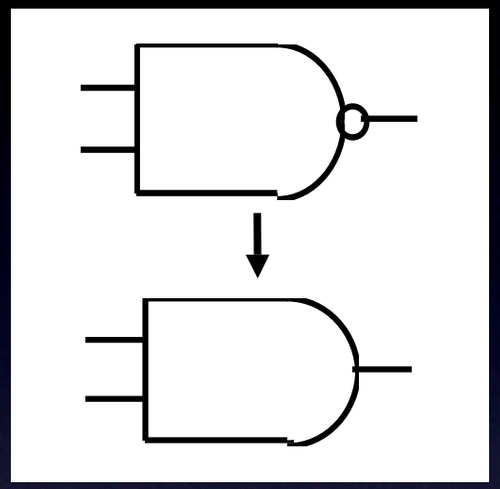
Fragile



Classical Noise

$$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

Classical Faults



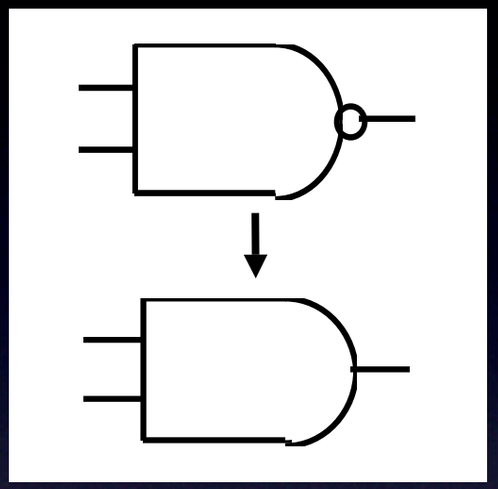
Fragile



Classical Noise

$$\begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix}$$

Classical Faults



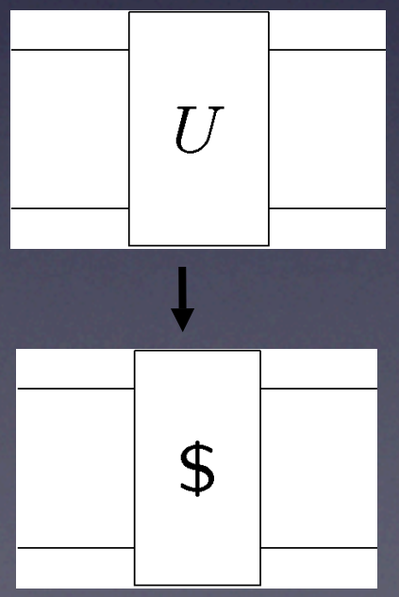


Quantum Noise

$$\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix}$$

$$\begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix}$$

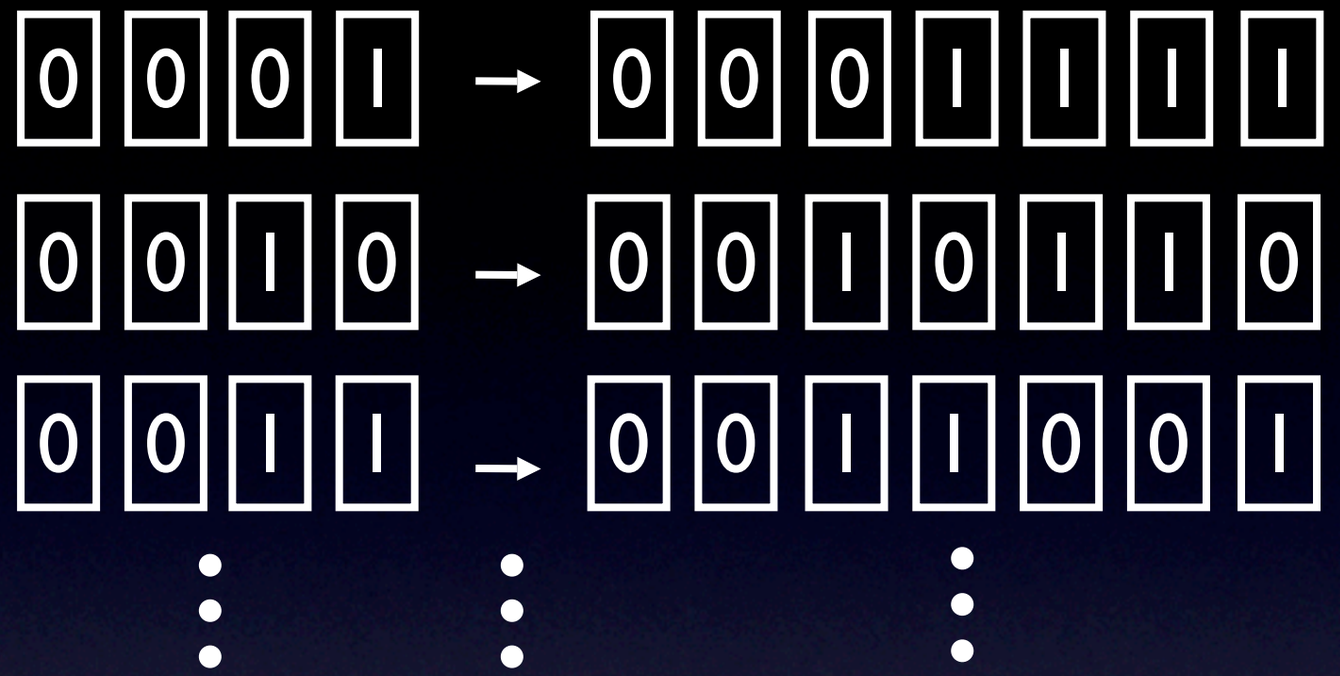
Quantum Faults



Error Correction

classical error correcting code:

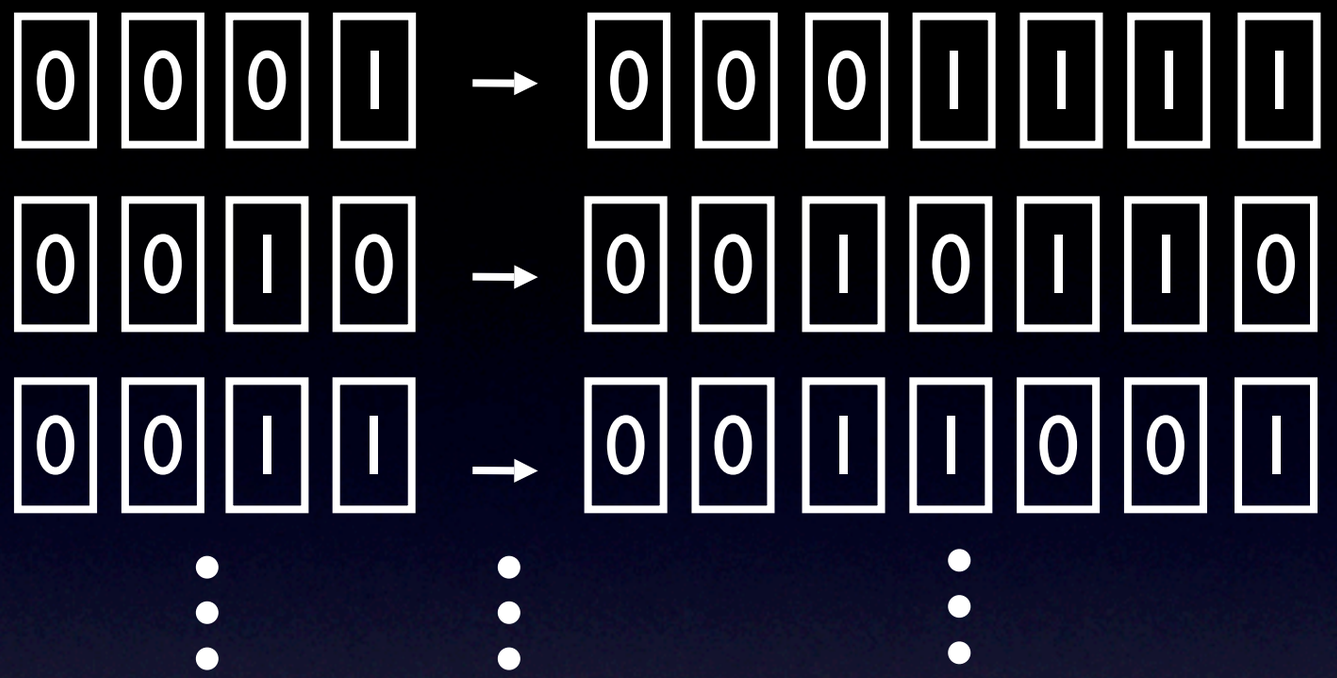
(bits and pbits)

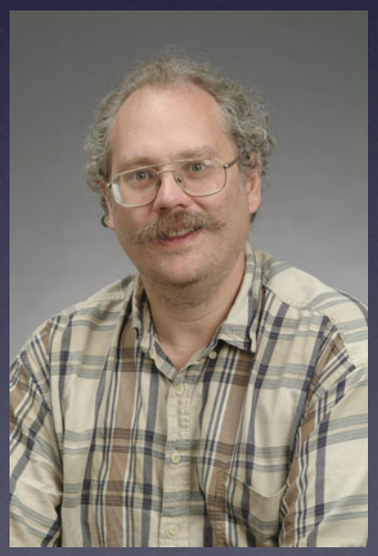


Error Correction

classical error correcting code:

(bits and pbits)





Peter Shor



Andrew Steane

1995: quantum error correcting codes

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \square\square\square\square\square$$

encode quantum information across multiple quantum subsystems

Perfection Through Concatenation

failure probability

$$\frac{}{p}$$

Perfection Through Concatenation



failure probability

p

cp^2

Perfection Through Concatenation



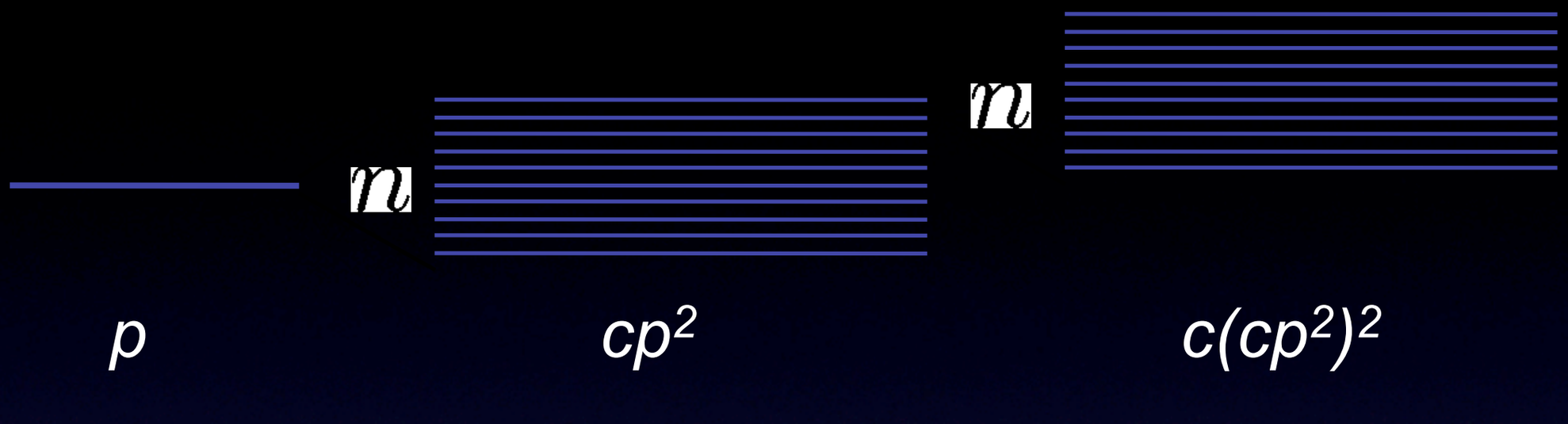
failure probability

p

cp^2

$c(cp^2)^2$

Perfection Through Concatenation



At k^{th} level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c}$$

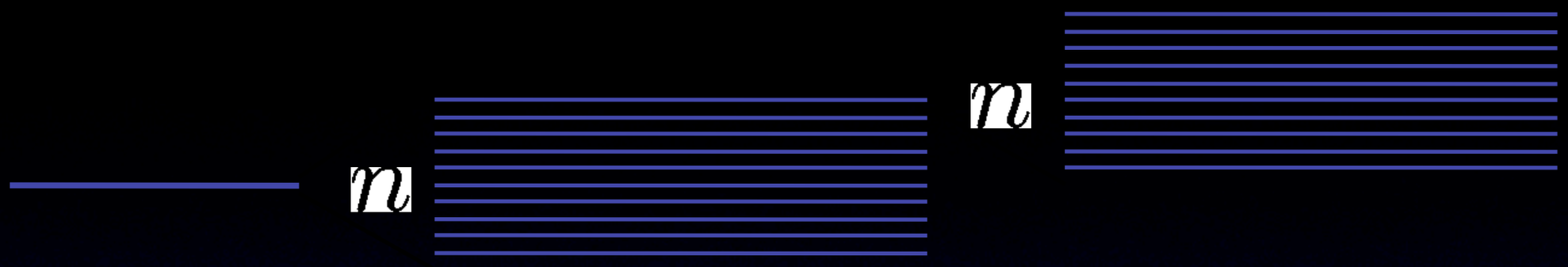
$$\frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k}$$

$$p_{th} = \frac{1}{c}$$

$$N_{qubits} = n^k$$

qubits

Perfection Through Concatenation



At k^{th} level of concatenation, probability of an error which destroys the encoded quantum information is

$$p_{fail} = \frac{(cp)^{2^k}}{c}$$

$$\frac{p_{fail}}{p_{th}} = \left(\frac{p}{p_{th}}\right)^{2^k}$$

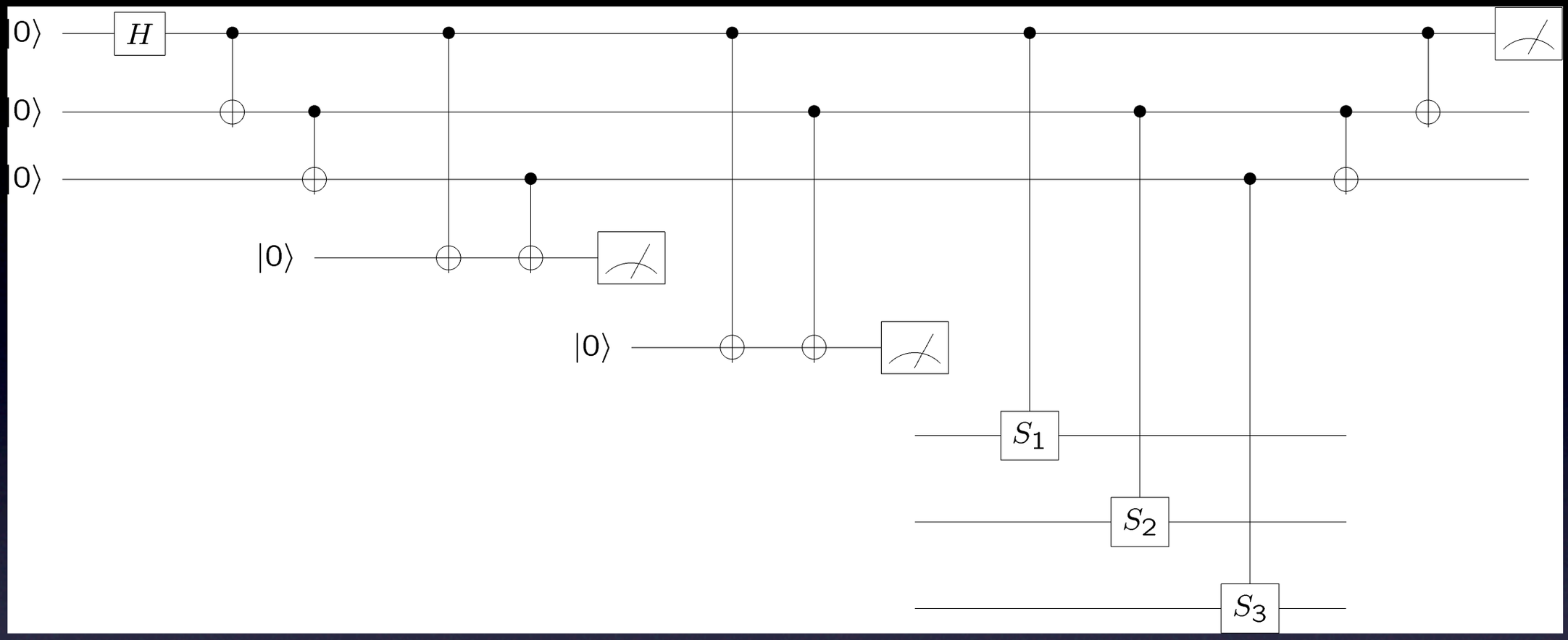
$$p_{th} = \frac{1}{c}$$

$$N_{qubits} = n^k$$

qubits

$$N_{qubits} = \left[\log_{\frac{p}{p_{th}}} \left(\frac{p_{th}}{p_{fail}}\right)\right]^{\log_2 n} = O\left(\text{poly}\log \frac{1}{p_{fail}}\right)$$

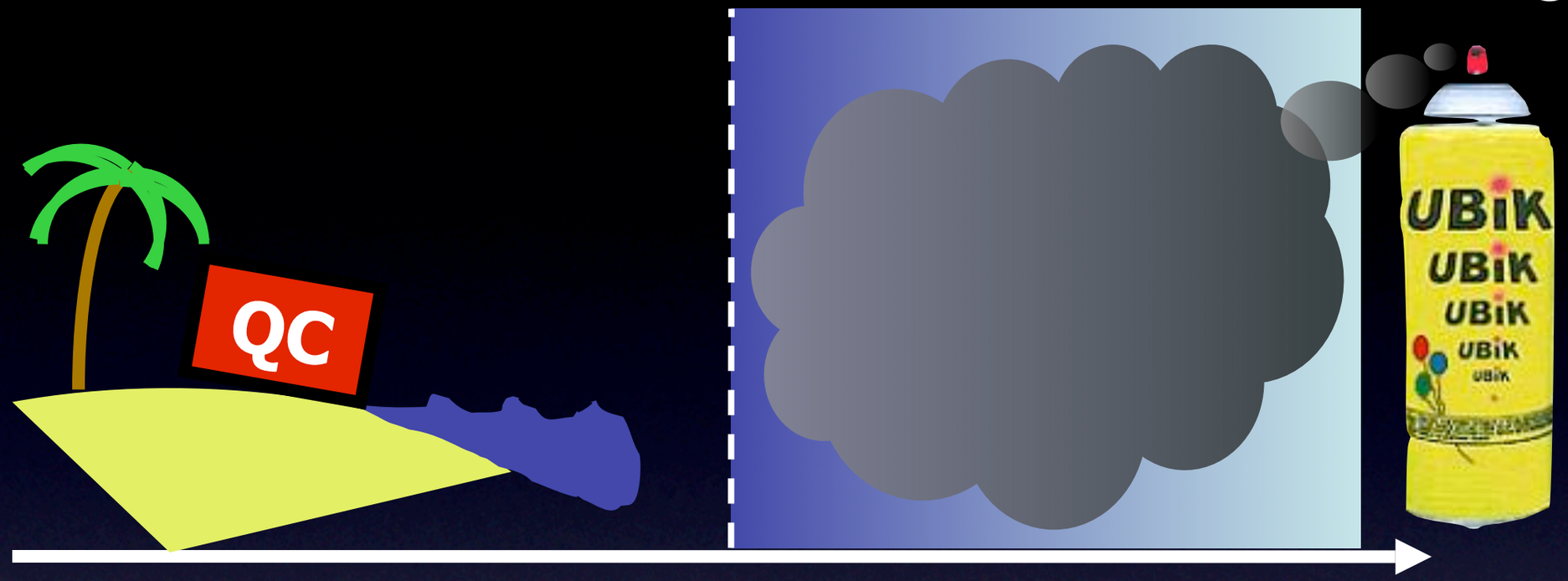
Fault-Tolerant Quantum Computation



Must design quantum circuits which succeed even when every element of the circuit (preparation, unitary gates, Measurements) can fail.

Quantum error correction + Fault-tolerant procedures

Threshold Theorem, 1996



“decoherence”

“imprecision”

Threshold Theorem for Fault-Tolerant Quantum Computation

[Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Gottesman, Preskill]

If noise is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

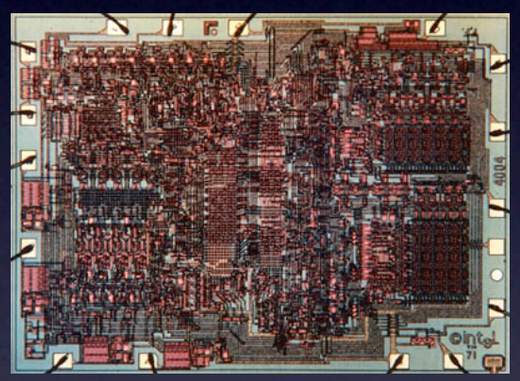
Reflection

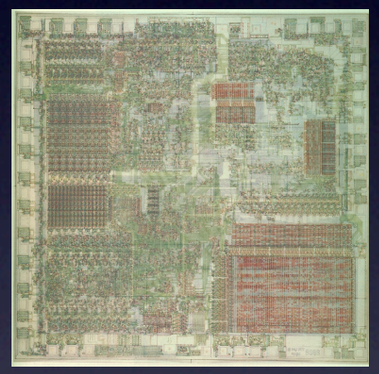
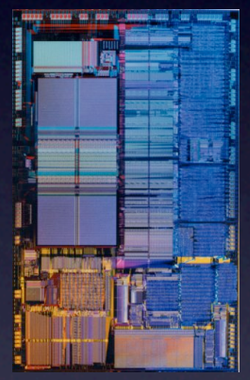
Even with the theory of fault-tolerant quantum computation

⇓

Is quantum computation a viable technology?





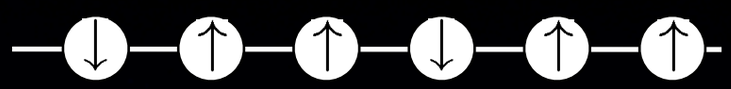
?

Can we write down Moore's law for a quantum computer
....and be correct?

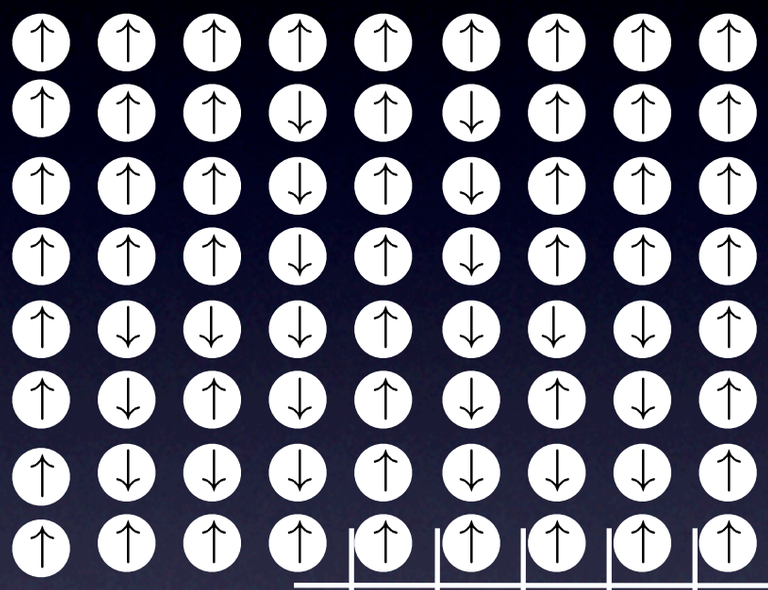
What was it that led us to be able to build classical computers in the first place...

Three Storage Testbeds

Ferromagnetic Ising models:

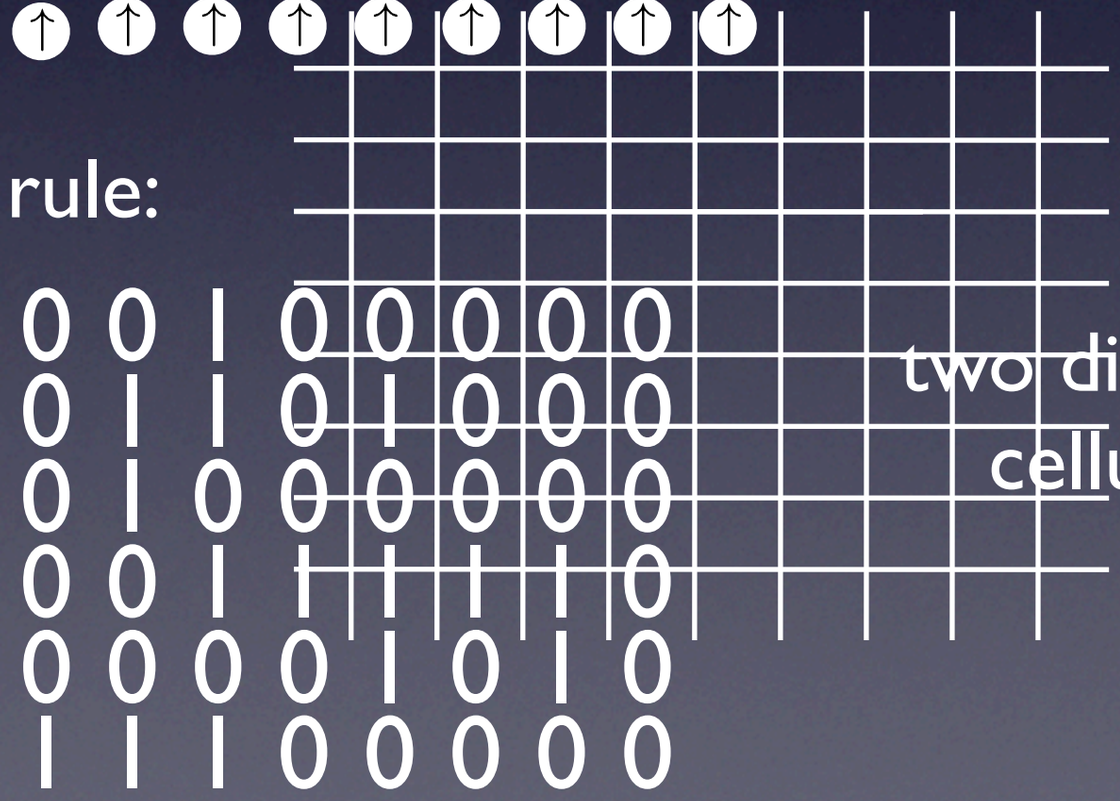


one dimensional



two (and >) dimensional

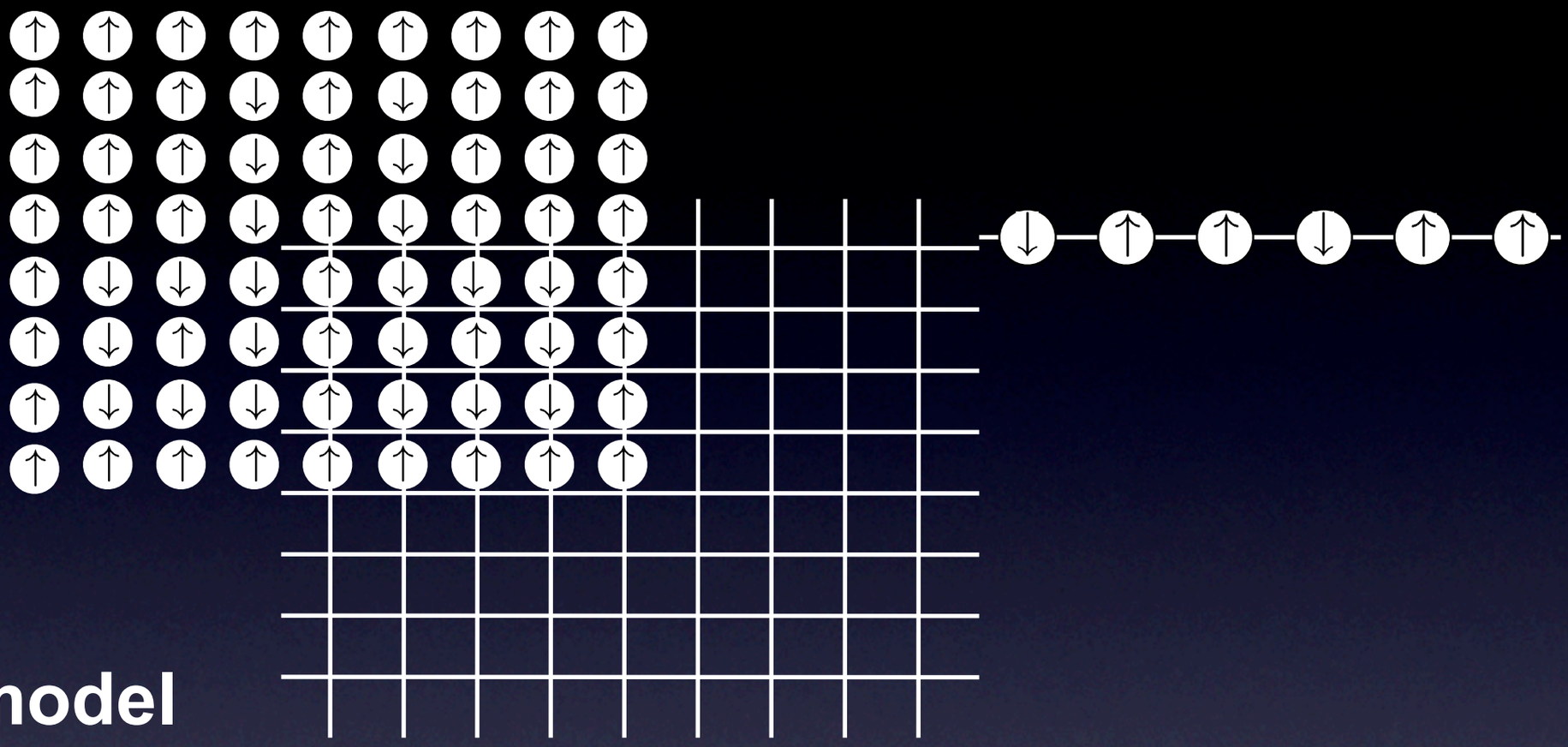
Toom's rule:



two dimensional noisy cellular automata

Ferromagnetic Ising Models





Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

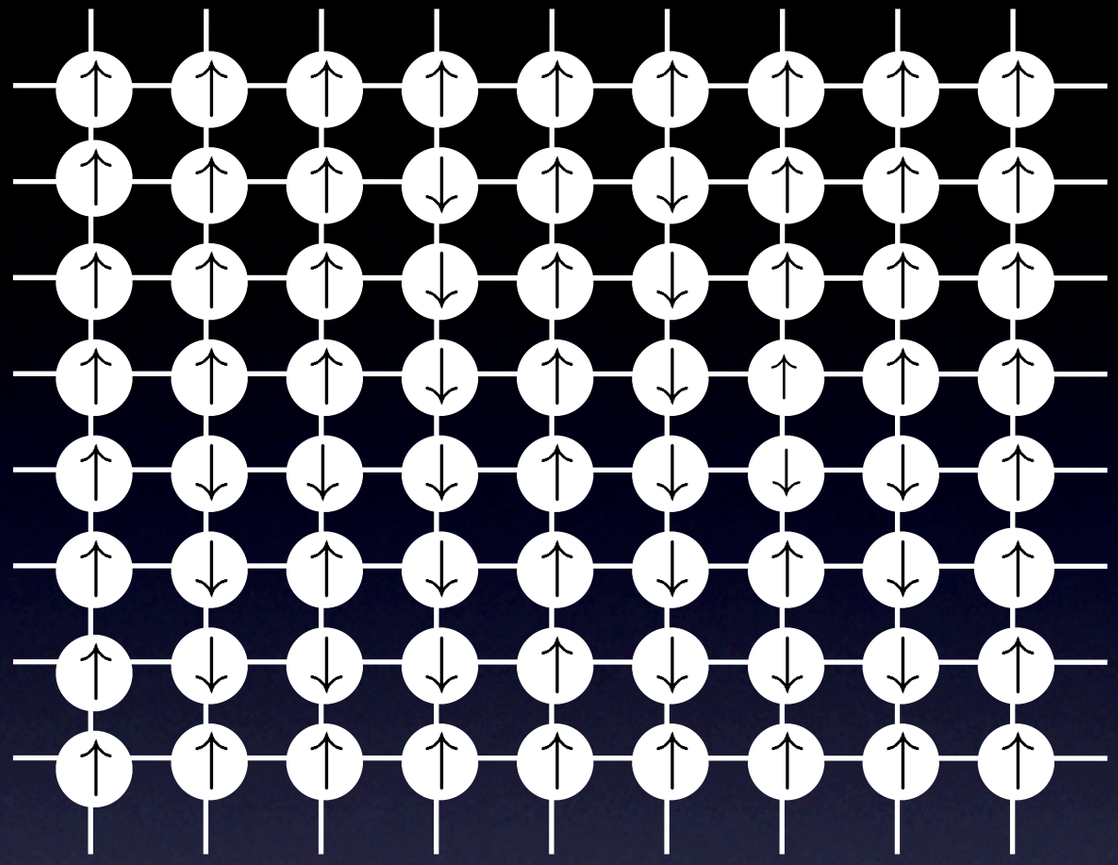
$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

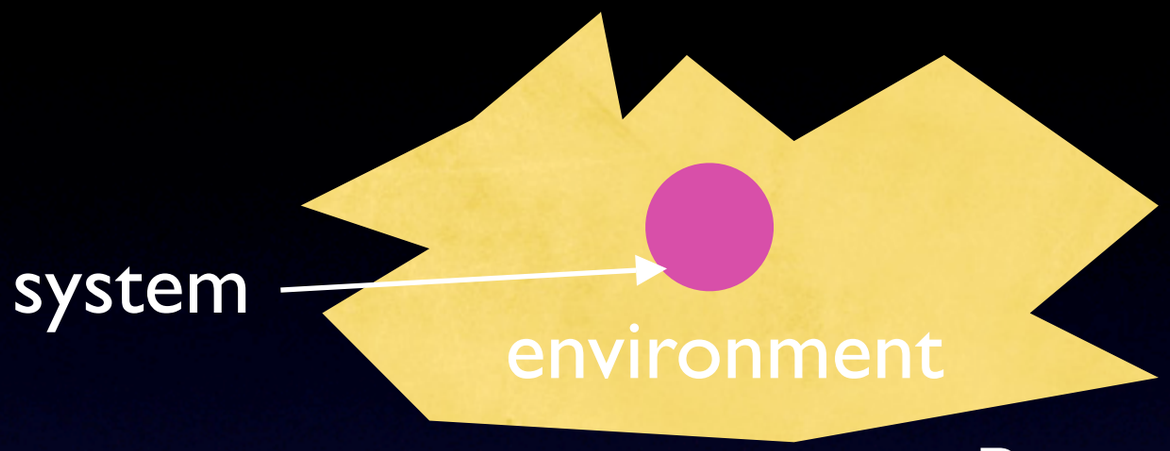
Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Energetics of Decoherence



$$H = H_{sys} + H_{int} + H_{env}$$

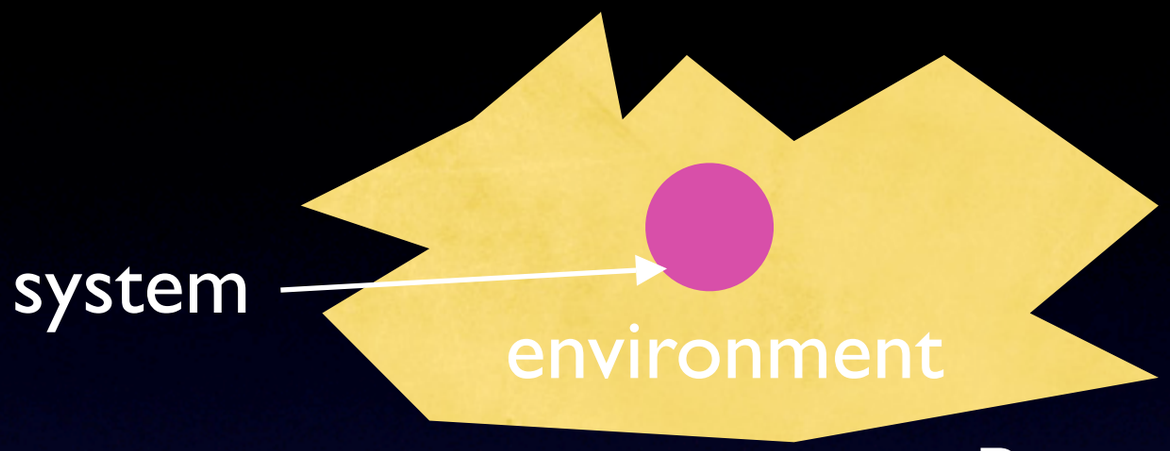
E_{sys} E_{env}

Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.

Energetics of Decoherence

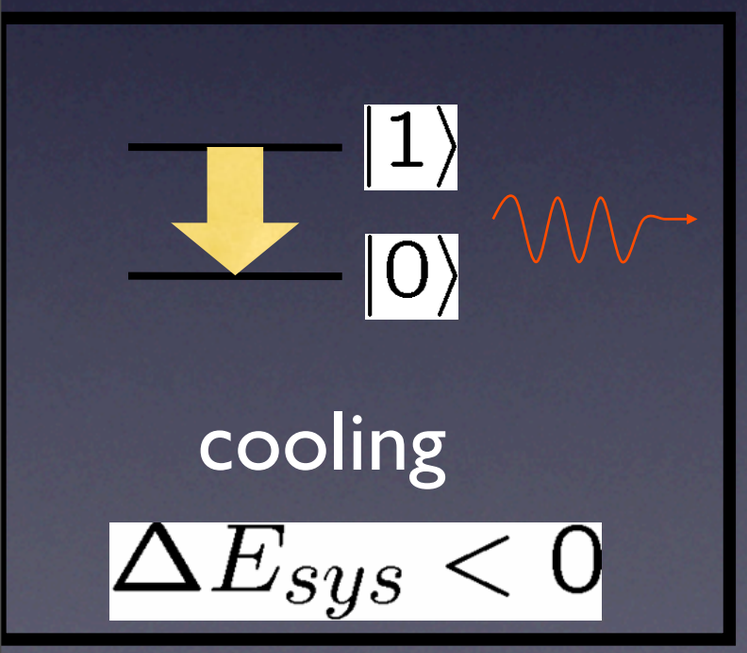
$$H = H_{sys} + H_{int} + H_{env}$$



E_{sys} E_{env}

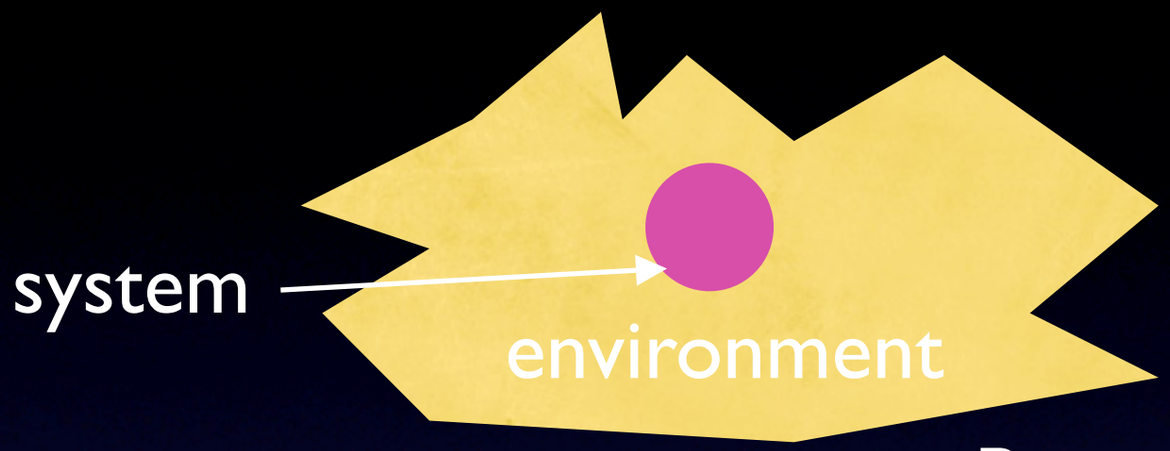
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Energetics of Decoherence

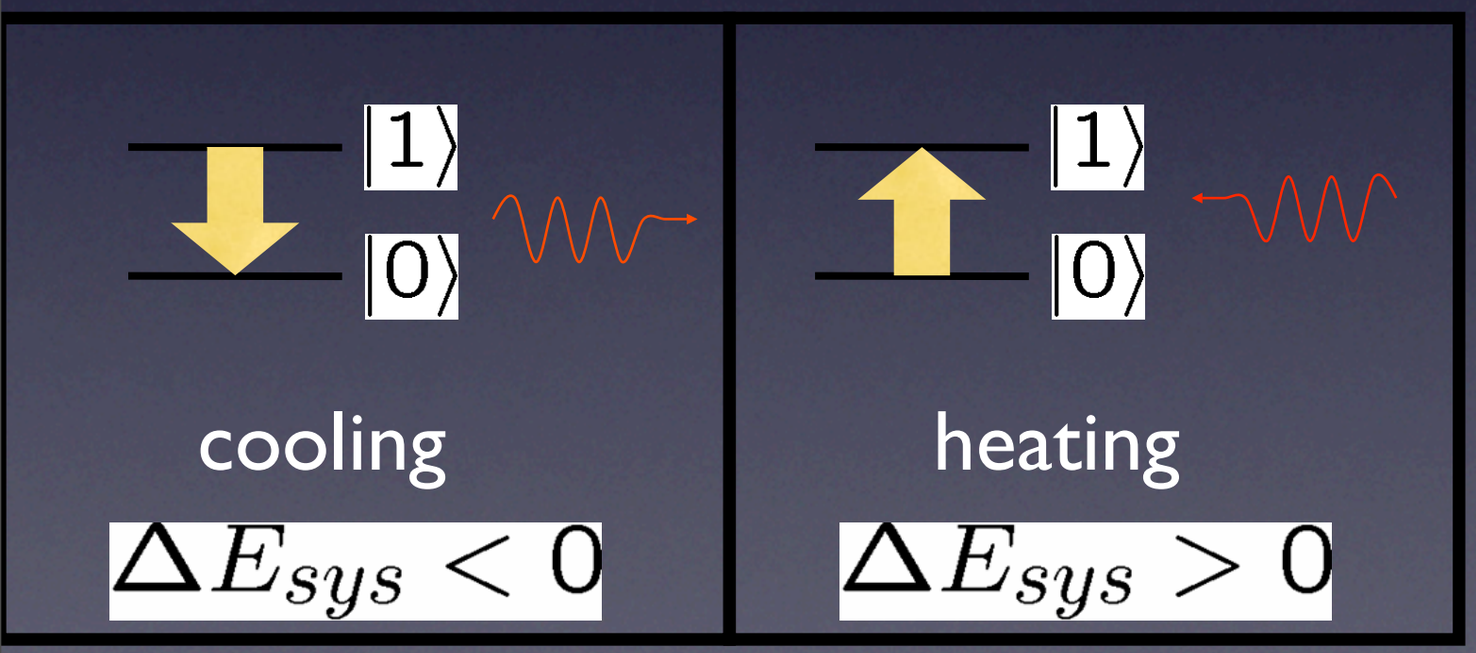
$$H = H_{sys} + H_{int} + H_{env}$$



E_{sys} E_{env}

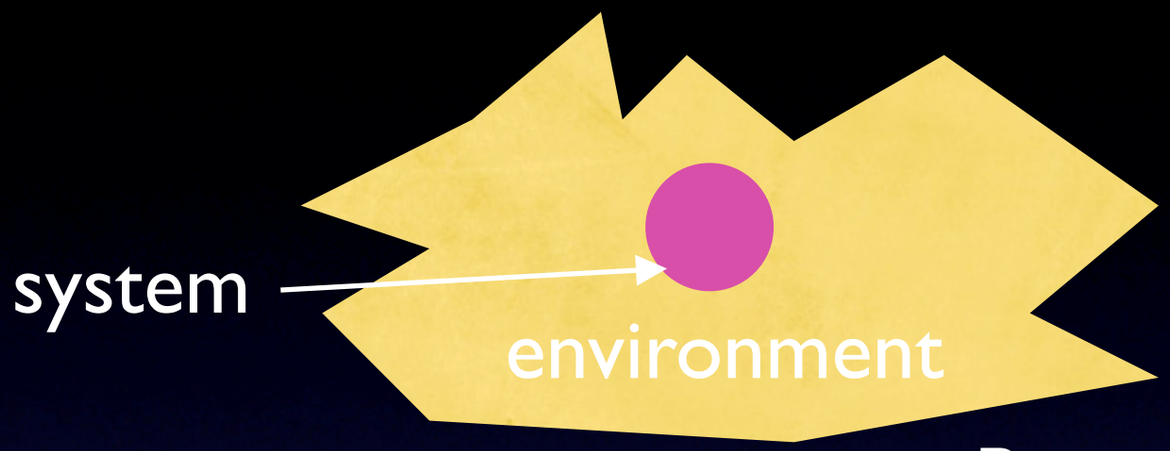
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.



Energetics of Decoherence

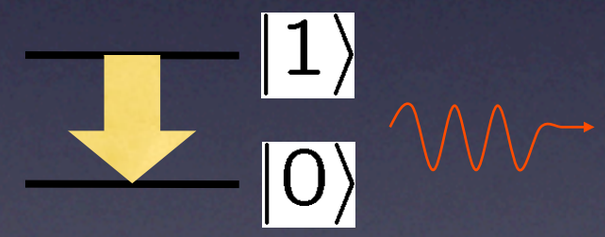
$$H = H_{sys} + H_{int} + H_{env}$$



E_{sys} E_{env}

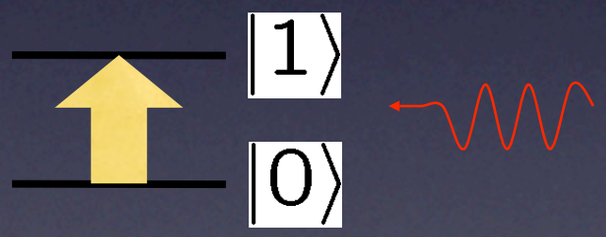
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.



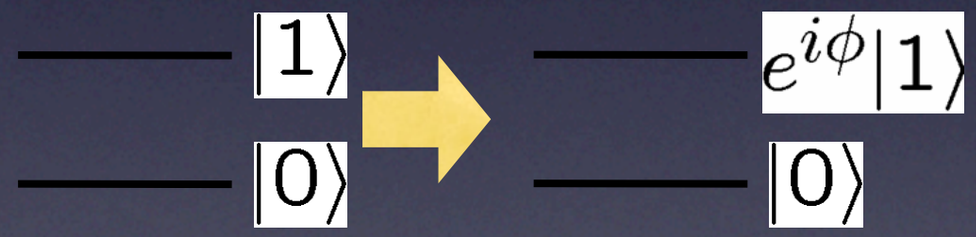
cooling

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$

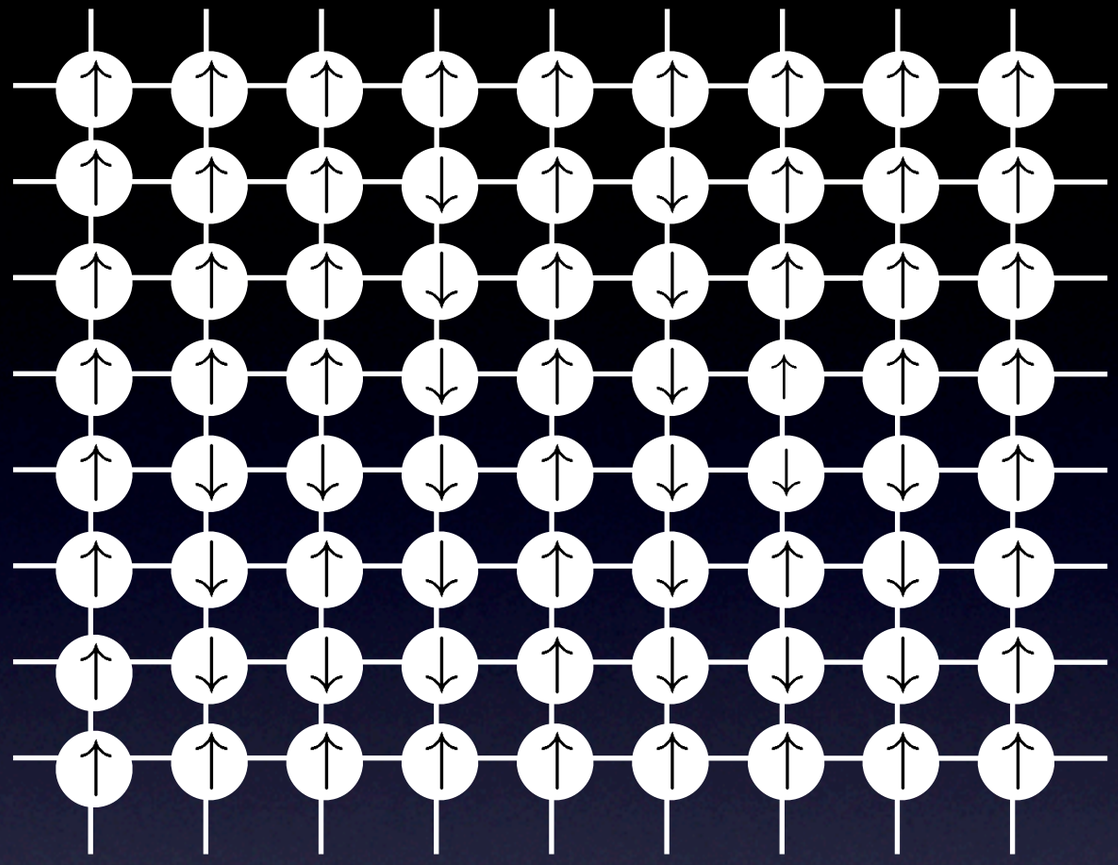


dephasing

$\Delta E_{sys} = 0$

by **cooling environment** we can eliminate heating

Dynamic Ising Models



Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

Update rule:

If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Toom's Rule



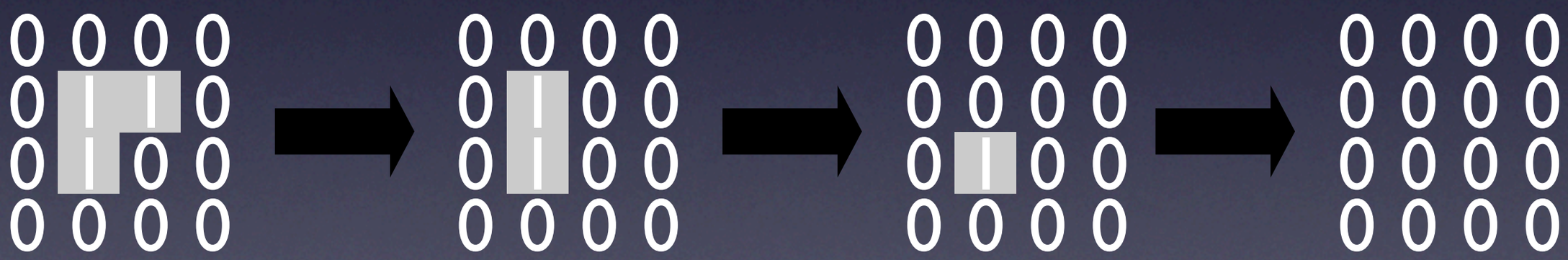
0 0 I 0 0 0 0 0
0 I I 0 I 0 0 0
0 I 0 0 0 0 0 0
0 0 I I I I I 0
0 0 0 0 I 0 I 0
I I I 0 0 0 0 0

two dimensional noisy cellular automata

Update rule:



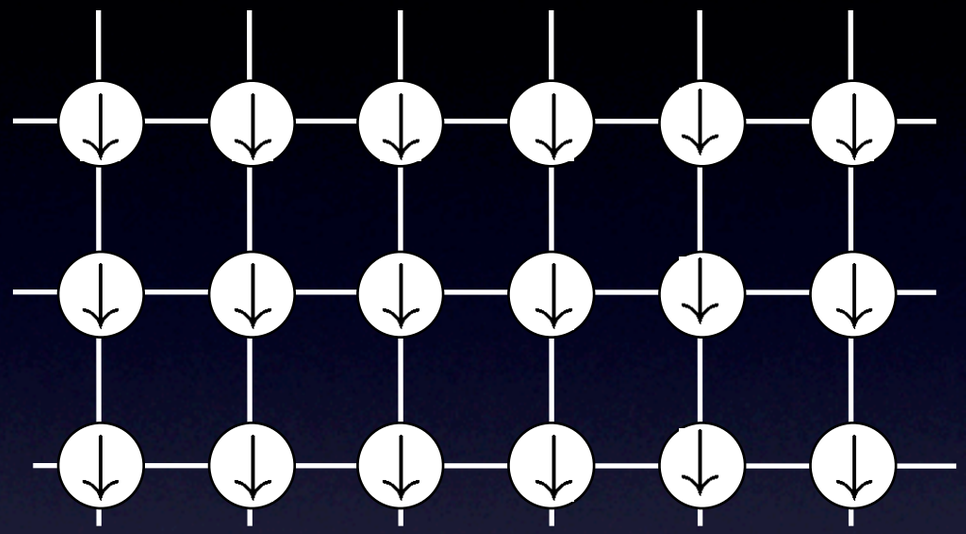
b=majority(n,e,b)



Update rule with noise: with some probability the update fails and a random state is assigned

Compare and Contrast

Encode information into the system and watch what happens



0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0



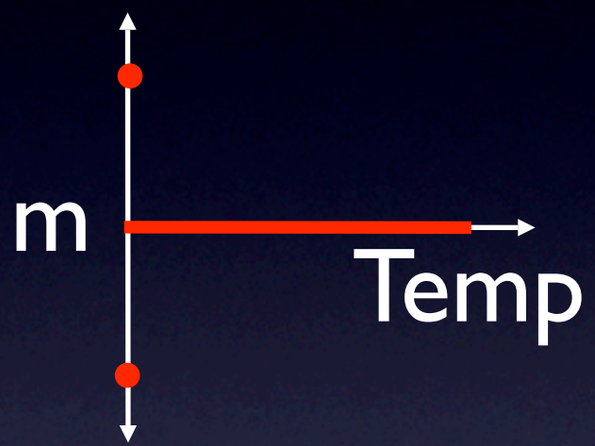
Redundancy code.

"Order parameter" is magnetization: $$m = \frac{1}{N}\sum_i s_i$$

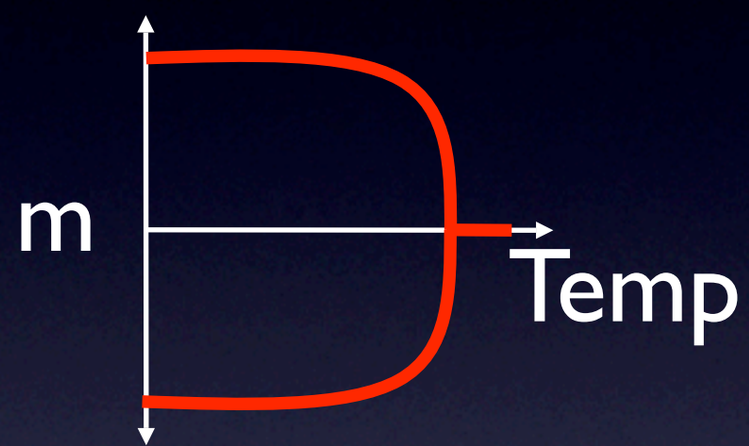
Storage

Long time behavior (not infinite time, but effectively)

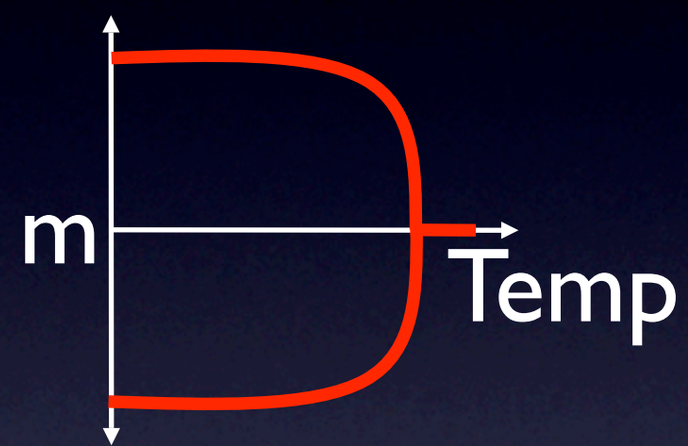
1D Ising



2D Ising



Toom's Rule



Criticism: (1) thermodynamic limit taken
(2) what if relaxation to equilibrium takes a long time?

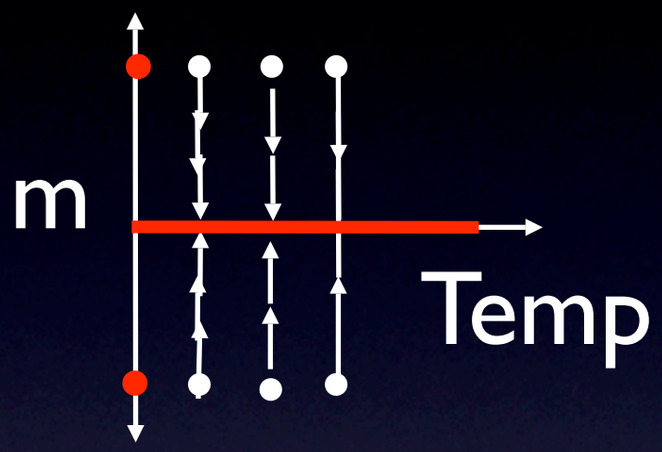
Storage



Relaxation to thermal

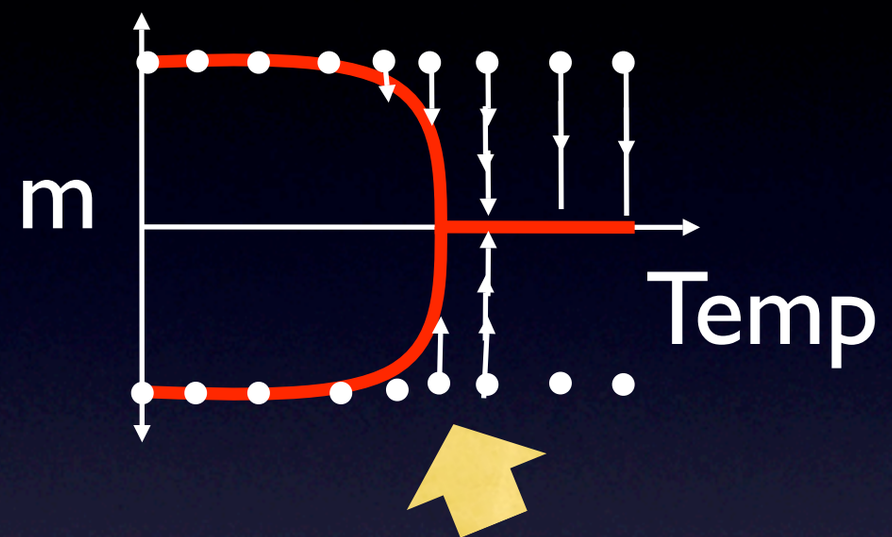
2nd attempt:

1D Ising



decay rate $\propto e^{-\beta J}$

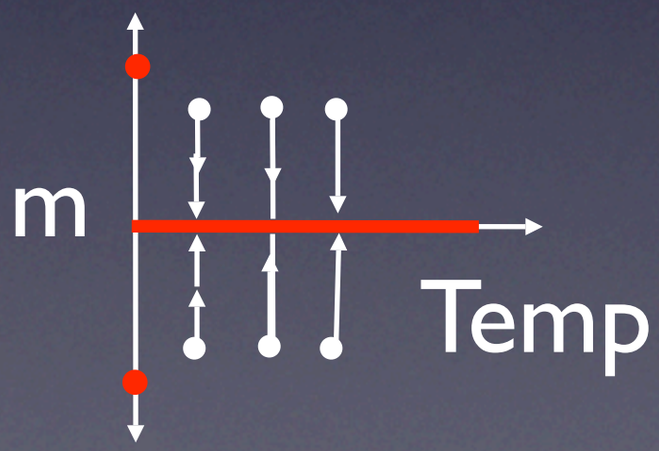
2D Ising, Toom's Rule

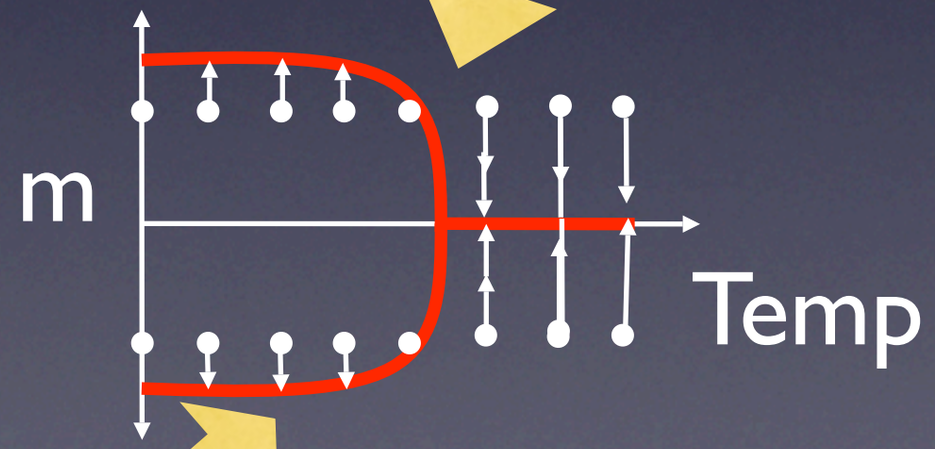


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





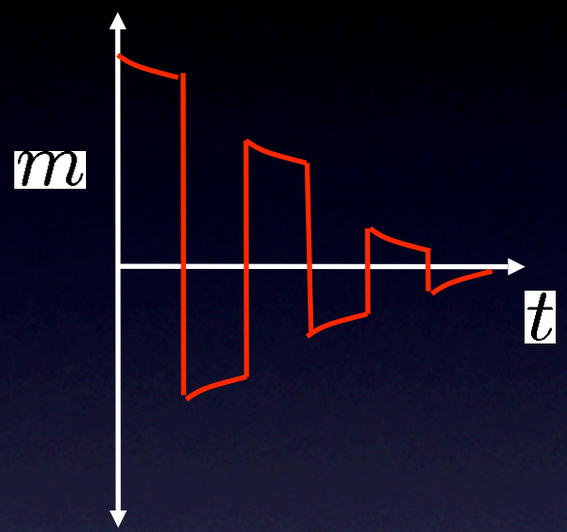
fast relaxation to ordered state

Manipulation

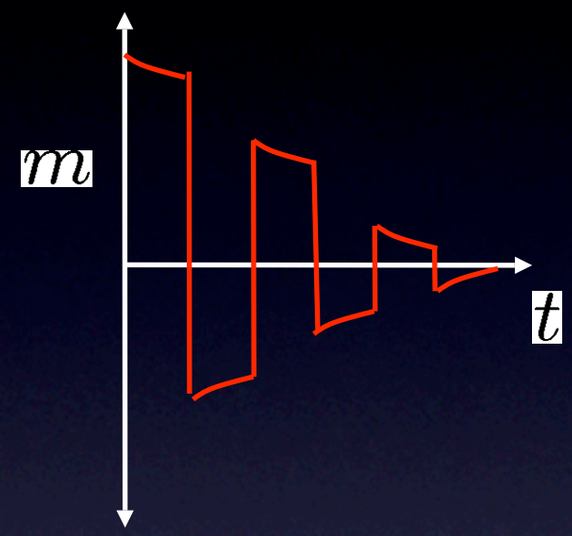


Flipping spins is imperfect

1D Ising



2D Ising
Toom's Rule



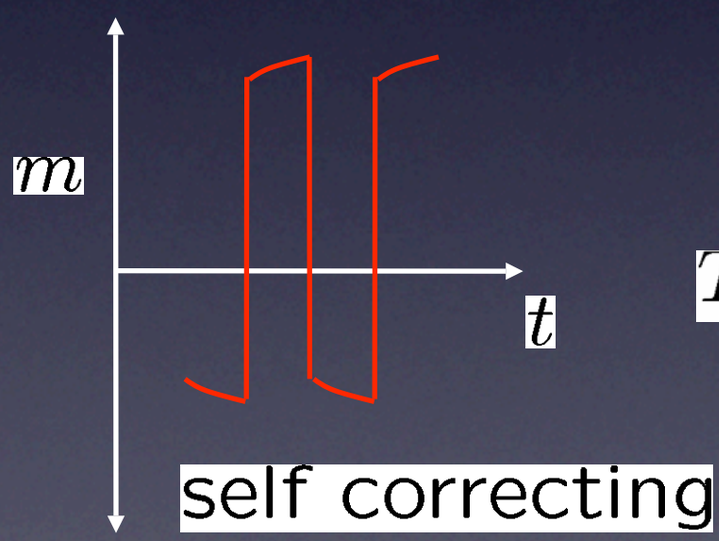
$T > T_C$

1D Ising

- temperature less than gap
- manipulation error must be small (i.e. NOT fault-tolerant)

2D Ising

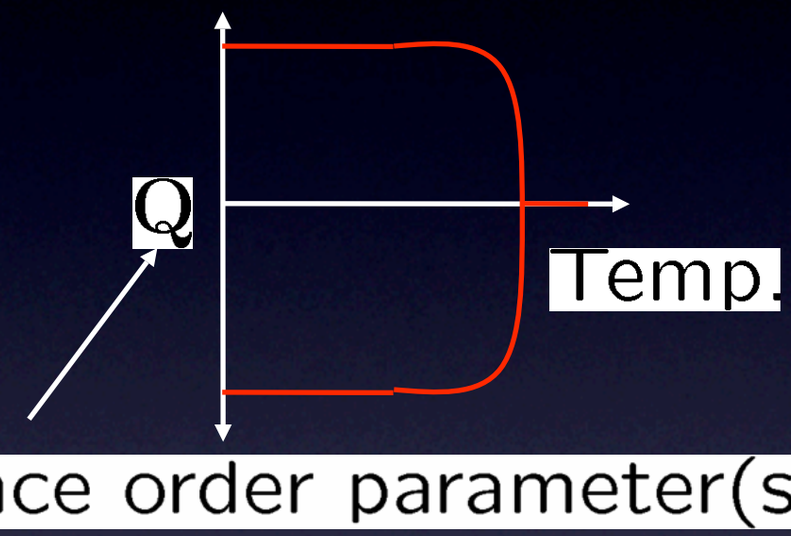
- temperature less than critical temp



$T < T_C$

Quantum Hard Drive?

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?

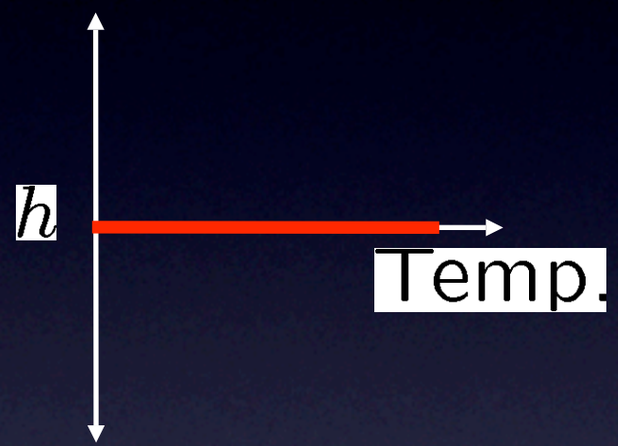


Today in QC: we have theory (like von Neumann, Shannon) but not devices which implement these idea (without severe device complexity overhead.)

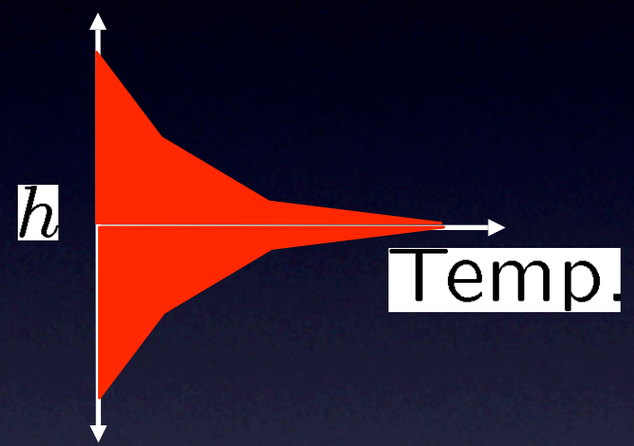
Toom's Rule Versus 2D Ising

Applied magnetic field (corresponds to a bias in noise)

2D Ising



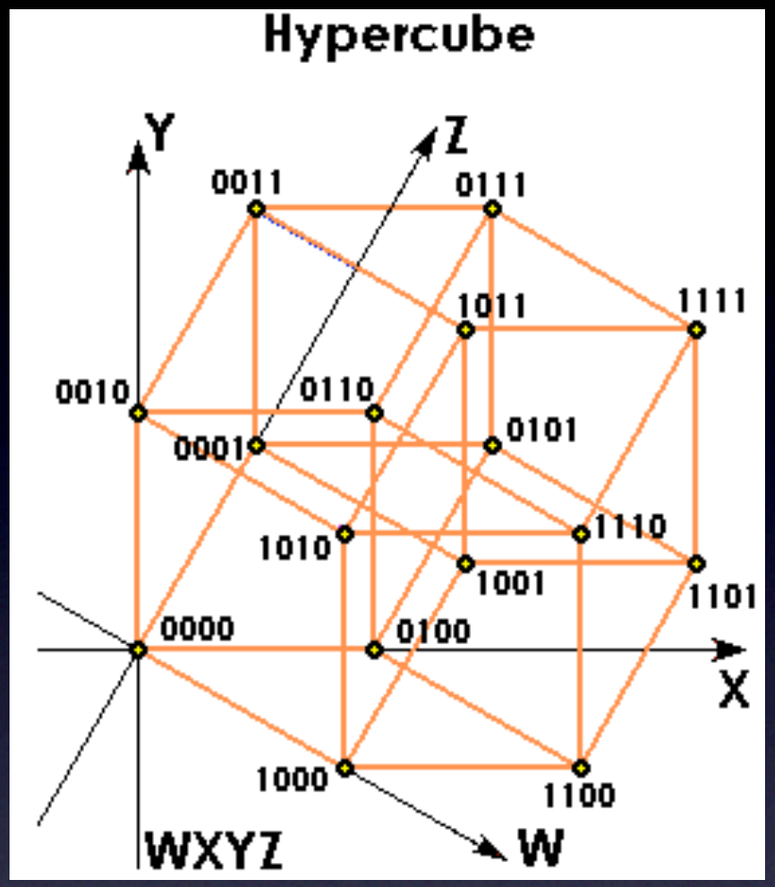
Toom's Rule



Regions with two coexisting phases

External magnetic fields disorder magnetic media

Into the Fourth Dimension



Toric code in four dimension

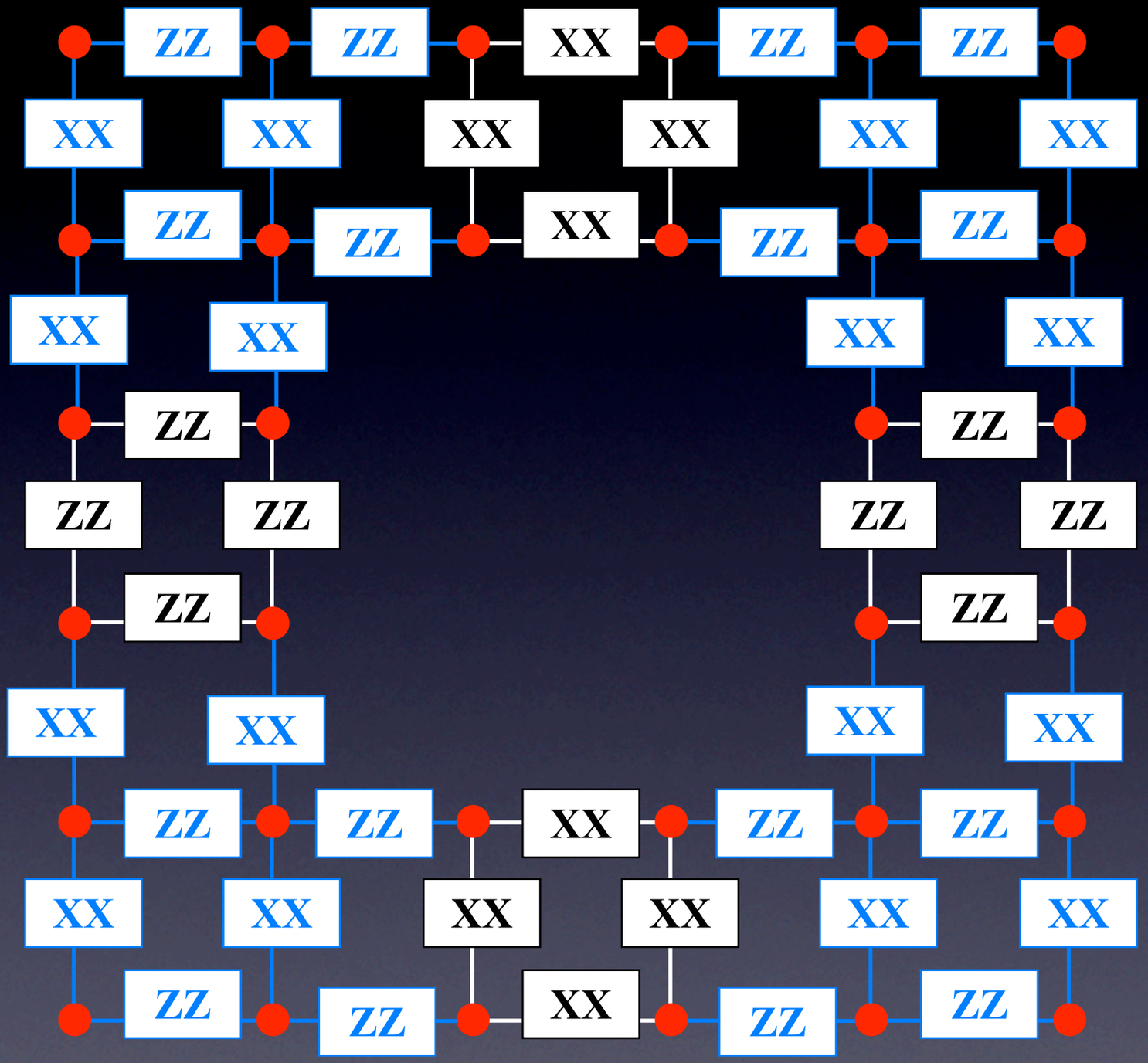
$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

Excitations require energy proportional to perimeter of erred region.

Major open question: are there three dimensional systems which possess a similar stability to quantum errors?

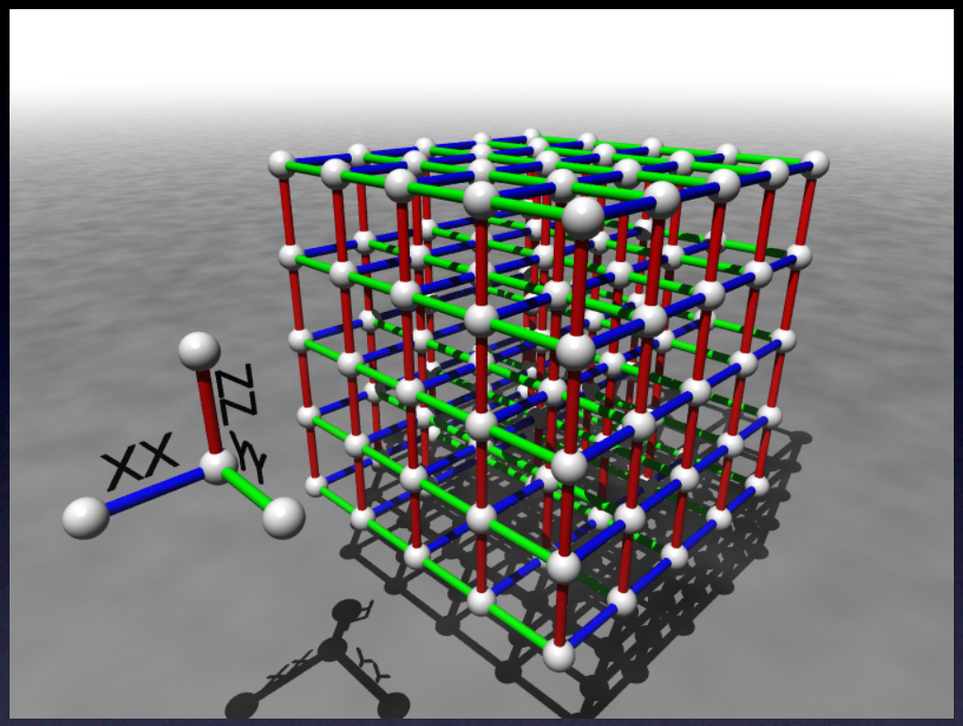
Dennis, Kitaev, Landahl, Preskill, quant-ph/0110143, J. Math. Phys. 43, 4452 (2002)

Without Four Dimensions



Implementable using superconducting qubits

To Build a Quantum Computer....



physics motivated methods
for error correction

Landscape

Quantum Entanglement

Quantum Error Correction

Experimental Quantum Computing

Quantum Information Theory

Quantum Computer Science

ion traps

quantum dots

superconducting circuits

linear optics

Quantum Cryptography

Quantum Algorithms